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SPECIAL RELATIVITY & RELATIVISTIC ELECTRODYNAMICS

A set of Lectures for the first semester of Master's degree in Physics at the University of Delhi, India.

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1 A brief review of Classical Electrodynamics

Classical electromagnetic theory, commonly known as the *classical electromagnetism* deals with physical phenomena involving the interactions of electric charges or (and) currents, or (and) their distributions, within the domain of classical Newtonian physics. Specifically, this implies dealing with electricity and magnetism at length scales much higher than the atomic scale ($\sim 10^{-8} - 10^{-6}$ cm). There is actually the inherent assumption of the validity of certain mathematical limiting processes in which the charge, as well as the current, distributions could be localized in infinitesimally small volumes of space. In other words, the classical limit assumes that an elementary charge or current loop has a continuous density distribution, and the limiting processes would yield results that agree with experiments on non-atomic scales.

Maxwell's theory of *classical electrodynamics*, sometimes called *Maxwell's electromagnetism*, is a “unified” generalization of the classical static theories of electricity and magnetism, when the electric and magnetic field vectors as well as their sources are allowed to vary with both time and space. The entire formulation, i.e. the generalization and in particular the unification, is based on certain empirically established results and some idealistic postulates discussed below.

1.1 Gauss Law and the Non-existence of Magnetic Monopoles

In electrostatics, the *Gauss law* describes the variation of the static electric field in the way that the total electric charge contained in a given volume is the resultant of the electric lines of force entering into or getting out of the volume,. As this happens to be the case at every instant of time, one may straightaway postulate that the Gauss law would continue to describe the field variation in space even in electrodynamics, i.e. when the electric field $\vec{E}$ and the source charges with density ρ are time-dependent. The formal statement of the Gauss law in electrodynamics is therefore the following:

- The net electric flux Φ_E through a closed surface S is equal to 4π times the total net electric charge q in a volume V which is enclosed by the surface S :

$$\Phi_E := \oint_S d\vec{S} \cdot \vec{E}(t, \vec{x}) = 4\pi q. \quad (1.1)$$

Now the total charge q being the integral of the charge density ρ over the volume V , i.e.

$$q = \int_V d^3x \rho(t, \vec{x}), \quad (1.2)$$

one uses the divergence theorem to get

$$\oint_S d\vec{S} \cdot \vec{E}(t, \vec{x}) = \int_V d^3x \nabla \cdot \vec{E}(t, \vec{x}) = 4\pi \int_V d^3x \rho(t, \vec{x}). \quad (1.3)$$

Since the volume V is arbitrary, one therefore has the Gauss law in differential form as

$$\nabla \cdot \vec{E}(t, \vec{x}) = 4\pi \rho(t, \vec{x}). \quad (1.4)$$

For the magnetic field $\vec{B}$, the corresponding statement, often called the *Gauss law for magnetism*, is that

- The net magnetic flux Φ_B through a closed surface S would vanish identically if a volume V which is enclosed by the surface S consists only of free magnetic charges or *monopoles*:

$$\Phi_B := \oint_S d\vec{S} \cdot \vec{B}(t, \vec{x}) \stackrel{(\text{free})}{=} 0. \quad (1.5)$$

It is in fact more convenient to use this equation in differential form

$$\nabla \cdot \vec{B}(t, \vec{x}) = 0, \quad (1.6)$$

which physically implies that the effect of the magnetic field lines entering in any elementary volume of space is exactly canceled by that of the field lines exiting that volume element, at a particular instant of time. Thus, no net “magnetic monopole” can build up anywhere in space, i.e. one can equivalently state that:

- Magnetic monopoles do not exist.

1.2 Equation of Continuity

With the general (time-dependent) charge density and current density denoted respectively by $\rho(t, \vec{x})$ and $\vec{j}(t, \vec{x})$, the conservation of charge demands that at any point in space $\rho(t, \vec{x})$ and $\vec{j}(t, \vec{x})$ should satisfy the so-called *equation of continuity*:

$$\boxed{\nabla \cdot \vec{j}(t, \vec{x}) + \frac{\partial}{\partial t} \rho(t, \vec{x}) = 0} . \quad (1.7)$$

In alternate form, integrating Eq. (1.7) over a small volume V and using the divergence theorem, we have

$$- \int_V d^3x \frac{\partial \rho}{\partial t} = - \frac{\partial q}{\partial t} = \int_S d^2x \hat{n} \cdot \vec{j} , \quad (1.8)$$

where S is the surface (with unit normal $\hat{n}$) that encloses the volume V which contains the charge q , corresponding to the density ρ . The physical interpretation of Eq. (1.8) is that

- *The time rate of reduction of charge within a volume, however small, should balance the outward flux of charge through the surface enclosing that volume, in order to conserve the total charge.*

For charges (with density ρ) moving freely with velocity $\vec{v}$, the resulting “free” current density (sometimes called the *conduction current density*) is $\vec{j} = \rho \vec{v}$, and the above continuity equation (1.7) reduces to

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0 . \quad (1.9)$$

1.3 Ohm’s Law

A purely observational fact is summed up in the *Ohm’s law* which states that

- *For most conductors, a linear relationship holds between the current density and the applied electric field that produces it:*

$$\boxed{\vec{j}(t, \vec{x}) = \sigma \vec{E}(t, \vec{x})} , \quad (1.10)$$

where the proportionality factor σ is known as the *electric conductivity*, which is a characteristic of the material of the conductor. Perfect conductors have $\sigma \rightarrow \infty$ and for them $\vec{E} = \vec{j}/\sigma \rightarrow 0$. In other words, a conductor is said to be “good” (if not perfect) if the electric field required to drive current in it is negligible. Ohm’s law (1.10) is actually an approximate, but effective, realization of the statement that

- *The density $\vec{j}$ of the current flow, due to the motion of charges, is proportional to the force per unit charge $\vec{f}$ in most substances, i.e. $\vec{j} = \sigma \vec{f}$.*

If the force is electromagnetic then for small enough velocities of the charges, $\vec{f}$ is approximately equal to $\vec{E}$ — the applied electric field, and hence we have the Ohm’s law.

1.4 Faraday’s Law and Lenz’s Law

The inference drawn by Faraday on the basis of a series of experiments is that

- *A changing magnetic flux linked by a closed circuit C causes a transient current flow in the circuit.*

Actually, the changing magnetic flux induces an electric field $\vec{E}$ in the circuit C . The line integral of $\vec{E}$ is the so-called *electromotive force* (EMF) that is induced around C :

$$\mathfrak{E} := \oint_C \vec{dl} \cdot \vec{E} . \quad (1.11)$$

The EMF $\mathfrak{E}$ causes the flow of the transient current in C , by virtue of the Ohm’s law.

The formal statement of *Faraday's law* is that

- The induced EMF around any closed circuit is proportional to the time rate of decrease of the magnetic flux linking the circuit.

That it is indeed *decrease* is specified by *Lenz's law*, which states that

- The EMF induced in a closed circuit gives rise to an electric current, which is in such a direction that the magnetic field due to it opposes the change in the magnetic flux that had caused the induction.

In mathematical terms, Faraday's law is expressed as

$$\boxed{\mathfrak{E} = - \frac{1}{c} \frac{d\Phi_B}{dt}}, \quad (1.12)$$

where Φ_B is the magnetic flux through the surface S (with outward normal $\hat{\mathbf{n}}$) enclosed by the circuit C . By definition, the magnetic flux is given by

$$\Phi_B := \int_S d\vec{S} \cdot \vec{B} = \int_S d^2x \hat{\mathbf{n}} \cdot \vec{B}, \quad (1.13)$$

where $\vec{B}$ is the magnetic induction in the neighbourhood. Substituting Eqs. (1.11) and (1.13) in Eq. (1.12) we get

$$\boxed{\oint_C d\vec{\ell} \cdot \vec{E} = - \frac{1}{c} \int_S d^2x \hat{\mathbf{n}} \cdot \frac{d\vec{B}}{dt}}, \quad (1.14)$$

which is valid as long as the circuit C is *fixed* in space.

For a circuit C which is moving with a velocity $\vec{v}$ in some direction w.r.to a coordinate system (say, the laboratory frame of reference, as is generally the case), the equivalent statement of the Faraday's law is

$$\boxed{\oint_C d\vec{\ell} \cdot \vec{E} = - \frac{1}{c} \int_S d^2x \hat{\mathbf{n}} \cdot \frac{\partial \vec{B}}{\partial t}}, \quad (1.15)$$

where $\vec{E}$ is the electric field in the laboratory. Using the Stokes theorem one can convert the line integral in the l.h.s. to a surface integral over the surface S bounded by the circuit C , whence

$$\int_S d^2x \hat{\mathbf{n}} \cdot \left(\nabla \times \vec{E} + \frac{1}{c} \frac{\partial \vec{B}}{\partial t} \right) = 0, \quad (1.16)$$

and since the surface S is arbitrary, one has Faraday's law in differential form

$$\boxed{\nabla \times \vec{E} + \frac{1}{c} \frac{\partial \vec{B}}{\partial t} = 0}. \quad (1.17)$$

1.5 Lorentz Force Equation

While working out the mathematical equation for the Faraday's law in the general case of a circuit moving with a velocity $\vec{v}$ w.r.to the laboratory, one demands that the equation should be *form-invariant*, i.e. the Faraday's law should be *covariant*, under the frame transformations in *Galilean relativity*. Such a demand implies that one should have the relation between the electric field $\vec{E}'$ in the moving circuit and that in the laboratory, $\vec{E}$, given by

$$\vec{E}' = \vec{E} + \frac{1}{c} \vec{v} \times \vec{B}. \quad (1.18)$$

For a free particle with charge q at rest w.r.to the moving circuit, the force due to the electric field $\vec{E}'$ is thus

$$\vec{F} = q\vec{E}' = q\left(\vec{E} + \frac{1}{c}\vec{v} \times \vec{B}\right). \quad (1.19)$$

That is, in the laboratory frame, we have in addition to the force $\vec{F}_E = q\vec{E}$ due to the corresponding electric field $\vec{E}$, there is an additional force $\vec{F}_B = (q/c)\vec{v} \times \vec{B}$ due to the magnetic induction $\vec{B}$:

$$\vec{F} = \vec{F}_E + \vec{F}_B; \quad \vec{F}_E = q\vec{E}, \quad \vec{F}_B = \frac{q}{c}\vec{v} \times \vec{B}. \quad (1.20)$$

From the laboratory point of view, $\vec{F}_B$ can be interpreted as the force experienced by a current, of density $\vec{j} = \rho\vec{v}$, which is due to the motion with velocity $\vec{v}$ of the free charge q ($= \int d^3x \rho$, with ρ as the charge density). This is a consequence of the postulate that the *Ampère's force relation* in magnetostatics, which follows from the *Biot-Savart law*, can be generalized in a straightforward way in electrodynamics, so that for the free charge q and a fixed velocity $\vec{v}$ one has

$$\begin{aligned} \vec{F}_B(t, \vec{x}) &= \frac{1}{c} \int d^3x' \vec{j}(t, \vec{x}') \times \vec{B}(t, \vec{x}) = \frac{1}{c} \int d^3x' [\rho(t, \vec{x}') \vec{v}] \times \vec{B}(t, \vec{x}) \\ &= \frac{1}{c} \left[\int d^3x' \rho(t, \vec{x}') \right] \vec{v} \times \vec{B}(t, \vec{x}) = \frac{q}{c} \vec{v} \times \vec{B}(t, \vec{x}). \end{aligned} \quad (1.21)$$

Of course, as one may notice, here we have made allusion only to the Galilean covariance of the Faraday's law. Strictly speaking, this seems to be an approximation because, as we shall discuss later, the Galilean principle of relativity is only valid for frame speeds $v \ll c$. Faraday's law is no approximation though — its validity is beyond that of the Galilean relativity, and so is the validity of the governing force equation (1.19), referred to as the *Lorentz force* on a particle of charge q moving with a velocity $\vec{v}$ in an electromagnetic field:

$$\boxed{\vec{F} \equiv \vec{F}_L = q\left(\vec{E} + \frac{1}{c}\vec{v} \times \vec{B}\right)}. \quad (1.22)$$

1.6 Maxwell's generalization of the Ampère's Law and the Displacement Current

The *Ampère's law*, which is of central importance in magnetostatics, determines the static magnetic field associated with a steady current, or vice versa. The formal statement of the Ampère's law is that

- The line integral of the static magnetic field $\vec{B}(\vec{x})$ around a closed curve C is equal to $(4\pi)/c$ times the total current I , with density $\vec{j}(\vec{x})$, through a surface S enclosed by the curve C :

$$\oint_C d\vec{\ell} \cdot \vec{B}(\vec{x}) = \frac{4\pi}{c} \int_S d\vec{S} \cdot \vec{j}(\vec{x}) = \frac{4\pi}{c} I, \quad (1.23)$$

or, in the differential form

$$\nabla \times \vec{B}(\vec{x}) = \frac{4\pi}{c} \vec{j}(\vec{x}). \quad (1.24)$$

A straightforward generalization of this in electrodynamics, however, leads to a contradiction. This could be seen quite easily: as one tries to generalize Eq. (1.24) by making the substitution $\vec{B}(\vec{x}) \rightarrow \vec{B}(t, \vec{x})$, $\vec{j}(\vec{x}) \rightarrow \vec{j}(t, \vec{x})$, the l.h.s. of the resulting equation is identically divergence-free whereas the r.h.s. fails to be so, by virtue of the continuity equation (1.7).

Maxwell's resolved this problem via his famous postulate of the so-called *displacement current density* $\vec{j}_D(t, \vec{x})$, which in addition to the usual (conduction or/and convection) current density $\vec{j}(t, \vec{x})$ composes the r.h.s. of the generalized Ampère's law:

$$\boxed{\nabla \times \vec{B}(t, \vec{x}) = \frac{4\pi}{c} [\vec{j}(t, \vec{x}) + \vec{j}_D(t, \vec{x})]}. \quad (1.25)$$

For consistency with the continuity equation (1.7), the displacement current density is deduced to be

$$\vec{j}_D(t, \vec{x}) = \frac{1}{4\pi} \frac{\partial}{\partial t} \vec{E}(t, \vec{x}) . \quad (1.26)$$

In S.I. units,

$$\vec{j}_D(t, \vec{x}) = \frac{\partial}{\partial t} [\epsilon_0 \vec{E}(t, \vec{x})] \equiv \frac{\partial}{\partial t} \vec{D}(t, \vec{x}) , \quad (1.27)$$

where ϵ_0 is the free space *permittivity* and $\vec{D} = \epsilon_0 \vec{E}$ is known as the *electric displacement* vector. The displacement current behaves like a current even in absolutely empty space (without any free or bound electric charges or current loops). The proportionality of $\vec{j}_D$ and the time derivative of the electric field $\vec{E}$ demonstrates that

- A time-varying electric field induces a time-varying magnetic field — the converse of the Faraday's law.

1.7 Maxwell's Electrodynamic Equations

The equations governing the space or (and) time variation of the electric and magnetic fields, $\vec{E}$ and $\vec{B}$, with or without the space(time) variation of the source densities, ρ and $\vec{j}$, are grouped together as a set of four equations, known as the *Maxwell's equations* in electrodynamics:

$$\left. \begin{array}{ll} \text{(i)} \quad \nabla \cdot \vec{E} = 4\pi \rho , & \text{(iii)} \quad \nabla \times \vec{E} + \frac{1}{c} \frac{\partial \vec{B}}{\partial t} = 0 , \\ \text{(ii)} \quad \nabla \cdot \vec{B} = 0 , & \text{(iv)} \quad \nabla \times \vec{B} - \frac{1}{c} \frac{\partial \vec{E}}{\partial t} = \frac{4\pi}{c} \vec{j} \end{array} \right\} . \quad (1.28)$$

In S.I. units,

$$\left. \begin{array}{ll} \text{(i)} \quad \nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} , & \text{(iii)} \quad \nabla \times \vec{E} + \frac{\partial \vec{B}}{\partial t} = 0 , \\ \text{(ii)} \quad \nabla \cdot \vec{B} = 0 , & \text{(iv)} \quad \nabla \times \vec{B} - \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t} = \mu_0 \vec{j} \end{array} \right\} , \quad (1.29)$$

where ϵ_0, μ_0 are respectively the free space *permittivity* and *permeability*, which are related as $\epsilon_0 \mu_0 = 1/c^2$.

1.8 Electrodynamic Potentials

Since $\nabla \cdot \vec{B} = 0$, we can always express

$$\vec{B}(t, \vec{x}) = \nabla \times \vec{A}(t, \vec{x}) , \quad (1.30)$$

where $\vec{A}(t, \vec{x})$ is called the *magnetic vector potential* (or, simply the *vector potential* in electrodynamics). Substituting Eq. (1.30) in the r.h.s. of the Faraday's law equation (1.28 iii) we have

$$\nabla \times \left[\vec{E}(t, \vec{x}) + \frac{1}{c} \frac{\partial}{\partial t} \vec{A}(t, \vec{x}) \right] = 0 . \quad (1.31)$$

As the curl of the gradient of any scalar is identically zero, the quantity within the brackets [] in the l.h.s. of the above equation can be customarily expressed as the negative of the gradient of a scalar function $\phi(t, \vec{x})$, known as the *electromagnetic scalar potential* (or, simply the *scalar potential* in electrodynamics). The electric field is therefore

$$\vec{E}(t, \vec{x}) = - \nabla \phi(t, \vec{x}) - \frac{1}{c} \frac{\partial}{\partial t} \vec{A}(t, \vec{x}) . \quad (1.32)$$

A detailed account of the potential formulation of electrodynamics will be given later on. However, for the time being, we shall discuss in the next section the conflict that arose between Maxwell's theory of electrodynamics, right after its formulation, and then prevalent Galilean relativity. We will also elucidate on what were the possible measures to resolve such a conflict, and what measure was actually adopted for the resolution.

2 Confrontation of Classical Electrodynamics with Galilean Relativity

2.1 Galilean Transformations

Consider two systems of reference Σ and Σ' . Σ is characterized by coordinates (x, y, z) and time t , whereas Σ' is characterized by coordinates (x', y', z') and time t' . Let the origins of the two systems coincide at $t = t' = 0$ and Σ' is moving relative to Σ with constant velocity $\vec{v}$ in an arbitrary direction (see Fig. 1), although for simplicity we consider that the axes of the two frames are parallelly oriented.

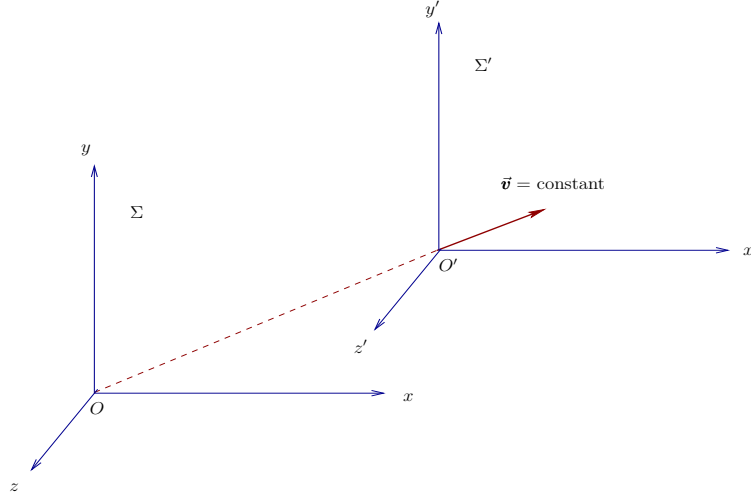


Figure 1: Galilean transformations for frame motion in an arbitrary direction, preserving the axes orientation.

The *Galilean transformation* equations represent a set of linear coordinate transformations given by

$$\begin{array}{l} \boxed{\begin{array}{l} t' = t \\ x' = x - v_x t \\ y' = y - v_y t \\ z' = z - v_z t \end{array}} \quad \text{or, in vector notation:} \quad \boxed{\begin{array}{l} t' = t \\ \vec{x}' = \vec{x} - \vec{v}t \end{array}}, \end{array} \quad (2.1)$$

where v_x, v_y, v_z are the components of $\vec{v}$ along x, y and z respectively.

2.2 Galilean Principle of Relativity

“All laws of mechanics are invariant under the Galilean transformations.”

2.2.1 Illustration: System of particles interacting via two-body central potentials

Equation of motion of the i^{th} particle in the reference frame Σ' :

$$m_i \frac{d\vec{v}'_i}{dt'} = - \nabla'_i \sum_j V_{ij} |\vec{x}'_i - \vec{x}'_j| \quad (2.2)$$

m_i being the mass of the i^{th} particle, which interacts with the j^{th} particle via the potential V_{ij} . From Eqs. (2.1) it follows:

$$t' = t \quad , \quad \vec{x}'_i - \vec{x}'_j = \vec{x}_i - \vec{x}_j \quad , \quad (2.3)$$

so that

$$\vec{v}'_i = \frac{d\vec{x}'_i}{dt'} = \frac{d\vec{x}_i}{dt} - \vec{v} = \vec{v}_i - \vec{v} \quad \implies \quad \frac{d\vec{v}'_i}{dt'} = \frac{d\vec{v}_i}{dt} \quad (2.4)$$

Now, to work out the relations between the coordinate differentials in the two frames Σ and Σ' , let us consider a scalar function ψ [$\psi = \psi(t, x, y, z)$ in Σ , and $\psi = \psi(t', x', y', z')$ in Σ']. Then

$$d\psi = \frac{\partial\psi}{\partial t}dt + \frac{\partial\psi}{\partial x}dx + \frac{\partial\psi}{\partial y}dy + \frac{\partial\psi}{\partial z}dz = \frac{\partial\psi}{\partial t'}dt' + \frac{\partial\psi}{\partial x'}dx' + \frac{\partial\psi}{\partial y'}dy' + \frac{\partial\psi}{\partial z'}dz'. \quad (2.5)$$

Or,

$$\frac{\partial\psi}{\partial t}dt + \frac{\partial\psi}{\partial x}dx + \frac{\partial\psi}{\partial y}dy + \frac{\partial\psi}{\partial z}dz = \frac{\partial\psi}{\partial t'}dt + \frac{\partial\psi}{\partial x'}(dx - v_x dt) + \frac{\partial\psi}{\partial y'}(dy - v_y dt) + \frac{\partial\psi}{\partial z'}(dz - v_z dt). \quad (2.6)$$

Equating separately the coefficients of dt, dx, dy and dz on both sides, we have

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial x'} \quad , \quad \frac{\partial}{\partial y} = \frac{\partial}{\partial y'} \quad , \quad \frac{\partial}{\partial z} = \frac{\partial}{\partial z'} \quad , \quad \text{and} \quad \frac{\partial}{\partial t} = \frac{\partial}{\partial t'} - v_x \frac{\partial}{\partial x'} - v_y \frac{\partial}{\partial y'} - v_z \frac{\partial}{\partial z'}. \quad (2.7)$$

Or, in vectorial form

$$\nabla = \nabla' \quad ; \quad \frac{\partial}{\partial t} = \frac{\partial}{\partial t'} - \vec{v} \cdot \nabla', \quad (2.8)$$

and conversely,

$$\nabla' = \nabla \quad , \quad \frac{\partial}{\partial t'} = \frac{\partial}{\partial t} + \vec{v} \cdot \nabla. \quad (2.9)$$

Using Eqs. (2.3), (2.4) and the first relation of (2.9), we can write Eq. (2.2) as

$$m_i \frac{d\vec{v}_i}{dt} = - \nabla_i \sum_j V_{ij} |\vec{x}_i - \vec{x}_j| \quad (2.10)$$

which is the equation of motion of the i^{th} particle in the reference frame Σ . This demonstrates that the Galilean transformation equations preserve the form of the Newtonian equations of motion.

2.2.2 Counter-illustration: Homogeneous wave equation for a scalar field

Let us again consider a scalar field ψ , and suppose that it satisfies the (homogeneous) wave equation in the reference frame Σ :

$$\left(\frac{1}{u^2} \frac{\partial^2}{\partial t^2} - \nabla^2 \right) \psi(t, \vec{x}) = 0, \quad (2.11)$$

where the speed of propagation of the wave is u .

Under the Galilean transformations, the use of the Eqs. (2.8) at once shows that the form of the above wave equation (2.11) is not preserved in the reference frame Σ' , if one imposes the condition that the propagation speed u should remain unchanged under such transformations:

$$\left[\frac{1}{u^2} \left\{ \frac{\partial^2}{\partial t'^2} - 2\vec{v} \cdot \nabla' \frac{\partial}{\partial t'} + (\vec{v} \cdot \nabla') (\vec{v} \cdot \nabla') \right\} - \nabla'^2 \right] \psi(t', \vec{x}') = 0. \quad (2.12)$$

2.3 The general incompatibility of Maxwell's Electromagnetism with Galilean Relativity

The above counter-illustration is an indication that the Galilean relativity is in general conflicting with the electrodynamic theory formulated by Maxwell. To see this, let us first show that the scalar ψ in (2.11) could be identified as any of the components of the electric field vector $\vec{E}$ or the magnetic field vector $\vec{B}$, which satisfy the Maxwell's equations in empty space:

$$\left[\begin{array}{ll} \text{(i)} \quad \nabla \cdot \vec{E} = 0 & \text{(iii)} \quad \nabla \times \vec{E} + \frac{1}{c} \frac{\partial \vec{B}}{\partial t} = 0 \\ \text{(ii)} \quad \nabla \cdot \vec{B} = 0 & \text{(iv)} \quad \nabla \times \vec{B} - \frac{1}{c} \frac{\partial \vec{E}}{\partial t} = 0 \end{array} \right]. \quad (2.13)$$

Taking the curl of both sides of Eq. (2.13 iii), one gets

$$\nabla \times (\nabla \times \vec{E}) = -\frac{1}{c} \nabla \times \frac{\partial \vec{B}}{\partial t} \quad \Rightarrow \quad \nabla (\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\frac{1}{c} \frac{\partial}{\partial t} \nabla \times \vec{B}. \quad (2.14)$$

Then the use of Eqs. (2.13 i) and (2.13 iv) leads to

$$\left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 \right) \vec{E} = 0. \quad (2.15)$$

Similarly, taking the curl of Eq. (2.13 iv) and using Eqs. (2.13 ii) and (2.13 iii) we get

$$\left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 \right) \vec{B} = 0. \quad (2.16)$$

Each of the component equations of (2.15) and (2.16) are of the form of Eq. (2.11)] with $u \equiv c$:

$$\left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 \right) \psi(t, \vec{x}) = 0, \quad [\psi := \text{any component of } \vec{E} \text{ or } \vec{B}]. \quad (2.17)$$

This is regarded as the homogeneous wave equation for the following reason:

Assume a solution of Eq. (2.17) of the form

$$\psi(t, \vec{x}) = \psi_0 e^{i(\vec{k} \cdot \vec{x} - \omega t)}, \quad (2.18)$$

where $\vec{k}$ is a constant vector, and ω and ψ_0 are also constants. We have

$$\nabla \psi = i\vec{k} \psi \quad \Rightarrow \quad \nabla^2 \psi = -|\vec{k}|^2 \psi \quad \text{and} \quad \frac{\partial \psi}{\partial t} = -i\omega \psi \quad \Rightarrow \quad \frac{\partial^2 \psi}{\partial t^2} = -\omega^2 \psi, \quad (2.19)$$

which implies that

$$\left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 \right) \psi = \left(-\frac{\omega^2}{c^2} + |\vec{k}|^2 \right) \psi = 0, \quad (2.20)$$

under the stipulation

$$k^2 \equiv |\vec{k}|^2 = \frac{\omega^2}{c^2}. \quad (2.21)$$

The solution (2.18) represents a wave “traveling” along (or opposite to) $\vec{k}$ (known as the *wave vector*) in the sense that a point of constant phase $(\vec{k} \cdot \vec{x} - \omega t)$ moves in the direction of $\pm \vec{k}$ with a speed c , which is determined by

$$\vec{k} \cdot \vec{x} - \omega t = \text{constant} \quad \Rightarrow \quad \vec{k} \cdot \frac{d\vec{x}}{dt} = \omega \quad \text{or, by Eq. (2.21):} \quad c = \pm \frac{\omega}{k} = \pm \hat{\vec{k}} \cdot \frac{d\vec{x}}{dt}, \quad (2.22)$$

where $\hat{\vec{k}} = \vec{k}/|\vec{k}|$ is the unit vector along $\vec{k}$. Moreover, since a surface of constant phase, given by the equation $\vec{k} \cdot \vec{x} - \omega t = \text{constant}$, is a *plane* perpendicular to $\vec{k}$, the solution (2.18) represents a *plane wave*.

Now, the non-covariance of the wave equation under the Galilean transformations (as shown above) poses a problem for Maxwell's electromagnetism, unlike for instance the theory of acoustics. The acoustic (sound) waves are due to the compressions and rarefactions in a given medium (e.g. air), and the reference frame Σ in which the wave equation has its usual form [Eq. (2.11)] is obviously a preferred one in which the medium is at rest. In all other reference frames, in which the medium is moving, the direct influence of such motion on the propagation of sound (by virtue of mechanical vibrations described by Newtonian classical mechanics) can account for the extra terms in Eq. (2.12) that modify the usual form of the wave equation under the Galilean transformations. For the electromagnetic waves however, the situation is in general different, because there is no influence of any intervening medium on the electromagnetic wave propagation other than just supporting the latter. The Maxwell's equations, which govern the electromagnetic wave propagation, involve the parameter c , which according to Maxwell's theory should remain “fixed” for all frames of reference, and could be interpreted as the fixed speed of propagation of the electromagnetic waves. It is the presence of this fixed c which prevents any possible way to compensate for the non-preservation of the Maxwell's equations (and consequently the wave equation) under the Galilean transformations. Maxwell's theory thus seems to be *incompatible* with Galilean relativity.

2.4 Ways to look for a consistent Relativistic theory of Electromagnetism

The following are the possible alternatives that one may choose to resolve the conflict between Maxwell's electromagnetism and Galilean relativity:

- I. Maxwell's theory is incorrect and a proper theory of electromagnetism must have its equations invariant under the Galilean transformations.
- II. Galilean relativity applies to the laws of Newtonian mechanics, and Maxwell's theory is correct but bizarre, in the sense that the electromagnetic wave propagation always requires a preferred frame — the one in which the intervening medium, the so-called luminiferous “*Æther*”, is at rest.
- III. There should exist a relativity principle, other than the Galilean one, for the laws of mechanics as well as the equations of electrodynamics.

The first alternative is hardly viable, given the immense success of the Maxwell's theory otherwise.

The no-go establishment of the *Æther*-hypothesis, following the works of Michaelson and Morley in 1886, rules out the second alternative as well.

So, what remained is to find out a set of transformation relations and a new relativity principle, so that the Maxwell's equations are of the same form, with the same speed parameter c in all frames moving with uniform relative speeds (i.e. inertial frames).

The first step towards obtaining such transformation relations, and hence to formulate a new theory of relativity, is to abolish the Newtonian concept of “*absolute space*”, and to keep the time and the space in equal footing. That is to say, one should now invoke the concept of “*space-time*” (instead of treating space and time separately) represented by all the spatial coordinates, as well as time multiplied by a constant speed (from dimensional arguments). Such a speed parameter must be a universal constant, which is argued to be the speed c of electromagnetic signals in empty space, as we see below.

3 Formulation of Special Relativity

3.1 The Postulates

I. “All laws of physics (except that for gravitation) are independent of uniform translational motion of the system, i.e., the reference frame (RF), in which they operate” — *Principle of Relativity*:

$$\text{Physical Laws} \xLeftrightarrow{\text{Inv.}} \left[\Sigma_{(RF)} \xrightarrow{v=\text{const.}} \Sigma'_{(RF)} \right]. \quad (3.1)$$

Corollary: The mathematical equations expressing the laws of nature must be *covariant*, i.e., invariant in form under the transformations that leave invariant the infinitesimal separation between two events in space-time.

II. “The speed of propagation (c) of electromagnetic signals in empty space is finite and independent of the motion of the source”:

$$c \xLeftrightarrow{\text{Inv.}} \left[\Sigma_{(RF)} \xrightarrow{v=\text{const.}} \Sigma'_{(RF)} \right]. \quad (3.2)$$

Corollary: For physical entities in every inertial frame, there is a finite limiting speed, equal to c (as argued via certain thought experiments):

$$v_{lim} = c = \text{const.} \quad (3.3)$$

The most general set of coordinate transformations which preserve the infinitesimal separation between two space-time events, side-by-side maintaining the constancy of c , are the *Poincaré transformations*. A sub-class of these are the so-called (*homogeneous*) *Lorentz transformations* which preserve the separation even when it is not necessarily infinitesimal, i.e., the events are not necessarily close enough. We discuss below in detail the form and properties of both the Lorentz and the Poincaré transformation equations in certain simplified, as well as generic, scenarios.

3.2 Lorentz Transformations for a Boost along one axis

3.2.1 Derivation of the Lorentz Transformation Equations

Consider two reference frames Σ and Σ' represented by (ct, x, y, z) and (ct', x', y', z') respectively, as shown in Fig. 2. Let us suppose that

- The axes of the two frames are parallelly oriented, i.e, $ct' || ct, x' || x, y' || y, z' || z$.
- Frame Σ' moves with constant velocity $\vec{v}$ relative to the frame Σ along, say, the positive x axis.
- The origins of the two frames coincide at $t = t' = 0$.

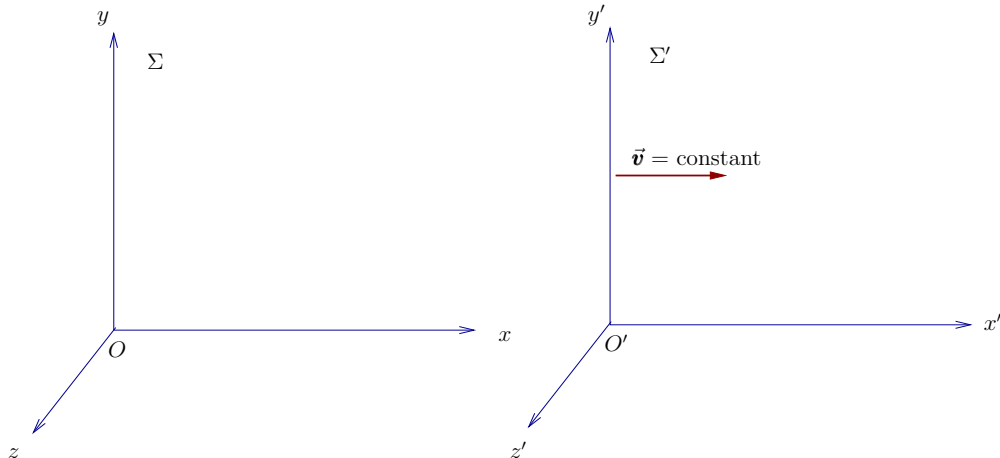


Figure 2: Lorentz transformations for frame motion along one axis, preserving the axes orientation.

Consider now an event P, whose coordinates are (ct', x', y', z') in frame Σ' and (ct, x, y, z) in frame Σ . As the origin of Σ' moves relative to Σ with speed v along positive x -direction

$$x' = 0 \quad \implies \quad x = vt. \quad (3.4)$$

Conversely, as the origin of Σ moves relative to Σ' with speed v along the negative x' -direction

$$x = 0 \quad \implies \quad x' = -vt. \quad (3.5)$$

Now, the transformation equations between the two systems must be *linear*, so that for an event in Σ there is a unique associated event in Σ' , and vice versa. The most general linear relations satisfying Eqs. (3.4) and (3.5) are

$$x' = \gamma(x - vt) \quad , \quad x = \gamma(x' + vt') \quad [\gamma = \text{dimensionless constant}]. \quad (3.6)$$

Or,

$$dx' = \gamma(dx - vdt) \quad , \quad dx = \gamma(dx' + vdt') . \quad (3.7)$$

Therefore,

$$\frac{dx'}{dt'} = \gamma \left(\frac{dx}{dt} - v \frac{dt}{dt'} \right) = \gamma \left(\frac{dx}{dt} - v \right) \frac{dt}{dt'} \quad , \quad \frac{dx}{dt} = \gamma \left(\frac{dx'}{dt} + v \frac{dt'}{dt} \right) = \gamma \left(\frac{dx'}{dt} + v \right) \frac{dt'}{dt} . \quad (3.8)$$

In the limit $dx'/dt' \rightarrow c$ and $dx/dt \rightarrow c$, Eqs. (3.8) reduce to

$$\frac{dt}{dt'} = \frac{c}{\gamma(c - v)} \quad , \quad \frac{dt'}{dt} = \frac{c}{\gamma(c + v)} . \quad (3.9)$$

Solving for γ , subject to the requirement that the transformation equations we seek must reduce to the Galilean transformation equations in the limit $v/c \rightarrow 0$, we get

$$\boxed{\gamma = (1 - \beta^2)^{-1/2}} \quad , \quad \text{where} \quad \boxed{\beta = \frac{v}{c}} . \quad (3.10)$$

One can now write down the full set of transformation equations by solving Eqs.(3.6) for x' and t' :

$$\boxed{\begin{array}{ll} \text{(i)} & ct' = \gamma(ct - \beta x) \\ \text{(ii)} & x' = \gamma(-\beta ct + x) \\ \text{(iii)} & y' = y \\ \text{(iv)} & z' = z \end{array}} \quad (3.11)$$

These are the *Lorentz transformation* (LT) equations in a simple scenario where there is a frame motion, or *Lorentz boost*, along one spatial direction (x here, note that there is no frame motion along y or z).

3.2.2 The Lorentz Transformation Matrix

Let us re-write the above LT equations (3.11) in matrix form as

$$\begin{pmatrix} x'^0 \\ x'^1 \\ x'^2 \\ x'^3 \end{pmatrix} = \begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix} . \quad (3.12)$$

where we denote the coordinates as:

$$x^0 \equiv ct \quad , \quad x^1 \equiv x \quad , \quad x^2 \equiv y \quad , \quad \text{and} \quad x^3 \equiv z \quad (3.13)$$

In compact form the LT equations are given by

$$\boxed{x'^{\mu} = \Lambda^{\mu}_{\nu} x^{\nu}} \quad , \quad [\mu, \nu = 0, 1, 2, 3], \quad (3.14)$$

where x^{μ} and Λ^{μ}_{ν} are the elements of

$$\underline{\mathbf{x}} \equiv \{x^{\mu}\} = \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix} : \quad 1 \times 4 \text{ Position (column) vector in } (3+1)\text{-dimensional spacetime}, \quad (3.15)$$

$$\underline{\Lambda} \equiv \{\Lambda^{\mu}_{\nu}\} = \begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} : \quad \begin{array}{l} 4 \times 4 \text{ (homogeneous) LT matrix in a } (3+1)\text{-dimensional} \\ \text{space-time for single Lorentz boost along positive } x^1\text{-axis;} \\ \mu \text{ denotes the columns and } \nu \text{ denotes the rows.} \end{array} \quad (3.16)$$

In the Eq. (3.14), *Einstein's summation convention* is implied, i.e., repeated indices are summed over:

$$\Lambda^{\mu}_{\nu} x^{\nu} \equiv \sum_{\nu=0}^3 \Lambda^{\mu}_{\nu} x^{\nu} \quad [\forall \mu = 0, 1, 2, 3]. \quad (3.17)$$

The inverse transformation equations are given by

$$\boxed{x_{\mu} = \Lambda_{\mu}^{\nu} x'_{\nu}} \quad , \quad [\mu, \nu = 0, 1, 2, 3], \quad (3.18)$$

where x^{μ} and Λ^{μ}_{ν} are the elements of

$$\underline{\mathbf{x}}^{-1} \equiv \{x_{\mu}\} = (x_0, x_1, x_2, x_3) \quad : \quad 4 \times 1 \text{ Position (row) vector in } (3+1)\text{-dimensional spacetime}, \quad (3.19)$$

$$\underline{\Lambda}^{-1} \equiv \{\Lambda_{\mu}^{\nu}\} = \{[\Lambda^{\mu}_{\nu}]^{-1}\} \quad : \quad 4 \times 4 \text{ Inverse LT matrix for a boost along negative } x^1 \text{ axis.} \quad (3.20)$$

The distinction between x^0 and x_0 , x^1 and x_1 , and so forth, will be explained later on, when we will make a general study of tensors and their components. For the time being, however, let's stick to the matrix terminology and focus on the properties of the LT matrix Λ^{μ}_{ν} , where μ denotes columns and ν denotes rows. Following are the two important properties of the LT matrix for a single Lorentz boost:

$$(i) \quad \boxed{\underline{\Lambda}^T = \underline{\Lambda}} \implies \boxed{\underline{\Lambda}^T \underline{\Lambda}^{-1} = \underline{\mathbf{I}}} \implies \boxed{\Lambda^{\mu}_{\nu} \Lambda_{\mu}^{\alpha} = \delta_{\nu}^{\alpha}}, \quad \text{and} \quad (ii) \quad \boxed{\det \underline{\Lambda} = 1}. \quad (3.21)$$

3.2.3 Lorentz Transformation as a Rotation: The Rapidity parameter

To observe the geometrical meaning of a “boost”, let us make the following parametrization:

$$\boxed{\beta = \frac{v}{c} = \tanh \zeta \quad , \quad \gamma = (1 - \beta^2)^{-1/2} = \cosh \zeta} \quad , \quad (3.22)$$

where ζ is a dimensionless constant, called the ‘Boost’ or ‘Rapidity’ parameter. Under the above parameterization, the LT equations (3.11) take the form:

$$\begin{aligned} x'^0 &= x^0 \cosh \zeta - x^1 \sinh \zeta, \\ x'^1 &= -x^0 \sinh \zeta + x^1 \cosh \zeta, \\ x'^2 &= x^2, \quad \text{and} \quad x'^3 = x^3, \end{aligned} \quad (3.23)$$

i.e., the LT matrix (due to a *boost* along the positive x^1 -direction with constant speed v) takes the form:

$$\underline{\Lambda} \equiv \{\Lambda^{\mu}_{\nu}\} = \begin{pmatrix} \cosh \zeta & -\sinh \zeta & 0 & 0 \\ -\sinh \zeta & \cosh \zeta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (3.24)$$

Under a further transformation, and reparameterization:

$$x^0 \rightarrow \tilde{x}^0 = i x^0 \text{ (i.e., } t \rightarrow \tilde{t} = i t) ; \quad x^k \rightarrow \tilde{x}^k = x^k \quad [k = 1, 2, 3] , \quad \zeta = i \theta , \quad (3.25)$$

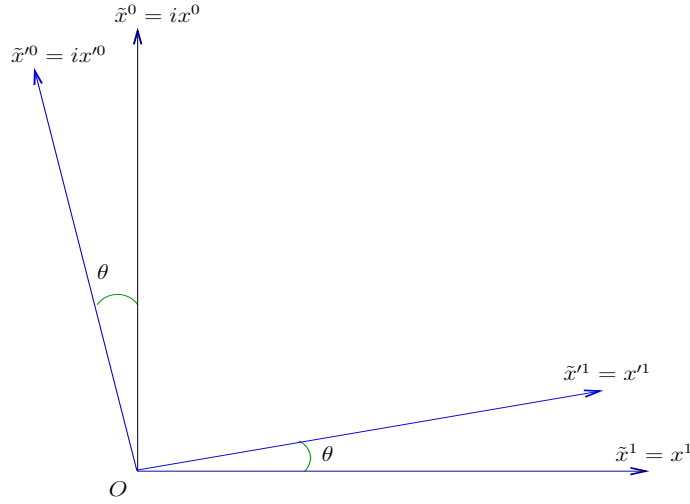


Figure 3: Simple Lorentz transformation as an (orthonormal) rotation of the $ix^0 - x^1$ plane.

the above LT matrix (3.24) reduces to

$$\underline{\Lambda} \equiv \{\Lambda^\mu_\nu\} = \begin{pmatrix} \cos \theta & \sin \theta & 0 & 0 \\ -\sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \{\mathcal{R}^\mu_\nu\} , \quad (3.26)$$

where $\mathcal{R}^\mu_\nu$ are the elements of the matrix corresponding to the ordinary (orthonormal) rotation of the $\tilde{x}^0 - \tilde{x}^1$ plane in the $(3+1)$ -dimensional space-time (see Fig. 3).

A *Lorentz Boost* along, say the positive x^1 direction, with speed v , may thus be viewed as a rotation of the complex $ix^0 - x^1$ plane by a real angle θ . In principle, there could be *three* Lorentz boosts along each of the three spatial axes. The transformations (3.25) imply the *Euclidianization* of space-time.

3.3 The invariant Line element

Refer back to the LT equations (3.11) for a boost along the positive x -direction with constant speed v . In differential form, they are given by

$$\begin{array}{ll} \text{(i)} & cdt' = \gamma(cdt - \beta dx) \\ \text{(ii)} & dx' = \gamma(-\beta cdt + dx) \\ \text{(iii)} & dy' = dy \\ \text{(iv)} & dz' = dz \end{array} \quad (3.27)$$

Now,

$$\begin{array}{lll} \text{Eq. (3.27 i)} + \text{Eq. (3.27 ii)} & \implies & cdt' + dx' = \gamma(1 - \beta)(cdt + dx) , \\ \text{Eq. (3.27 i)} - \text{Eq. (3.27 ii)} & \implies & cdt' - dx' = \gamma(1 + \beta)(cdt - dx) . \end{array}$$

Multiplying these two equations, and using the expressions (3.10) for β and γ , we get

$$c^2 dt'^2 - dx'^2 = c^2 dt^2 - dx^2 .$$

Using further, the Eqs. (3.27 iii & iv):

$$c^2 dt'^2 - dx'^2 - dy'^2 - dz'^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2 , \quad (3.28)$$

i.e., the Lorentz transformations leave invariant the differential quantity

$$ds^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2 \equiv (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2. \quad (3.29)$$

Performing a similar operation as above on the set of LT equations (3.11) themselves, we could have also found that the Lorentz transformations leave invariant the algebraic quantity

$$s^2 = c^2 t^2 - x^2 - y^2 - z^2 \equiv (x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2. \quad (3.30)$$

As in the ordinary Euclidean (three-)space, the elementary distance between two points is given by

$$|d\vec{x}| = \sqrt{dx^2 + dy^2 + dz^2} \equiv \sqrt{(dx^1)^2 + (dx^2)^2 + (dx^3)^2}, \quad (3.31)$$

the quantity ds^2 given by Eq. (3.29) is analogically referred to as the *Line element*, or the (squared) elementary separation between two events in a $(3 + 1)$ -dimensional space-time. Galilean transformations leave $|d\vec{x}|^2$ invariant, whereas Lorentz transformations leave ds^2 invariant.

Naïvely, the relative minus sign between the temporal and spatial parts of $ds^2 = c^2 dt^2 - |d\vec{x}|^2$ may be looked upon as follows: For signals travelling with the invariant speed c , which may also be identified with the speed of light in free space, Galilean relativity alludes to instantaneous reception after transmission, i.e., $\Delta t = 0$, which implies

$$\lim_{\Delta t \rightarrow 0} \frac{|\Delta \vec{x}|}{\Delta t} = \frac{|d\vec{x}|}{dt} = c \rightarrow \infty. \quad (3.32)$$

However, for finite c (as demanded in Maxwell's theory), the reception and the transmission should be separated in time by $\Delta t \neq 0$. One therefore imposes (instead of $dt = 0$)

$$ds^2 = c^2 dt^2 - |d\vec{x}|^2 = 0, \quad \text{so that} \quad \frac{|d\vec{x}|}{dt} = c, \quad \text{automatically.}$$

3.4 Poincaré Transformations for a Boost along one axis

The necessary condition for the constancy of c , in a relativity theory governed by certain transformation equations, is that the line element ds^2 must remain invariant under such transformations. Lorentz transformations (LT) preserve not only ds^2 , but s^2 [Eq. (3.30)] as well, which is not necessarily required. The invariance of s^2 is due to the third assumption made in the derivation of the LT equations in subsection 3.2, viz., the frames of reference Σ and Σ' coincide at $t = t' = 0$. If we relax this assumption, then the most general linear transformations that leave ds^2 invariant are the Poincaré transformations (PT), which in the simple case of a single Lorentz boost along (say) the x -axis, with constant speed v , are given by the equations:

$$\begin{aligned} \text{(i)} \quad & ct' = \gamma(ct - \beta x) + a_t \\ \text{(ii)} \quad & x' = \gamma(-\beta ct + x) + a_x \\ \text{(iii)} \quad & y' = y + a_y \\ \text{(iv)} \quad & z' = z + a_z \end{aligned} \quad (3.33)$$

where (a_t, a_x, a_y, a_z) are constant shifts between the origins of Σ and Σ' at $t = 0$, along the t, x, y and z directions respectively. The above equations are expressed in compact notation as

$$x'^\mu = \Lambda^\mu_\nu x^\nu + a^\mu, \quad (3.34)$$

where the elements of a^μ are

$$a^0 \equiv a_t, \quad a^1 \equiv a_x, \quad a^2 \equiv a_y, \quad a^3 \equiv a_z. \quad (3.35)$$

In differential form, however, the Lorentz and the Poincaré transformation equations are identical, viz.

$$dx'^\mu = \Lambda^\mu_\nu dx^\nu. \quad (3.36)$$

3.5 Lorentz Transformations for a Boost in an arbitrary direction

Consider again two frames Σ and Σ' with coordinates (x^0, x^1, x^2, x^3) and (x'^0, x'^1, x'^2, x'^3) respectively.

- The axes of Σ and Σ' are parallelly oriented, i.e. $x'^0 \parallel x^0$, $x'^1 \parallel x^1$, $x'^2 \parallel x^2$, $x'^3 \parallel x^3$.
- Frame Σ' moves with constant velocity $\vec{v}$ relative to the frame Σ in an arbitrary direction.
- The origins of Σ and Σ' coincide at $t = t' = 0$.

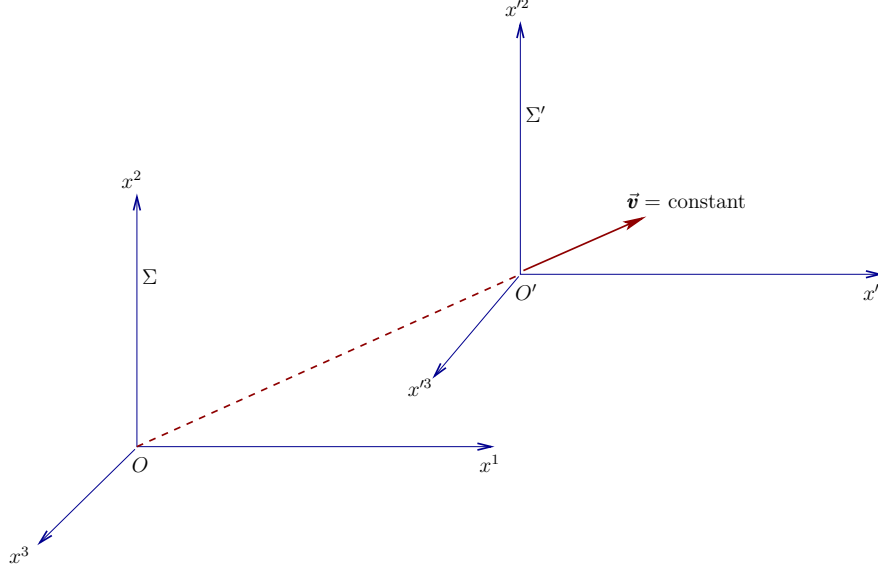


Figure 4: Lorentz transformation for frame motion in an arbitrary direction, preserving the axes orientation.

The corresponding Lorentz transformation equations involve three boost parameters, viz. the components (v^1, v^2, v^3) of the velocity $\vec{v}$ along respectively the three axes (x^1, x^2, x^3) . The elements of the general Lorentz boost transformation matrix are found in the Appendix A (see Eqs. (A.6) and (A.8)), one may verify that the corresponding transformation equations can be expressed in the ordinary (three-)vectorial notation as

$$x'^0 = \gamma (x^0 - \vec{\beta} \cdot \vec{x}), \quad \vec{x}' = \vec{x} + \frac{(\gamma - 1)}{|\vec{\beta}|^2} (\vec{\beta} \cdot \vec{x}) \vec{\beta} - \vec{\beta} \gamma x^0, \quad (3.37)$$

where

$$\vec{\beta} = (\beta^1, \beta^2, \beta^3) = \frac{\vec{v}}{c} = \left(\frac{v^1}{c}, \frac{v^2}{c}, \frac{v^3}{c} \right), \quad \gamma = (1 - |\vec{\beta}|^2)^{-1/2}. \quad (3.38)$$

3.6 Lorentz Transformations for arbitrary Boosts and Rotations

If, in addition to the scenario in previous subsection, the axes of the two frames Σ and Σ' are not taken to be parallelly oriented (see Fig. 5), then the corresponding LT matrix is in its most general form, which involves not only the three boost parameters (speeds), but three spatial rotation parameters (angles) as well. Even for this most general LT matrix, the first of the properties (3.21), viz.,

$$\underline{\Lambda}^T \underline{\Lambda}^{-1} = \underline{\mathbb{I}} \quad \Longleftrightarrow \quad \Lambda^\mu_\nu \Lambda_\mu^\alpha = \delta_\nu^\alpha, \quad (3.39)$$

can be shown to hold. As to the second property, taking determinant of both sides of Eq.(3.39), we have

$$\det \underline{\Lambda} = \pm 1. \quad (3.40)$$

Our interest is however in the so-called *proper* Lorentz transformations for which

$$\Lambda^0_0 \geq 0, \quad \text{and} \quad \det \underline{\Lambda} = +1, \quad (3.41)$$

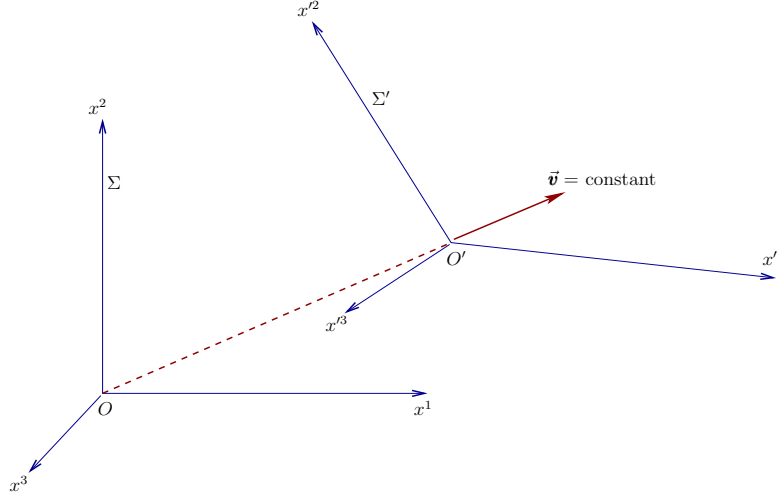


Figure 5: Lorentz transformation for frame motion in an arbitrary direction, and not preserving the axes orientation.

and which can always be converted to the identity transformation by continuous variation of parameters of Λ^μ_ν . Actually one can decompose the full LT into an ordinary spatial rotation, followed by a boost, followed by a further ordinary rotation. The first rotation lines up one of the spatial axes (say the x^1 axis) of Σ with the velocity $\vec{v}$ of Σ' . Then a boost in this direction with speed $v = |\vec{v}|$ transforms Σ to a frame which is at rest relative to Σ' . Finally, another rotation lines up the coordinate frame of Σ with that of Σ' .

3.7 Successive Lorentz Boosts

Two frames Σ and Σ' , with coordinates (x^0, x^1, x^2, x^3) and (x'^0, x'^1, x'^2, x'^3) , have their axes parallelly oriented and origins coincident at $x^0 = x'^0 = 0$. Frame Σ' moves with constant velocity $\vec{v}$ in an arbitrary direction, relative to the frame Σ .

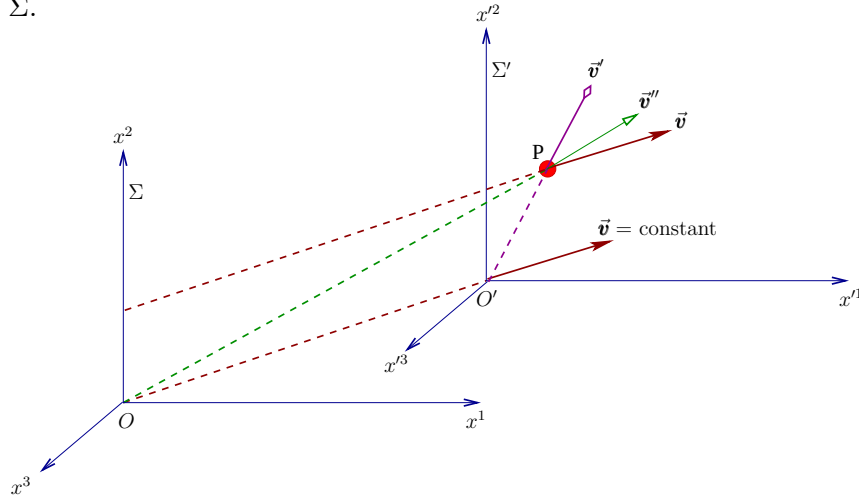


Figure 6: Successive Lorentz Boosts

Consider now a particle P which moves with constant velocity $\vec{v}'$ relative to Σ' . Let $\vec{v}''$ be the particle's velocity relative to Σ , and Σ'' be the particle's rest frame, i.e, the coordinate system $(x''^0, x''^1, x''^2, x''^3)$ attached to the particle. Then

I. For the relative motion between Σ' and Σ :

$$x'^\mu = \Lambda^\mu_\nu(\vec{\beta}) x^\nu, \quad [\vec{\beta} = \vec{v}/c]. \quad (3.42)$$

where $\Lambda^\mu_\nu(\vec{\beta})$ are the corresponding LT matrix elements.

II. For the relative motion between Σ'' and Σ' :

$$x''^\mu = \Lambda^\mu_\nu(\vec{\beta}') x'^\nu, \quad [\vec{\beta}' = \vec{v}'/c]. \quad (3.43)$$

where $\Lambda^\mu_\nu(\vec{\beta}')$ are the corresponding LT matrix elements.

III. For the relative motion between Σ'' and Σ :

$$x''^\mu = \Lambda^\mu_\nu(\vec{\beta}'') x^\nu, \quad [\vec{\beta}'' = \vec{v}''/c]. \quad (3.44)$$

where $\Lambda^\mu_\nu(\vec{\beta}'')$ are the corresponding LT matrix elements.

Substituting for x'^ν from Eq. (3.42) in Eq. (3.43), and comparing with Eq. (3.44), we have

$$x''^\mu = \Lambda^\mu_\lambda(\vec{\beta}') \Lambda^\lambda_\nu(\vec{\beta}) x^\nu = \Lambda^\mu_\nu(\vec{\beta}'') x^\nu \implies \boxed{\Lambda^\mu_\nu(\vec{\beta}'') = \Lambda^\mu_\lambda(\vec{\beta}') \Lambda^\lambda_\nu(\vec{\beta})}. \quad (3.45)$$

Thus a Lorentz boost along the direction of $\vec{\beta}''$ is the product of the Lorentz boosts along $\vec{\beta}'$ and $\vec{\beta}$.

3.7.1 Non-commutativity of the products of Boosts

The product of the boosts is in general non-commutative:

$$\Lambda^\mu_\lambda(\vec{\beta}') \Lambda^\lambda_\nu(\vec{\beta}) \neq \Lambda^\mu_\lambda(\vec{\beta}) \Lambda^\lambda_\nu(\vec{\beta}'). \quad (3.46)$$

To verify this, let us introduce a boost parameter $\vec{\beta}''$ such that

$$\Lambda^\mu_\nu(\vec{\beta}'') = \Lambda^\mu_\lambda(\vec{\beta}) \Lambda^\lambda_\nu(\vec{\beta}'). \quad (3.47)$$

We need to check under what circumstances $\Lambda^\mu_\nu(\vec{\beta}'') \neq \Lambda^\mu_\nu(\vec{\beta}'')$. Let us consider the following simple cases:

Case I: $\vec{\beta} \parallel \vec{\beta}'$

Let us choose $\vec{\beta}$ and $\vec{\beta}'$ to be directed along the positive x^1 axis, i.e.

$$\vec{\beta} = (\beta, 0, 0) \quad \text{and} \quad \vec{\beta}' = (\beta', 0, 0), \quad (3.48)$$

whence the non-vanishing elements of $\Lambda^\mu_\nu(\vec{\beta})$ and $\Lambda^\mu_\nu(\vec{\beta}')$ are given by

$$\Lambda^\mu_\nu(\vec{\beta}) : \begin{cases} \Lambda^0_0(\beta) = \Lambda^1_1(\beta) = \gamma, \\ \Lambda^1_0(\beta) = \Lambda^0_1(\beta) = -\beta\gamma, \\ \Lambda^2_2(\beta) = \Lambda^3_3(\beta) = 1, \end{cases} \quad \left[\gamma = \frac{1}{\sqrt{1-\beta^2}} \right], \quad (3.49)$$

$$\Lambda^\mu_\nu(\vec{\beta}') : \begin{cases} \Lambda^0_0(\beta') = \Lambda^1_1(\beta') = \gamma', \\ \Lambda^1_0(\beta') = \Lambda^0_1(\beta') = -\beta'\gamma', \\ \Lambda^2_2(\beta') = \Lambda^3_3(\beta') = 1, \end{cases} \quad \left[\gamma' = \frac{1}{\sqrt{1-\beta'^2}} \right]. \quad (3.50)$$

Using Eqs. (3.45) and (3.47) the non-vanishing elements of $\Lambda^\mu_\nu(\vec{\beta}'')$ and $\Lambda^\mu_\nu(\vec{\beta}'')$ are computed as follows:

$$\text{0-0 element: } \begin{cases} \Lambda^0_0(\vec{\beta}'') = \Lambda^0_\lambda(\vec{\beta}') \Lambda^\lambda_0(\vec{\beta}) \\ \quad = \Lambda^0_0(\beta') \Lambda^0_0(\beta) + \Lambda^1_1(\beta') \Lambda^1_0(\beta) + 0 + 0 \\ \quad = \gamma'\gamma(1 + \beta'\beta) \\ \Lambda^0_0(\vec{\beta}'') = \Lambda^0_\lambda(\vec{\beta}) \Lambda^\lambda_0(\vec{\beta}') \\ \quad = \Lambda^0_0(\beta) \Lambda^0_0(\beta') + \Lambda^1_1(\beta) \Lambda^1_0(\beta') + 0 + 0 \\ \quad = \gamma\gamma'(1 + \beta\beta') \end{cases} \implies \Lambda^0_0(\vec{\beta}'') = \Lambda^0_0(\vec{\beta}''). \quad (3.51)$$

$$\text{1-0 element: } \begin{cases} \Lambda^1_0(\vec{\beta}'') &= \Lambda^1_\lambda(\vec{\beta}') \Lambda^\lambda_0(\vec{\beta}) \\ &= \Lambda^1_0(\beta') \Lambda^0_0(\beta) + \Lambda^1_1(\beta') \Lambda^1_0(\beta) + 0 + 0 \\ &= -\gamma' \gamma (\beta' + \beta) \\ \Lambda^1_0(\vec{\beta}'') &= \Lambda^1_\lambda(\vec{\beta}) \Lambda^\lambda_0(\vec{\beta}') \\ &= \Lambda^1_0(\beta) \Lambda^0_0(\beta') + \Lambda^1_1(\beta) \Lambda^1_0(\beta') + 0 + 0 \\ &= -\gamma \gamma' (\beta + \beta') \end{cases} \implies \Lambda^1_0(\vec{\beta}'') = \Lambda^1_0(\vec{\beta}'') . \quad (3.52)$$

Similarly, the other elements of $\Lambda^\mu_\nu(\vec{\beta}'')$ and $\Lambda^\mu_\nu(\vec{\beta}'')$ also match. Thus Eq. (3.46) does not hold in this case.

Case II: $\vec{\beta} \perp \vec{\beta}'$

Let us choose $\vec{\beta}$ and $\vec{\beta}'$ to be directed along the positive x^1 axis and the positive x^2 respectively, i.e.

$$\vec{\beta} = (\beta, 0, 0) \quad \text{and} \quad \vec{\beta}' = (0, \beta', 0) , \quad (3.53)$$

whence the non-vanishing elements of $\Lambda^\mu_\nu(\vec{\beta})$ would be the same as in Eq. (3.49), whereas the non-vanishing elements of $\Lambda^\mu_\nu(\vec{\beta}')$ are given by

$$\Lambda^\mu_\nu(\vec{\beta}') : \quad \begin{cases} \Lambda^0_0(\beta') = \Lambda^2_2(\beta') = \gamma' , \\ \Lambda^2_0(\beta') = \Lambda^0_2(\beta') = -\beta' \gamma' , \\ \Lambda^1_1(\beta') = \Lambda^3_3(\beta') = 1 , \end{cases} \quad \left[\gamma' = \frac{1}{\sqrt{1 - \beta'^2}} \right] . \quad (3.54)$$

Using Eqs. (3.45) and (3.47) the non-vanishing elements of $\Lambda^\mu_\nu(\vec{\beta}'')$ and $\Lambda^\mu_\nu(\vec{\beta}'')$ are now computed as:

$$\text{0-0 element: } \begin{cases} \Lambda^0_0(\vec{\beta}'') &= \Lambda^0_\lambda(\vec{\beta}') \Lambda^\lambda_0(\vec{\beta}) \\ &= \Lambda^0_0(\beta') \Lambda^0_0(\beta) + 0 + 0 + 0 = \gamma' \gamma \\ \Lambda^0_0(\vec{\beta}'') &= \Lambda^0_\lambda(\vec{\beta}) \Lambda^\lambda_0(\vec{\beta}') \\ &= \Lambda^0_0(\beta) \Lambda^0_0(\beta') + 0 + 0 + 0 = \gamma \gamma' \end{cases} \implies \Lambda^0_0(\vec{\beta}'') = \Lambda^0_0(\vec{\beta}'') . \quad (3.55)$$

But

$$\text{1-0 element: } \begin{cases} \Lambda^1_0(\vec{\beta}'') &= \Lambda^1_\lambda(\vec{\beta}') \Lambda^\lambda_0(\vec{\beta}) \\ &= 0 + \Lambda^1_1(\beta') \Lambda^1_0(\beta) + 0 + 0 = -\beta \gamma \\ \Lambda^1_0(\vec{\beta}'') &= \Lambda^1_\lambda(\vec{\beta}) \Lambda^\lambda_0(\vec{\beta}') \\ &= \Lambda^1_0(\beta) \Lambda^0_0(\beta') + 0 + 0 + 0 = -\beta \gamma \gamma' \end{cases} \implies \Lambda^1_0(\vec{\beta}'') \neq \Lambda^1_0(\vec{\beta}'') . \quad (3.56)$$

Similarly, a few other elements, such as 2-0, 1-2, ..., do not match as well, and hence verify Eq. (3.46).

3.7.2 Relativistic Velocity Addition

Refer back to Fig. 6, and the scenario it depicts. If the relative frame velocity $\vec{v}$ and the particle's velocity $\vec{v}'$ w.r.to the frame Σ' are known, then what would be the particle's velocity $\vec{v}''$ w.r.to the frame Σ ?

To determine this, let us consider the simple case in which $\vec{v}$ and $\vec{v}'$ are parallel. If we choose to take $\vec{v} = c\vec{\beta}$ and $\vec{v}' = c\vec{\beta}'$ to be directed along x^1 , then this is the case I considered in §3.7.1. From Fig. 6 it is obvious that $\vec{v} \parallel \vec{v}' \parallel x^1$ implies $\vec{v}''$ is also directed along x^1 . The problem thus becomes an (1+1)-dimensional one, rather than (3+1)-dimensional, as the $(x^0 - x^1)$ plane is of relevance only.

The non-vanishing elements of the transformation matrices $\Lambda^\mu_\nu(\vec{\beta})$ and $\Lambda^\mu_\nu(\vec{\beta}')$ are given in Eqs. (3.49) and (3.50) respectively. Since $\vec{\beta}'' = \vec{v}''/c = (\beta'', 0, 0)$, the non-zero elements of the transformation matrix

$\Lambda^\mu_\nu(\vec{\beta}'')$ are similar:

$$\Lambda^\mu_\nu(\vec{\beta}'') : \quad \begin{cases} \Lambda^0_0(\beta'') = \Lambda^1_1(\beta'') = \gamma'' , \\ \Lambda^1_0(\beta'') = \Lambda^0_1(\beta'') = -\beta'' \gamma'' , \\ \Lambda^2_2(\beta'') = \Lambda^3_3(\beta'') = 1 , \end{cases} \quad \left[\gamma'' = \frac{1}{\sqrt{1 - \beta''^2}} \right] . \quad (3.57)$$

Comparing the $\Lambda^0_0(\beta'')$ and $\Lambda^1_0(\beta'')$ with those in Eqs. (3.51) and (3.52) [case I of §3.7.1], we have

$$\gamma'' = \gamma' \gamma (\beta' + \beta) , \quad \beta'' \gamma' = \gamma' \gamma (\beta' + \beta) . \quad (3.58)$$

Solving simultaneously gives the *velocity addition formula* in this simple case in which the relative velocity $\vec{v}$ of the two frames Σ and Σ' is parallel to the particle's velocity $\vec{v}'$ w.r.to Σ' :

$$\boxed{\beta'' = \frac{\beta' + \beta}{1 + \beta' \beta}} \quad \Longrightarrow \quad \boxed{v'' = \frac{v' + v}{1 + v' v / c^2}} . \quad (3.59)$$

For the general formula of velocity addition (when $\vec{v}$ is not parallel to $\vec{v}'$), see the derivation in Appendix B.

4 Space-time Diagrams, Lightcone, Invariant Intervals and Proper Time

4.1 The Minkowski diagram

One intriguing aspect of Special Relativity is the schematic representation of the dynamics of space-time *events* via the so-called *Minkowski diagram*. As a simplification, we shall illustrate below the essential features of such a diagram in the $(1+1)$ -dimensional scenario, i.e. with one axis of the plot taken to be $x^0 = ct$, and the other axis is one of the space axes, say $x^1 = x$, the remaining axes $x^2 = y$ and $x_3 = z$ being suppressed.

- **World-line:** The loci of all events associated with a particle, trace out a trajectory $x(t)$, referred to as the *world-line*, in the space-time diagram. Various types of world-lines are shown in Figs. 7(a) and (b).

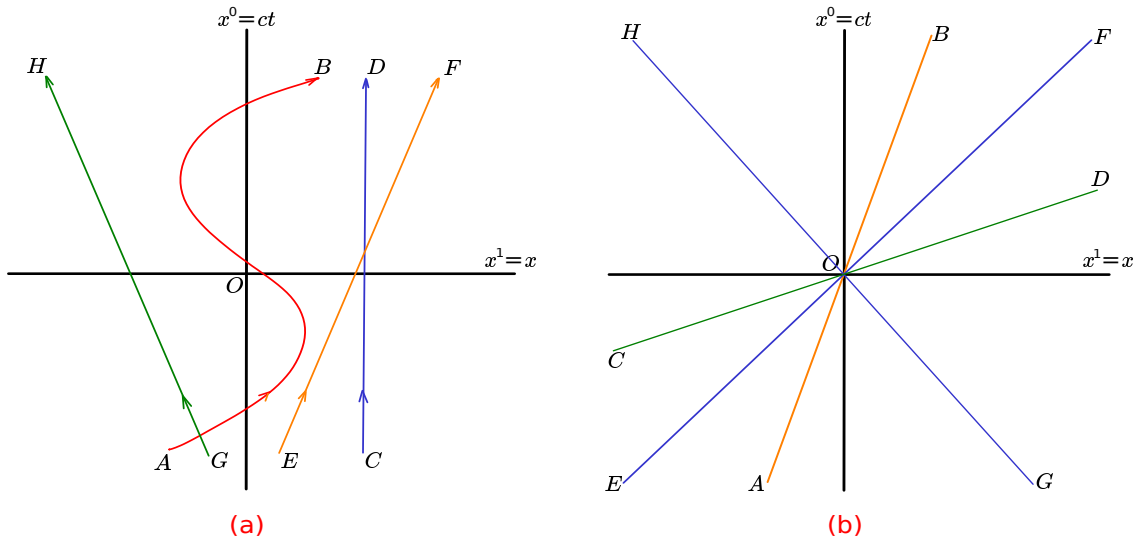


Figure 7: Illustration of world-lines in the $(1+1)$ -dimensional space-time diagram. Whereas, the lines for the events associated with stationary, uniformly moving and accelerated particles are shown in (a), in (b) one has the lines representing trajectory of events including that at the origin O , for particles moving with constant speed greater than, less than or equal to c in magnitude.

In Fig. 7(a), the line AB represents the world-line of events associated with a generally accelerated particle, that for a particle at rest ($x = \text{constant}$) by the line CD , that for a particle moving with constant positive (negative) speed by the line EF (GH).

In Fig. 7(b), the lines AB and CD represent the trajectories of events including that at $(0,0)$, i.e. the origin O , associated with particles moving with constant positive speeds. The lines EF and GH , also passing through the origin O , have equations $ct = \pm x$, i.e. slopes ± 1 , are for particles moving with speed equal to $\pm c$. These lines (EF and GH), which make 45° angles with the axes, are known as the *null lines* and are of immense importance in Special Relativity. They actually let us identify the *causal* connection of events, which we shall discuss later on. Meanwhile, note that if the particle's speed (which is constant for all the lines in Fig. 7(b)) is denoted by v , then the equations of AB and CD are

$$x = vt \quad \Rightarrow \quad ct = \frac{c}{v} x = \frac{1}{\beta} x, \quad \left[\text{with } \beta = \frac{v}{c} \right], \quad (4.1)$$

i.e. the slope of these lines are $\beta^{-1} = c/v$. Therefore the line AB , which has a slope > 1 , implies that it is the world-line of events associated with a particle moving with constant speed $v < c$. On the other hand, the line CD , whose slope < 1 , is the world-line of events associated with a particle moving with constant speed $v < c$.

- **Scaling of axes:** Refer now to the Fig. 8(a). We consider two inertial frames represented by coordinates (t, x) and (t', x') , the axes being parallelly oriented. The (t', x') system is moving with constant speed $v (= c\beta)$ relative to the (t, x) system along the parallel x and x' axes. In the space-time diagram shown in Fig. 8(b), however, the axes of the (t', x') system is not parallel to the corresponding axes of the (t, x) system, but identified as follows:

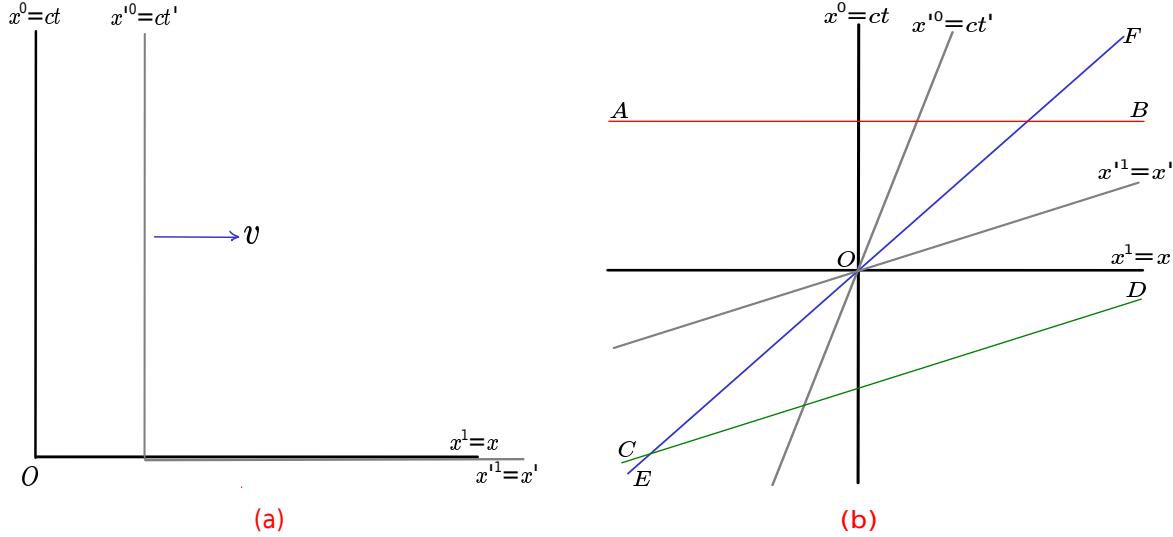


Figure 8: Two inertial coordinate systems (t, x) and (t', x') , with parallelly oriented axes are shown in (a); the (t', x') system is moving with constant speed v with respect to the (t, x) system along the x direction. In (b) we have the space-time diagram in which the moving frame axes ct' and x' are inclined, and symmetrically placed about the null line EF . The lines of simultaneity in the two frames are also shown.

By the Lorentz transformation equations

$$ct' = \gamma(ct - \beta x) \quad ; \quad x' = \gamma(-\beta ct + x) \quad , \quad (4.2)$$

ct' axis: which defines the trajectory of a moving observer with $x' = 0$, i.e. the observer is at rest in the coordinate system (t', x') , is given by the straight line

$$ct = \beta^{-1} x \quad , \quad [\text{with slope} = \beta^{-1}] \quad , \quad (4.3)$$

passing through the origin O , i.e. the point $(ct = 0, x = 0)$.

x' axis: which defines the world-line of events *simultaneous* with the event at the origin O in the coordinate system (t', x') , is given by the straight line $t' = 0$, i.e.

$$ct = \beta x \quad , \quad [\text{with slope} = \beta] \quad , \quad (4.4)$$

passing through the origin O .

Thus, in the space-time diagram if the ct and x axes are shown respectively by vertical and horizontal straight lines intersecting at O , as in Fig. 8(b), then the ct' and x' axes are inclined straight lines, intersecting at O , with slopes scaled as β^{-1} and β respectively, and symmetric about the null line EF .

- **Lines of simultaneity:** These are straight lines parallel to the space axis in the $(1 + 1)$ -dimensional space-time diagram, which represent the world-lines of events occurring at different spatial locations, but at the same time, in a given reference frame. For instance, in Fig. 8(b) the line AB is a simultaneity line for the (t, x) system, whereas the line CD is that for the (t', x') system.
- **Scaled axes ticks:** For the coordinate systems (t, x) and (t', x') , it is easy to show that the tick lengths differ in the ct and ct' axes and in the x and x' axes. The conversion factor is $\gamma = (1 - \beta^2)^{-1/2}$, as illustrated in Fig. 9(a), which can be verified as follows:

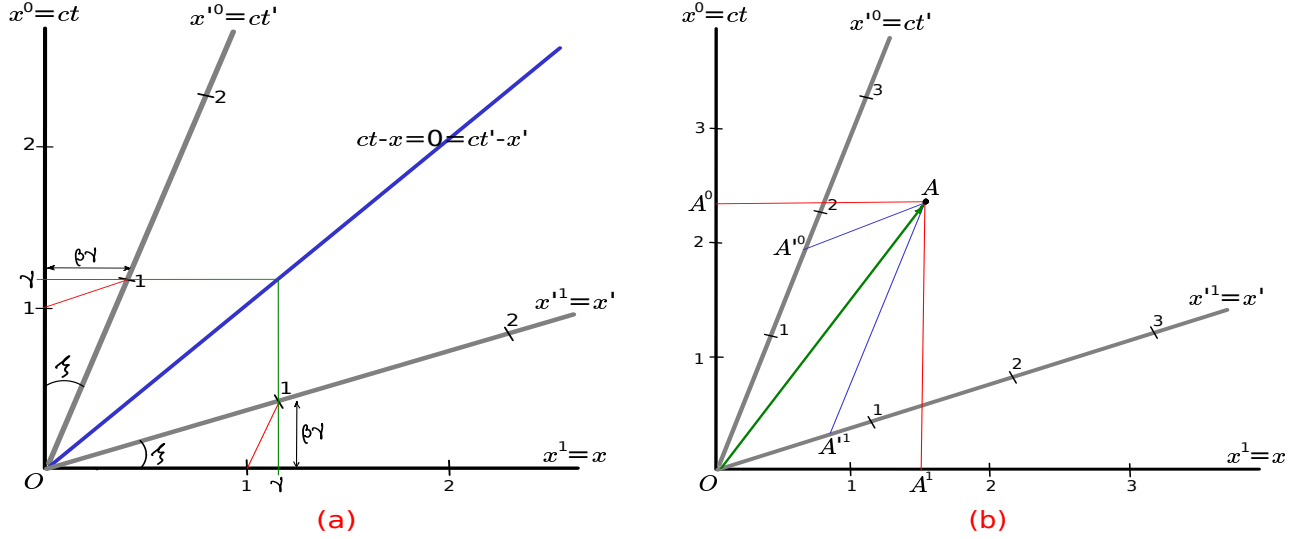


Figure 9: Comparison of axes tick lengths of two coordinate systems (t, x) and (t', x') are shown in the space-time diagram in (a). In (b), the determination of the coordinates of an event at A , by taking parallel projections on the ct and x axes and on the ct' and x' axes are illustrated.

Let us suppose that we have unit ticks on the ct and x axes. But by the Lorentz transformation equations (4.2), we have

ct' unit tick: $ct' = 1, x' = 0$ implies $1 = \gamma(ct - \beta x)$, $\beta ct = x$. Solving simultaneously,

$$ct = \gamma^{-1} (1 - \beta^2)^{-1} = \gamma, \quad x = \beta \gamma. \quad (4.5)$$

x' unit tick: $ct' = 0, x' = 1$ implies $ct = \beta x$, $1 = \gamma(-\beta ct + x)$. Solving simultaneously,

$$x = \gamma^{-1} (1 - \beta^2)^{-1} = \gamma, \quad ct = \beta \gamma. \quad (4.6)$$

Thus unit ticks on the ct' and x' axes correspond to lengths different from unity on the ct and x axes. There is an enhancement precisely by a factor of γ in both the cases.

- **Rapidity:** Eqs. (4.5) and (4.6) also imply that the spatial separation between the unit tick on the ct' axis and the corresponding mark γ on the ct axis along the (ct, x) observers line of simultaneity, and the temporal separation between the unit tick on the x' axis and the corresponding mark γ on the x axis along the (ct, x) observers line of rest position, are both equal to $\beta\gamma$, as shown in Fig. 9(a). Therefore, the angle between the ct' axis and the ct axis is given by

$$\zeta = \tan^{-1} \beta, \quad (4.7)$$

and the same is the tilt of the x' axis w.r.to the x axis. The angle ζ is the rapidity parameter that we had discussed in §3.2.3.

- **Coordinates of an event:** Consider an event at some point A on the space-time diagram as shown in Fig. 9(b). The coordinates of the event, in the (t, x) and (t', x') systems, i.e. (A^0, A^1) and (A'^0, A'^1) , are determined by taking parallel projections of $\vec{OA}$ on the $x^0 = ct$ and $x^1 = x$ axes and on the $x'^0 = ct'$ and $x'^1 = x'$ axes respectively. The space-time distance $|\vec{OA}|$, between the event at A and some other event at the origin O , is a Lorentz invariant quantity, given by

$$|\vec{OA}|^2 = (A^0)^2 - (A^1)^2 = (A'^0)^2 - (A'^1)^2 \implies \text{Inv.} \quad (4.8)$$

It is important to note that $|\vec{OA}|^2$ can have all possible values — positive, negative and zero — in Minkowski space-time, unlike in Euclidean space which has a positive definite squared distance between any two points (as determined by applying the Pythagorus theorem on the parallel projections of the vectorial separation of those points on the axes). We will discuss more on this later on.

- **Time dilation and Length contraction:** The above verification of the elongation of unit ticks of both the time and space axes of a moving frame, is a clear demonstration of the phenomena of *time dilation* and *length contraction*. Specifically, the statements are as follows:

Time dilation is the effect that a moving clock ticks at a rate slower by a factor of γ compared to an identical clock at rest.

Length contraction is the effect that a moving stick appears shorter by a factor of γ than an identical stick at rest.

- **Temporal ordering of events:**

The *time-reversal* invariance of physical laws demands that in a theory of relativity a sequence of events $\{P_1, P_2, \dots, P_{n-1}, P_n\}$ occurring at times $t_{P_1} < t_{P_2} < \dots < t_{P_{n-1}} < t_{P_n}$ must be described by the same set of physical laws which describe the time reversed sequence of events $\{P_n, P_{n-1}, \dots, P_2, P_1\}$ occurring at $t_{P_n} < t_{P_{n-1}} < \dots < t_{P_2} < t_{P_1}$. Special relativistic implementation of the time-reversal invariance is that the *time ordering* of events must be Lorentz invariant. There are however certain restrictions.

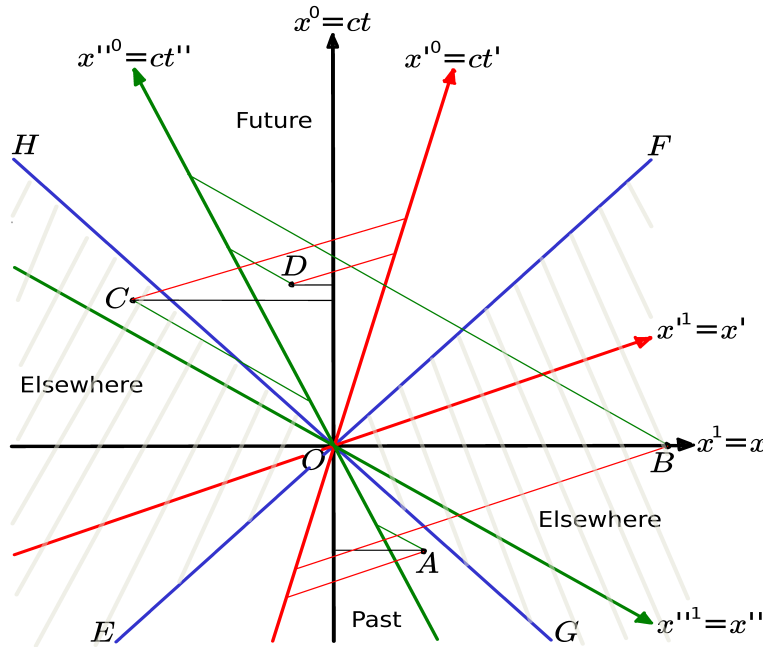


Figure 10: Temporal ordering of events, which decides the past, present and future, are illustrated for three coordinate systems (t, x) , (t', x') and (t'', x'') .

In Fig. 10, we have three inertial systems of reference (t, x) , (t', x') and (t'', x'') , and a set of events at space-time locations A, B, C, D and at O , the common origin of all the systems. These events however occur in different sequence in the three systems:

In system (t, x) : the event at A , is followed simultaneously by the events at O and B , followed by the event at C , followed by the event at D .

In system (t', x') : the event at A , is followed by the event at B , followed by the event at O , followed by the event at D , followed by the event at C .

In system (t'', x'') : the event at A , is followed by the event at O , followed by the event at C , followed by the event at D , followed by the event at B .

For world-lines passing through, say, the origin O , the requirement however is that the time ordering of events should be maintained in all the systems. This is fulfilled only by the events at A, O, D — the event at A happens before that at O whereas the event at D happens after that at O . If we say that the event at O is the event occurring at “*present*”, then the events at A and D are respectively in the “*past*”

and “future” of O in all the systems. On the other hand, whether the events at B and C are at past or at future of the event at O , or simultaneously occurring with the latter, is ambiguous (i.e. depends on the choice of the reference frame). Therefore the world-lines connecting the events at B and C with O have no physical significance.

- **Super-luminal signals and Causality violation:**

One may note that the world-lines connecting an event at O with those at B and C , in Fig. 10, have slopes less than 1 (in magnitude), i.e. they represent the signals propagating from O to B and C , or vice versa, with speeds greater than c . Such signals are referred to as “super-luminal”, as it is well-known that the speed c can be identified with that of the light rays in vacuum (although we shall not prove this statement here).

It is in fact globally acknowledged that the concept of “past” and “future” of a given space-time location O is uniquely specified only for world-lines with slopes greater than 1, terminating at or starting from O . In other words, the signals connecting an event at O with another event of physical viability must not propagate with speed greater than c , and must be represented by world-lines that are contained within the unshaded regions formed by the null lines EF and GH in Fig. 10. World-lines “elsewhere” (shaded regions), which represent events connected to O by super-luminal signals, violate *causality*, as demonstrated by the space-time diagram shown in Fig. 11.

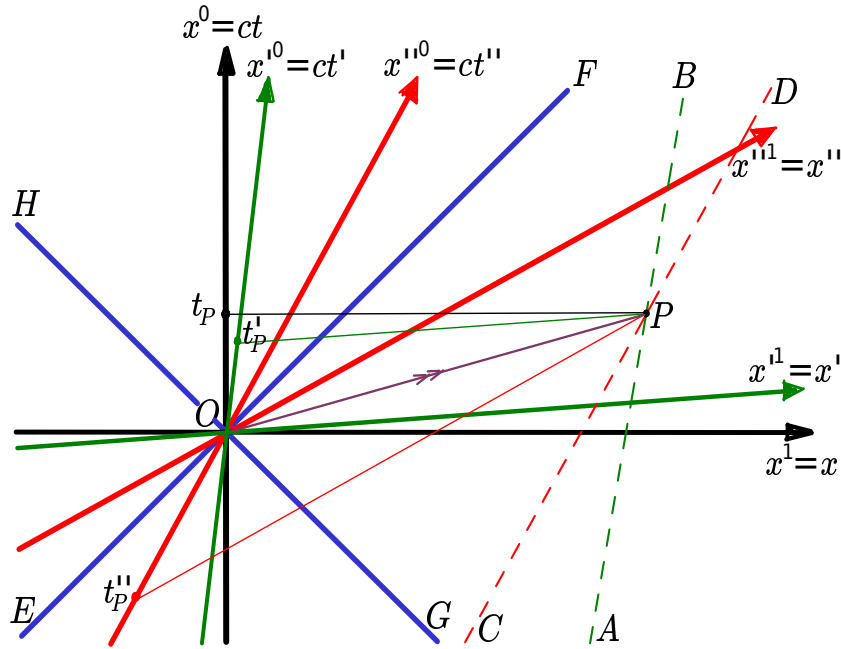


Figure 11: Illustration of the causality violation of an event connected to the origin by a super-luminal signal. A counter-example in the coordinate system (t', x') is also shown.

Consider first a stationary transmitter in the (t, x) system, and a receiver at rest in the (t', x') system. The world-line of the receiver is AB , as shown in Fig. 11. Suppose that at $t = 0$ the transmitter, located at the origin O , is turned on so that a signal is sent with a super-luminal speed along the line OP . After an elapsed time t_P , the signal gets received at the location P situated on the world-line AB of the receiver, whence the latter’s clock reads t'_P . In the (t, x) system, the event at O (viz. the transmitter being turned on) causes the event at P (viz. the signal is received), since $t_P > 0$. The same is the sequence of cause and effect in the (t', x') system, since $t'_P > 0$ as well. Therefore, the causal relationship is maintained in both the systems (t, x) and (t', x') , although the signal propagates with a speed greater than c .

Consider now the transmitter to be stationary in the (t, x) system, while the receiver is at rest in the (t'', x'') system. The world-line of the receiver is CD , as shown in Fig. 11. At $t = 0$ the transmitter

sends a super-luminal signal from the origin O , along the line OP . After an elapsed time t_P , the signal gets received at the location P situated on the world-line CD of the receiver, whence the latter's clock reads t_P'' . But since $t_P'' < 0$, in this case we have the cause and effect getting reversed in the (t'', x'') system — the signal is received at P *before* the transmitter is turned on at O !

What would be our inference then? Super-luminal signals violate causality? The answer is *yes*, despite the counter-illustration of the (t', x') system in Fig. 11. The argument in support of the causality violation is that since both the time and space axes of the receiver's frame could be aligned as close as the null line EF , therefore *for a world-line representing the propagation of a super-luminal signal, it is always possible to find a Lorentz frame in which the sequence of cause and effect gets reversed*. Such a Lorentz frame could be the one represented by the ct'' and x'' axes in Fig. 11, or with axes making even more acute angle between them. Existence of one such frame is enough to conclude *causality violation by super-luminal signals*, no matter how many other frames are there (e.g. the (t, x) and (t', x') systems in Fig. 11) in which cause and effect are not interchanged for super-luminal signals.

4.2 The Lightcone

So far we have discussed only about the $(1 + 1)$ -dimensional Minkowski space-time diagram. If we generalize such a diagram in, say, a $(2 + 1)$ -dimensional space-time (with two spatial coordinates x^1, x^2 and one time coordinate x^0), then the intersecting null lines would become two dimensional conical surfaces intersecting at the origin O . That is, these so-called *null surfaces* would form the curved surfaces of two cones whose axes are along the positive and negative time axis $x^0 = ct$, the (null) surfaces being at 45° to the axis, as shown in Fig. 12.

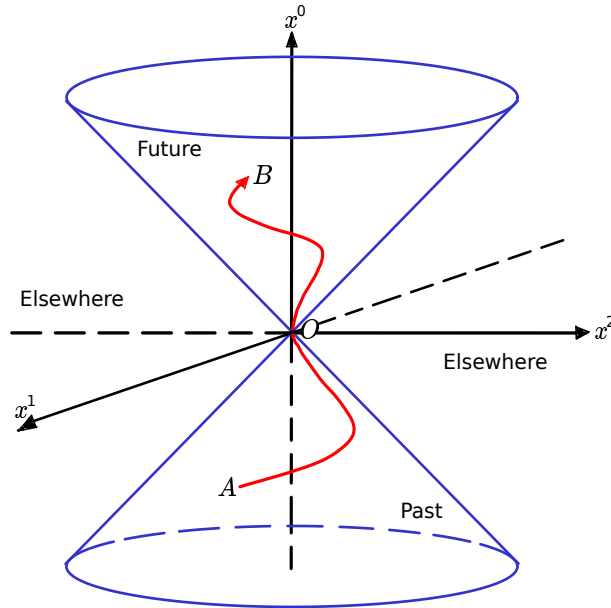


Figure 12: The Lightcone in a $(2 + 1)$ -dimensional space-time with the time axis x^0 and space axes x^1 and x^2 , the third space axis is suppressed. The world-line of a material particle with speed less than c is also shown.

The entire construction with the two cones above and below the origin O is known as a $(2 + 1)$ -dimensional *Lightcone*, as any two points on its surfaces (viz. the null surfaces) could be connected only by signals propagating with the speed of light c . The top and base of this Lightcone are the circles with equation $(x^1)^2 + (x^2)^2 = (x^0)^2$. The cone below the origin O , which defines the region in the past of O , is known as the “*past cone*”, whereas the cone above the origin O , which defines the region in the future of O , is known as the “*future cone*”. The regions outside the Lightcone are referred to as “*elsewhere*”. The line AOB confined within the Lightcone could be the world-line of a material particle whose speed $|\vec{v}| = |d\vec{x}/dt|$ always remains less than c . As time proceeds (from zero onwards) the events associated with the particle trace out the sector OB of the world-line, which lies within the future cone of O (i.e. the $t = 0$ point). On the other hand, the

history of all events associated with the particle, upto the chosen origin O , is represented by the sector AO of the world-line, which lies within the past cone of O , as shown in Fig. 12.

The complete $(3 + 1)$ -dimensional version of the Minkowski diagram takes into account the $(3 + 1)$ -dimensional Lightcone, which is a higher double-cone whose top and base are three-spheres described by the equation $(x^1)^2 + (x^2)^2 + (x^3)^2 = (x^0)^2$. The edges of such a Lightcone are 3-dimensional conical hypersurfaces, which make 45° with the axes.

All the features of the $(1 + 1)$ -dimensional Minkowski diagram (discussed in the previous sub-section), and the arguments thereof, hold for the $(2 + 1)$ and $(3 + 1)$ -dimensional versions of the Lightcone.

4.3 Time-like, Space-like and Null Intervals

Consider now the space-time separation between two events $P_A(t_A, \vec{x}_A)$ and $P_B(t_B, \vec{x}_B)$ in the $(t, \vec{x})$ system:

$$s_{AB}^2 = c^2 (t_A - t_B)^2 - |\vec{x}_A - \vec{x}_B|^2. \quad (4.9)$$

For any two such events, there are three possibilities:

1. $s_{AB}^2 > 0$: In which case, the events are said to have a “time-like” separation.

- Since s_{AB}^2 is invariant under Lorentz transformation (LT), it is always possible to find a reference system $(t', \vec{x}')$ in which $\vec{x}'_A = \vec{x}'_B$, whence $s_{AB}^2 = c^2 (t'_A - t'_B)^2 > 0$ (guaranteed). In other words, there always exists a Lorentz frame Σ' , in which the events occur at the same position, but are separated in time. For e.g. one event may be at the origin O , and the other in the past or future region of the Lightcone.
- If the events are infinitesimally close in time originally, then

$$s_{AB}^2 > 0 \Rightarrow \lim_{(t_A - t_B) \rightarrow 0} \frac{|\vec{x}_A - \vec{x}_B|^2}{(t_A - t_B)^2} < c^2 \Rightarrow \underline{\underline{v < c}}. \quad (4.10)$$

Accordingly, the particles moving with speed $v < c$ would have their event-trajectories inside the Lightcone, say for instance the world-line AOB in Fig. 12. Such particles are called *time-like particles*. All material particles are time-like.

2. $s_{AB}^2 < 0$: In which case, the events are said to have a “space-like” separation.

- It is always possible to find a reference system $(t', \vec{x}')$ in which $t'_A = t'_B$, whence $s_{AB}^2 = -|\vec{x}'_A - \vec{x}'_B|^2 < 0$ (guaranteed). In other words, there always exists a Lorentz frame Σ' , in which the events occur at different space points at the same instant of time. For e.g. one event may be at the origin O , and the other in the elsewhere region (outside the Lightcone).
- If the events are infinitesimally close in time originally, then

$$s_{AB}^2 < 0 \Rightarrow \lim_{(t_A - t_B) \rightarrow 0} \frac{|\vec{x}_A - \vec{x}_B|^2}{(t_A - t_B)^2} > c^2 \Rightarrow \underline{\underline{v > c}}. \quad (4.11)$$

Accordingly, the particles moving with speed $v > c$ would have their event-trajectories protruding outside of the Lightcone. Such particles are called *space-like particles*. No material particle could be space-like, only some hypothetical (unstable) particles arising in some effective theories are space-like.

3. $s_{AB}^2 = 0$: In which case, the events are said to have a “light-like” or “null” separation.

- The events occur on the Lightcone and can be connected only by light signals, propagating with speed c , since for infinitesimally close enough events A and B

$$s_{AB}^2 = 0 \Rightarrow \lim_{(t_A - t_B) \rightarrow 0} \frac{|\vec{x}_A - \vec{x}_B|^2}{(t_A - t_B)^2} = c^2 \Rightarrow \underline{\underline{v = c}}. \quad (4.12)$$

- The corresponding particles are called *null particles* or *light particles* (e.g. photons, neutrinos).

4.4 The Proper Time and Time Dilation

Consider a particle moving with instantaneous velocity $\vec{v}(t)$ relative to some inertial frame Σ :

- In time dt , the change in the position: $d\vec{x} = \vec{v}(t) dt$.
- Corresponding infinitesimal invariant interval:

$$ds^2 = c^2 dt^2 - |d\vec{x}|^2 = c^2 dt^2 \left[1 - |\vec{\beta}(t)|^2 \right], \quad \text{where } \vec{\beta}(t) = \frac{\vec{v}(t)}{c}. \quad (4.13)$$

- In another frame Σ' , in which the particle is supposed to be instantaneously at rest:

$$ds^2 = c^2 dt'^2 = c^2 d\tau^2, \quad \text{where } \boxed{d\tau = dt \sqrt{1 - |\vec{\beta}(t)|^2} = \frac{dt}{\gamma(t)}}: \text{ Lorentz invariant}. \quad (4.14)$$

- The time variable τ is called the *Proper time*. It is the time as measured in the rest frame of a particle, or the time measured in a frame moving synchronously with a particle.

Consider now two successive events P_A and P_B , associated with a particle, separated by the proper time interval $(\tau_B - \tau_A)$ in the particle's rest frame. In another frame Σ (with time coordinate t), relative to which the particle is moving with a velocity $\vec{v}$, the events are temporally separated by:

$$(t_B - t_A) = \int_{\tau_A}^{\tau_B} \gamma(\tau) d\tau. \quad (4.15)$$

Since $\gamma = (1 - |\vec{\beta}|^2)^{-1/2} > 1$ always, Eq. (4.15) implies that $(t_B - t_A) > (\tau_B - \tau_A)$, i.e., the temporal separation between the events is elongated in the moving frame Σ — a phenomenon called *Time dilation*.

5 Tensor Analysis

5.1 General Rule of Transformation of Coordinate Differentials

Consider two reference frames (RF) Σ and Σ' . If the coordinates $\{x'^\mu\} = (x'^0, x'^1, x'^2, x'^3)$ in Σ' be related to the coordinates $\{x^\mu\} = (x^0, x^1, x^2, x^3)$ in Σ by a coordinate transformation:

$$x'^\mu = x'^\mu(x^0, x^1, x^2, x^3), \quad \mu = 0, 1, 2, 3, \quad (5.1)$$

then the necessary and sufficient condition that the four components of x'^μ form a set of independent, real functions of (x^0, x^1, x^2, x^3) is that their *Jacobian*:

$$\mathcal{J}(x \rightarrow x') = \left| \frac{\partial x'}{\partial x} \right| = \begin{vmatrix} \frac{\partial x'^0}{\partial x^0} & \cdots & \frac{\partial x'^3}{\partial x^0} \\ \vdots & & \vdots \\ \frac{\partial x'^0}{\partial x^3} & \cdots & \frac{\partial x'^3}{\partial x^3} \end{vmatrix} \neq 0. \quad (5.2)$$

In such case, one can have the inverse coordinate transformations as:

$$x^\mu = x^\mu(x'^0, x'^1, x'^2, x'^3), \quad \mu = 0, 1, 2, 3. \quad (5.3)$$

At any point in a general spacetime, the direction (of a vector) in a coordinate frame, say Σ , is determined by the coordinate differentials dx^μ defined in that frame. In another frame, say Σ' , the direction is determined by the associated coordinate differentials dx'^ν . The relation between the coordinate differentials in the two frames is given by the transformation rules:

$$dx'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} dx^\nu, \quad \text{and} \quad dx^\mu = \frac{\partial x^\mu}{\partial x'^\nu} dx'^\nu. \quad (5.4)$$

While carrying out such a transformation, we have

$$\frac{\partial x'^\mu}{\partial x^\alpha} \frac{\partial x^\alpha}{\partial x'^\nu} = \delta^\mu_\nu, \quad \text{and} \quad \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x'^\mu}{\partial x^\beta} = \delta^\alpha_\beta. \quad (5.5)$$

So, if one considers $\partial x'^\mu / \partial x^\alpha$ as a matrix — the *transformation matrix*, for transformation from the x^α coordinate system to the x'^μ coordinate system — then $\partial x^\alpha / \partial x'^\mu$ is its inverse, i.e., the transformation matrix for the inverse transformation from the x'^μ coordinate system to the x^α coordinate system. Consequently, the Jacobians of the transformation and the inverse transformation satisfy:

$$\mathcal{J}(x \rightarrow x') \mathcal{J}(x' \rightarrow x) = \left| \frac{\partial x'}{\partial x} \right| \left| \frac{\partial x}{\partial x'} \right| = 1. \quad (5.6)$$

In Special Relativity (SR), the transformation matrices are the LT matrices, and therefore the Jacobian is identically equal to unity:

$$\frac{\partial x'^\mu}{\partial x^\nu} \stackrel{\text{SR}}{=} \Lambda^\mu_\nu, \quad \mathcal{J}(x \rightarrow x') = \left| \frac{\partial x'}{\partial x} \right| \stackrel{\text{SR}}{=} \det(\Lambda^\mu_\nu) = 1. \quad (5.7)$$

5.2 Definition and Categorization of tensors

Tensors (of specific *rank* and *type*) are physical entities associated with a spacetime point, which follow linear, homogeneous laws of transformation under the transformation of coordinates. A tensor carries superscript and subscript indices, the total number of which signifies the so-called *rank* of the tensor:

$$T^{\alpha_1 \alpha_2 \dots \alpha_m}_{\beta_1 \beta_2 \dots \beta_n}(x^\mu) \Rightarrow \text{Rank} = (m + n). \quad (5.8)$$

In general, each component of a tensor is a function of the coordinates x^μ ($\mu = 0, 1, 2, 3$).

5.2.1 Tensor of Rank 0 : Scalar

Rank 0 tensors are scalars which remain invariant under the transformation of coordinates:

$$\phi(x^\mu) \xrightarrow{\text{Inv.}} \text{Coordinate transformations } x^\mu \rightarrow x'^\mu. \quad (5.9)$$

In SR: a (space-time) scalar, invariant under the Lorentz transformations, is often called a *Lorentz scalar* or a *Four-scalar*, e.g., the invariant distance squared $s^2 = c^2 t^2 - |\vec{x}|^2$.

5.2.2 Tensor of Rank 1 : Vector

Rank 1 tensors are vectors, which are of two types:

I. Contravariant vector: denoted by

$$A^\alpha(x^\mu) = \left(A^0(x^\mu), A^1(x^\mu), A^2(x^\mu), A^3(x^\mu) \right),$$

transforms like the coordinate differentials dx^α , under a given coordinate transformation:

$$\boxed{A'^\mu = \frac{\partial x'^\mu}{\partial x^\alpha} A^\alpha} \xrightarrow[\text{to}]{\text{similar}} dx'^\mu = \frac{\partial x'^\mu}{\partial x^\alpha} dx^\alpha. \quad (5.10)$$

In SR: the above transformation rule becomes

$$\boxed{A'^\mu = \Lambda^\mu_\alpha A^\alpha}, \quad (5.11)$$

and a contravariant vector is often called a (space-time) *Four-vector*.

II. Covariant vector: denoted by

$$B_\beta(x^\mu) = \left(B_0(x^\mu), B_1(x^\mu), B_2(x^\mu), B_3(x^\mu) \right),$$

transforms like the partial derivatives $\partial/\partial x^\beta$ of a scalar $\phi(x^\mu)$, under a given coordinate transformation:

$$\boxed{B'_\nu = \frac{\partial x^\beta}{\partial x'^\nu} B_\beta} \xrightarrow[\text{to}]{\text{similar}} \frac{\partial \phi}{\partial x'^\nu} = \frac{\partial x^\beta}{\partial x'^\nu} \frac{\partial \phi}{\partial x^\beta}. \quad (5.12)$$

In SR: the above transformation rule becomes

$$\boxed{B'_\nu = \Lambda_\nu^\beta B_\beta}, \quad (5.13)$$

and a covariant vector is often called a (space-time) *co-vector*.

One may also note, by analogy with the ordinary three-dimensional Euclidean space, that the partial derivatives of a scalar ϕ w.r.to the coordinates x^μ constitute the gradient of the scalar, which points along the normal to the surface $\phi = \text{constant}$. Accordingly, the components of a covariant vector B_β may be thought of as the *normal*, or *perpendicular*, projections of the vector on the coordinate axes. By a similar reasoning, the components of a contravariant vector A^α may be thought of as the *tangential*, or *parallel*, projections of the vector on the coordinate axes. Our familiar perception is that the parallel projections define the components of a vector in the Euclidean three-dimensional space. Hence, it is convenient to consider the contravariant vectors to form a vector space in the usual sense, whereas the covariant vectors constitute a *dual* vector space. Such a duality is of course the duality of projections in the coordinate space, and not to be confused with the *tensor duality* which we shall discuss later on.

5.2.3 Tensors of Rank ≥ 2

These are of three types:

I. Contravariant tensor: of say rank $m (\geq 2)$, denoted by $T^{\alpha_1 \alpha_2 \dots \alpha_m}(x^\mu)$, transforms like a direct product of m contravariant vectors, under a given coordinate transformation:

$$\boxed{T'^{\mu_1 \dots \mu_m} = \frac{\partial x'^{\mu_1}}{\partial x^{\alpha_1}} \dots \frac{\partial x'^{\mu_m}}{\partial x^{\alpha_m}} T^{\alpha_1 \dots \alpha_m}} \xrightarrow[\text{to}]{\text{similar}} A'^{\mu_1} \dots A'^{\mu_m} = \frac{\partial x'^{\mu_1}}{\partial x^{\alpha_1}} \dots \frac{\partial x'^{\mu_m}}{\partial x^{\alpha_m}} A^{\alpha_1} \dots A^{\alpha_m}. \quad (5.14)$$

In SR: the above transformation rule becomes

$$\boxed{T'^{\mu_1 \dots \mu_m} = \Lambda^{\mu_1}_{\alpha_1} \dots \Lambda^{\mu_m}_{\alpha_m} T^{\alpha_1 \dots \alpha_m}}. \quad (5.15)$$

II. Covariant tensor: of say rank $n (\geq 2)$, denoted by $T_{\beta_1 \beta_2 \dots \beta_n}(x^\mu)$, transforms like a direct product of n covariant vectors, under a given coordinate transformation:

$$\boxed{T'_{\nu_1 \dots \nu_n} = \frac{\partial x^{\beta_1}}{\partial x'^{\nu_1}} \dots \frac{\partial x^{\beta_n}}{\partial x'^{\nu_n}} T_{\beta_1 \dots \beta_n}} \xrightarrow[\text{to}]{\text{similar}} B'_{\nu_1} \dots B'_{\nu_n} = \frac{\partial x^{\beta_1}}{\partial x'^{\nu_1}} \dots \frac{\partial x^{\beta_n}}{\partial x'^{\nu_n}} B_{\beta_1} \dots B_{\beta_n}. \quad (5.16)$$

In SR: the above transformation rule becomes

$$\boxed{T'_{\nu_1 \dots \nu_n} = \Lambda_{\nu_1}^{\beta_1} \dots \Lambda_{\nu_n}^{\beta_n} T_{\beta_1 \dots \beta_n}}. \quad (5.17)$$

III. Mixed tensor: of say rank $(m+n) (\geq 2, m \geq 1, n \geq 1)$, denoted by $T^{\alpha_1 \alpha_2 \dots \alpha_m}_{\beta_1 \beta_2 \dots \beta_n}(x^\mu)$, transforms like a direct product of m contravariant and n covariant vectors, under a given coordinate transformation:

$$\boxed{T'^{\mu_1 \dots \mu_m}_{\nu_1 \dots \nu_n} = \left(\frac{\partial x'^{\mu_1}}{\partial x^{\alpha_1}} \dots \frac{\partial x'^{\mu_m}}{\partial x^{\alpha_m}} \right) \left(\frac{\partial x^{\beta_1}}{\partial x'^{\nu_1}} \dots \frac{\partial x^{\beta_n}}{\partial x'^{\nu_n}} \right) T^{\alpha_1 \dots \alpha_m}_{\beta_1 \dots \beta_n}} \quad (5.18)$$

$$\xrightarrow[\text{to}]{\text{similar}} (A'^{\mu_1} \dots A'^{\mu_m}) (B'_{\nu_1} \dots B'_{\nu_n}) = \left(\frac{\partial x'^{\mu_1}}{\partial x^{\alpha_1}} \dots \frac{\partial x'^{\mu_m}}{\partial x^{\alpha_m}} \right) \left(\frac{\partial x^{\beta_1}}{\partial x'^{\nu_1}} \dots \frac{\partial x^{\beta_n}}{\partial x'^{\nu_n}} \right) (A^{\alpha_1} \dots A^{\alpha_m}) (B_{\beta_1} \dots B_{\beta_n}).$$

In SR: the above transformation rule becomes

$$\boxed{T'^{\mu_1 \dots \mu_m}_{\nu_1 \dots \nu_n} = (\Lambda^{\mu_1}_{\alpha_1} \dots \Lambda^{\mu_m}_{\alpha_m}) (\Lambda_{\nu_1}^{\beta_1} \dots \Lambda_{\nu_n}^{\beta_n}) T^{\alpha_1 \dots \alpha_m}_{\beta_1 \dots \beta_n}}. \quad (5.19)$$

5.3 Tensor Algebra

5.3.1 Linear Combination of Tensors

Only tensors of the same rank and type (contravariant, covariant or mixed) could be added to produce another tensor of identical rank and type. The same is true, in fact, for any linear combination of tensors of same rank and type. For e.g. given two covariant tensors of rank 2, $A_{\alpha\beta}$ and $B_{\alpha\beta}$, the linear combination

$$T_{\alpha\beta} = \phi A_{\alpha\beta} + \xi B_{\alpha\beta}, \quad [\text{where } \phi, \xi \text{ are space-time scalars}], \quad (5.20)$$

also transforms like a covariant tensor of rank 2.

5.3.2 Multiplication of Tensors

Unlike the linear combination, the multiplication of tensors of any type and rank, same or different, is possible. Accordingly, one can define three kinds of tensor products:

I. Direct Product: The aggregate obtained via multiplication of any two tensors of the same or different type, and of ranks m and n say, is another tensor of rank $(m + n)$. For e.g.

$$A^\alpha B_\beta = T^\alpha_\beta, \quad A^{\alpha\beta} B^\gamma = T^{\alpha\beta\gamma}, \quad A^\alpha B_{\beta\gamma}{}^\delta = T^\alpha_{\beta\gamma}{}^\delta, \quad \dots \quad (5.21)$$

II. Contracted Product: If two different types of tensors of ranks m and n say, have p number of matching contra and covariant indices, then the direct product of the tensors reduces to another tensor of rank $(m + n - 2p)$ with no matching contra and covariant indices. For e.g.

$$A^{\alpha\beta} B_{\beta\gamma} = T^\alpha_\gamma, \quad A^{\alpha\beta}{}_{\mu\nu} B_{\gamma\delta}{}^{\mu\nu} = T^{\alpha\beta}{}_{\gamma\delta}, \quad A^{\alpha\beta}{}_\gamma B_\beta{}^\gamma = T^\alpha, \quad \dots \quad (5.22)$$

Contraction implies the reduction of any mixed tensor of rank n (> 1), and with p matching contra and covariant indices, to a new tensor of rank $(n - 2p)$ after summing over those p indices. For e.g.

$$T^\mu{}_{\alpha\mu} = T_\alpha, \quad T^\mu{}_{\alpha\mu\beta} = T_{\alpha\beta}, \quad T^{\mu\alpha}{}_\mu{}^{\nu\beta}{}_{\nu\gamma} = T^{\alpha\beta}{}_\gamma, \quad \dots \quad (5.23)$$

III. Scalar Product: Scalar function which is the outcome of a fully contracted product of two tensors of the same rank and when the contra(co)variant indices of one is identical to the co(contra)variant indices of the other. For e.g.

$$A \cdot B := A^\alpha B_\alpha, \quad S \cdot T := S^{\alpha\beta} T_{\alpha\beta}, \quad Q \cdot R := Q^{\alpha\beta}{}_\gamma R_{\alpha\beta}{}^\gamma, \quad \dots \quad (5.24)$$

5.3.3 Norm and Trace of Tensors

The *norm* of a tensor of any rank is its scalar product with itself. For e.g.

$$\|A\|^2 := A_\alpha A^\alpha, \quad \|S\|^2 := S_{\alpha\beta} S^{\alpha\beta}, \quad \|T\|^2 := T_{\alpha\beta\gamma} T^{\alpha\beta\gamma}, \quad \dots \quad (5.25)$$

Unlike the norm, the *trace* is defined only for tensors of even rank (≥ 2). The trace is the full contraction of the indices of the even rank tensor in the mixed form. For e.g.

$$\text{tr}(S^{\alpha\beta}) := S^\alpha_\alpha, \quad \text{tr}(T^{\alpha\beta\gamma\delta}) := T^\alpha_\alpha{}^\beta_\beta{}^\gamma_\gamma{}^\delta_\delta, \quad \dots \quad (5.26)$$

5.4 The Metric Tensor

5.4.1 Definition and Properties

Let us recall the foremost criterion in the formulation of a relativity theory, that the corresponding transformation equations must leave invariant the line element ds^2 , i.e. the squared separation between two infinitesimally close points in space, or events in space-time. For e.g. in Galilean relativity, the corresponding Galilean transformation equations leave invariant the infinitesimal squared separation of points in the three-dimensional Euclidean space:

$$ds^2 \stackrel{\text{Euc}}{=} |d\vec{x}|^2 = (dx^1)^2 + (dx^2)^2 + (dx^3)^2, \quad (5.27)$$

whereas in special relativity, the corresponding Lorentz or Poincaré transformation equations leave invariant the infinitesimal squared separation of events in the four-dimensional Minkowski space-time:

$$ds^2 \stackrel{\text{SR}}{=} (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2. \quad (5.28)$$

Since ds^2 is quadratic in the differentials of the coordinates (dx^α , where $\alpha = 0, 1, 2, 3$ in general), one may express it (as any other quadratic):

$$\boxed{ds^2 = g_{\alpha\beta} dx^\alpha dx^\beta}, \quad (5.29)$$

where the coefficients $g_{\alpha\beta}$ ($\alpha, \beta = 0, 1, 2, 3$) form the elements of a covariant tensor of rank 2, known as the *metric tensor*, or sometimes simply the *metric*. In general, $g_{\alpha\beta}$ could be a function of the coordinates, and not necessarily diagonal, i.e. there could be cross terms in ds^2 , such as $dx^1 dx^2$, $dx^2 dx^3$, $dx^0 dx^1$, $\dots$ (unlike in the Euclidean space or the Minkowski space-time, with ds^2 given by Eqs. (5.27) and (5.28) respectively).

That the metric $g_{\alpha\beta}$ indeed transforms as a covariant tensor of rank 2, can be verified at once: if, let's say, the above relation (5.29) holds in one frame of reference Σ , then in another frame Σ' , characterized by coordinates x'^μ , the corresponding relation is

$$ds'^2 = g'_{\mu\nu} dx'^\mu dx'^\nu. \quad (5.30)$$

Since $ds'^2 = ds^2$ under the transformation $x^\alpha \longrightarrow x'^\alpha$, we have

$$g'_{\mu\nu} dx'^\mu dx'^\nu = g_{\alpha\beta} dx^\alpha dx^\beta \implies \underline{\underline{g'_{\mu\nu} = \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} g_{\alpha\beta}}}, \quad [\text{QED}]. \quad (5.31)$$

The following are the properties of the metric tensor:

- (i) It is *symmetric* under the interchange of indices, i.e.

$$\boxed{g_{\alpha\beta} = g_{\beta\alpha}}, \quad \forall \alpha, \beta. \quad (5.32)$$

- (ii) Its contracted product with the metric tensor in contravariant form, gives the Kronecker delta, which can be shown to transform as a mixed tensor of rank 2:

$$\boxed{g_{\mu\alpha} g^{\mu\beta} = \delta_\alpha^\beta}, \quad \delta_\alpha^\beta = \begin{cases} 1 & \text{for } \alpha = \beta \\ 0 & \text{otherwise} \end{cases}. \quad (5.33)$$

The symmetry property (5.32) ensures there are at most $4(4+1)/2 = 10$ independent components of $g_{\alpha\beta}$ in a $(3+1)$ -dimensional space-time. Accordingly, one can decompose the r.h.s. of Eq. (5.29) as[¶]:

$$ds^2 = \underbrace{g_{00} (dx^0)^2}_{\substack{\text{ONE term for} \\ \mu = \nu = 0.}} + \underbrace{2g_{0i} dx^0 dx^i}_{\substack{\text{Sum of THREE terms for} \\ \mu = 0, \nu = i. \text{ The factor of} \\ 2 \text{ is for symmetry } g_{0i} = g_{i0}.}} + \underbrace{g_{ij} dx^i dx^j}_{\substack{\text{Sum of SIX terms for} \\ \mu = i, \nu = j, \text{ due to} \\ \text{symmetry } g_{ij} = g_{ji}.}}. \quad (5.34)$$

The property (5.33), on the other hand, implies that the norm of the metric tensor $g_{\alpha\beta}$ is

$$\|g\|^2 = \delta_\alpha^\alpha = 4 \quad \text{in } (3+1)\text{-dimensions.} \quad (5.35)$$

5.4.2 Usage: Raising and Lowering of Tensor indices

Raising and lowering of tensor indices are done by contracting with the metric tensor $g_{\alpha\beta}$. For e.g.

$$\begin{aligned} A_\alpha &= g_{\alpha\beta} A^\beta, & A^\alpha &= g^{\alpha\beta} A_\beta; \\ T_{\mu\nu} &= g_{\mu\alpha} g_{\nu\beta} T^{\alpha\beta}, & T^{\mu\nu} &= g^{\mu\alpha} g^{\nu\beta} T_{\alpha\beta}; \\ T_{\mu\nu\lambda\dots}^{\alpha\beta\gamma\dots} &= \left(g^{\alpha\rho} g^{\beta\sigma} g^{\gamma\tau} \dots \right) T_{\mu\nu\lambda\dots\rho\sigma\tau\dots}, & \dots \end{aligned} \quad (5.36)$$

[¶]It is to be noted that in this section, and throughout, the Greek indices ($\alpha, \beta, \gamma, \dots$) run from 0 to 3, i.e. over the entire $(3+1)$ -dimensional space-time, whereas the Latin indices ($i, j, k, \dots$) run from 1 to 3, i.e. over the 3-dimensional spatial hypersurface.

Accordingly, the scalar products are expressed as

$$\begin{aligned} A \cdot B &:= A^\alpha B_\alpha = g^{\alpha\beta} A_\beta B_\alpha = A_\beta B^\beta \equiv A_\alpha B^\alpha, \\ S \cdot T &:= S^{\alpha\beta} T_{\alpha\beta} = g^{\mu\alpha} g^{\nu\beta} S_{\mu\nu} T_{\alpha\beta} = S_{\mu\nu} T^{\mu\nu} \equiv S_{\alpha\beta} T^{\alpha\beta}, \quad \dots \end{aligned} \quad (5.37)$$

One may also note that for the metric tensor itself the trace is equal to the norm:

$$g_{\alpha\beta} g^{\alpha\beta} = g_\alpha^\alpha = \delta_\alpha^\alpha = \|g\|^2. \quad (5.38)$$

5.4.3 The Minkowski Metric

Special Relativity (SR) alludes to a *special* form of the metric tensor: $g_{\alpha\beta} \equiv \eta_{\alpha\beta}$, referred to as the *Minkowski metric*, which is diagonal, and with each non-vanishing element normalized to unity in magnitude:

$$\eta_{\alpha\beta} = \text{diag}(1, -1, -1, -1) \quad \text{i.e.,} \quad |\eta_{\alpha\beta}| = \delta_{\alpha\beta}. \quad (5.39)$$

Explicitly:

$$\eta_{00} = 1, \quad \eta_{0i} = 0 \quad (\forall i = 1, 2, 3), \quad \eta_{ij} = -\delta_{ij} \quad (\forall i, j = 1, 2, 3), \quad (5.40)$$

and one may at once verify that the line element in Special Relativity is thus given above by Eq. (5.28).

5.4.4 Vectors and Tensors in Minkowski space-time

Let us now see how the Minkowski metric tensor $\eta_{\alpha\beta}$ helps in identifying or(and) correlating the contra and covariant components of a vector or a tensor in special relativity.

Components of a Vector: The components of a covariant vector A_α are related to the components of the corresponding contravariant vector A^α via

$$A_\alpha = \eta_{\alpha\beta} A^\beta \quad \implies \quad \begin{cases} A_0 = \eta_{0\beta} A^\beta = \eta_{00} A^0 = A^0, \\ A_i = \eta_{i\beta} A^\beta = \eta_{ij} A^j = -\delta_{ij} A^j. \end{cases} \quad (5.41)$$

Since in the ordinary Euclidean space, the parallel projections A^i define the components of a (three-)vector $\vec{A}$, we have

$$A^\alpha := (A^0, A^i) \equiv (A^0, \vec{A}) \quad \text{and} \quad A_\alpha := (A_0, A_i) = (A^0, -\delta_{ij} A^j) \equiv (A^0, -\vec{A}). \quad (5.42)$$

Components of a second rank Tensor: The covariant and contravariant components are related as

$$T_{\alpha\beta} = \eta_{\mu\alpha} \eta_{\nu\beta} T^{\mu\nu} \quad \implies \quad \begin{cases} T_{00} = \eta_{0\mu} \eta_{0\nu} T^{\mu\nu} = (\eta_{00})^2 T^{00} = T^{00}, \\ T_{0i} = \eta_{0\mu} \eta_{i\nu} T^{\mu\nu} = \eta_{00} \eta_{ij} T^{0j} = -\delta_{ij} T^{0j}, \\ T_{i0} = \eta_{i\mu} \eta_{0\nu} T^{\mu\nu} = \eta_{ij} \eta_{00} T^{j0} = -\delta_{ij} T^{j0}, \\ T_{ij} = \eta_{i\mu} \eta_{j\nu} T^{\mu\nu} = \eta_{ik} \eta_{jl} T^{kl} = \delta_{ik} \delta_{jl} T^{kl}. \end{cases} \quad (5.43)$$

Corollary: If we choose, $T_{\alpha\beta} \equiv \eta_{\alpha\beta}$, then for the Minkowski metric tensor $\eta_{\alpha\beta}$ itself:

$$\eta_{00} = \eta^{00} = 1, \quad \eta_{0i} = 0 = \eta^{0i}, \quad \eta_{ij} = -\delta_{ij} \iff \eta^{ij} = -\delta^{ij}. \quad (5.44)$$

This implies that $\eta_{\alpha\beta}$ and $\eta^{\alpha\beta}$ have the same set of components, viz. $\text{diag}(1, -1, -1, -1)$.

For higher rank tensors, the relationship between the contra and covariant components could be found in a similar manner as above.

5.5 Tensor Symmetry and Antisymmetry

5.5.1 Symmetric and Antisymmetric Tensors

- Symmetry or antisymmetry is defined only for contravariant and covariant tensors of rank ≥ 2 .
- Symmetry or antisymmetry could either be (i) *partial*, i.e. under the exchange of only a given pair of indices, or (ii) *complete*, i.e. under the exchange of any pair of indices.
- Symmetry or antisymmetry of some pattern (partial or complete) of a covariant tensor implies symmetry or antisymmetry of the same pattern of the corresponding contravariant tensor, and vice versa.

Let us illustrate these as follows:

Rank 2 tensors:

$$\text{Symmetric: } S_{\alpha\beta} = S_{\beta\alpha} \iff S^{\alpha\beta} = S^{\beta\alpha}, \quad (5.45)$$

$$\text{Antisymmetric: } A_{\alpha\beta} = -A_{\beta\alpha} \iff A^{\alpha\beta} = -A^{\beta\alpha} \implies A_{\alpha\alpha} = A^{\alpha\alpha} = 0. \quad (5.46)$$

Rank 3 tensors:

$$\text{Symmetric: } \begin{cases} \text{Partial: } S_{\alpha\beta\gamma} = S_{\beta\alpha\gamma}, \text{ or } S_{\alpha\beta\gamma} = S_{\alpha\gamma\beta}, \text{ or } S_{\alpha\beta\gamma} = S_{\gamma\beta\alpha}; \\ \text{Complete: } S_{\alpha\beta\gamma} = S_{\beta\gamma\alpha} = S_{\gamma\alpha\beta} = S_{\beta\alpha\gamma} = S_{\alpha\gamma\beta} = S_{\gamma\beta\alpha}, \end{cases} \quad (5.47)$$

$$\text{Antisymmetric: } \begin{cases} \text{Partial: } A_{\alpha\beta\gamma} = -A_{\beta\alpha\gamma}, \text{ or } A_{\alpha\beta\gamma} = -A_{\alpha\gamma\beta}, \text{ or } A_{\alpha\beta\gamma} = -A_{\gamma\beta\alpha}; \\ \text{Complete: } A_{\alpha\beta\gamma} = A_{\beta\gamma\alpha} = A_{\gamma\alpha\beta} = -A_{\beta\alpha\gamma} = -A_{\alpha\gamma\beta} = -A_{\gamma\beta\alpha}. \end{cases} \quad (5.48)$$

Similarly, one can have partial or complete symmetry or antisymmetry of higher rank tensors.

5.5.2 Symmetrization and Antisymmetrization of Tensors

Rank 2 tensors: Any (in general asymmetric, and let's say covariant) tensor $T_{\alpha\beta}$ could be split up as

$$\boxed{T_{\alpha\beta} = T_{(\alpha\beta)} + T_{[\alpha\beta]}}, \quad (5.49)$$

where

$$T_{(\alpha\beta)} = T_{(\beta\alpha)} = \frac{1}{2}(T_{\alpha\beta} + T_{\beta\alpha}), \quad T_{[\alpha\beta]} = -T_{(\beta\alpha)} = \frac{1}{2}(T_{\alpha\beta} - T_{\beta\alpha}). \quad (5.50)$$

The symmetric part $T_{(\alpha\beta)}$ is called the *symmetrization* of $T_{\alpha\beta}$, whereas the antisymmetric part $T_{[\alpha\beta]}$ is called the *antisymmetrization* of $T_{\alpha\beta}$. Both these parts transform like second rank covariant tensors. Such a splitting is also possible for a second rank contravariant tensor $T^{\alpha\beta}$. However, a splitting into a (completely) symmetrized part and a (completely) antisymmetrized part is not possible for tensors of rank > 2 .

Rank 3 tensors: Symmetrization and antisymmetrization of, let's say, a covariant tensor $T_{\alpha\beta\gamma}$ give

$$\begin{aligned} T_{(\alpha\beta\gamma)} &= \frac{1}{3!}(T_{\alpha\beta\gamma} + T_{\beta\gamma\alpha} + T_{\gamma\alpha\beta} + T_{\beta\alpha\gamma} + T_{\alpha\gamma\beta} + T_{\gamma\beta\alpha}), \\ T_{[\alpha\beta\gamma]} &= \frac{1}{3!}(T_{\alpha\beta\gamma} + T_{\beta\gamma\alpha} + T_{\gamma\alpha\beta} - T_{\beta\alpha\gamma} - T_{\alpha\gamma\beta} - T_{\gamma\beta\alpha}). \end{aligned} \quad (5.51)$$

By definition, $T_{(\alpha\beta\gamma)}$ and $T_{[\alpha\beta\gamma]}$ are completely symmetric and completely antisymmetric, respectively. One may also note that $T_{[\alpha\beta\gamma]}$ has non-vanishing components only for $\alpha \neq \beta \neq \gamma$. However, as mentioned above, addition (or any linear combination) of $T_{(\alpha\beta\gamma)}$ and $T_{[\alpha\beta\gamma]}$ does not give the original tensor $T_{\alpha\beta\gamma}$.

In principle, symmetrization and antisymmetrization of any general, let's say, covariant of rank $p (\geq 2)$, viz. $T_{\alpha_1\alpha_2\dots\alpha_p}$, in a D -dimensional spacetime (i.e. 1 time + $(D-1)$ space), give

$$T_{(\alpha_1\alpha_2\dots\alpha_p)} = \frac{1}{p!} \sum_P T_{\alpha_{P(1)}\alpha_{P(2)}\dots\alpha_{P(p)}}, \quad (5.52)$$

$$T_{[\alpha_1\alpha_2\dots\alpha_p]} = \frac{1}{p!} \sum_P (-1)^P T_{\alpha_{P(1)}\alpha_{P(2)}\dots\alpha_{P(p)}}, \quad (5.53)$$

where the summations are for all permutations P of the indices $(\alpha_1 \alpha_2 \dots \alpha_p)$ and the factor $(-1)^P$ indicates the parity — equal to -1 of odd permutations P , and $+1$ for even permutations P .

5.5.3 Characteristics of Symmetric and Antisymmetric Tensors

A few characteristics of symmetric and antisymmetric tensors, and of the operation of symmetrization and antisymmetrization of tensors, are in order:

1. Consider a tensor $T_{\alpha_1 \alpha_2 \dots \alpha_p}$ of rank p (say).
 - If it is completely symmetric then one can express $T_{\alpha_1 \alpha_2 \dots \alpha_p} = T_{(\alpha_1 \alpha_2 \dots \alpha_p)}$.
 - If it is partially symmetric, say, in the first n of the p indices, then $T_{\alpha_1 \alpha_2 \dots \alpha_p} = T_{(\alpha_1 \alpha_2 \dots \alpha_n) \alpha_{n+1} \dots \alpha_p}$.
 - If it is completely antisymmetric then one can express $T_{\alpha_1 \alpha_2 \dots \alpha_p} = T_{[\alpha_1 \alpha_2 \dots \alpha_p]}$.
 - If it is partially antisymmetric, say, in the first n of the p indices, then $T_{\alpha_1 \alpha_2 \dots \alpha_p} = T_{[\alpha_1 \alpha_2 \dots \alpha_n] \alpha_{n+1} \dots \alpha_p}$.
2. The scalar product of a symmetric tensor and an antisymmetric tensor vanishes identically. Consider for e.g. two second rank tensors $S_{\alpha\beta} = S_{(\alpha\beta)}$ and $A_{\mu\nu} = A_{[\mu\nu]}$. Their scalar product

$$\begin{aligned} S_{\alpha\beta} A^{\alpha\beta} &= S_{\beta\alpha} A^{\beta\alpha} & : & \text{by interchanging the dummy indices } \alpha, \beta \\ &= S_{\alpha\beta} \left[-A^{\alpha\beta} \right] & : & \text{by using the symmetry and antisymmetry of } S_{\alpha\beta} \text{ and } A^{\alpha\beta} \\ &= 0. \end{aligned} \tag{5.54}$$

In fact, the contracted product any two tensors of rank ≥ 2 vanishes identically, if one tensor is symmetric and the other tensor is antisymmetric in the pair of indices which are being contracted. This is irrespective of whether the tensors are partially or completely antisymmetric, whether they are of equal rank, and whether the contracted product involves more than two identical contra and covariant indices. For e.g.

$$S_{\alpha(\beta\gamma)} A^{[\beta\gamma]\delta} = 0, \quad S_{(\alpha\beta)} A^{[\alpha\beta\gamma\delta]} = 0, \quad S_{(\alpha\beta\gamma\delta)} A_{\mu}^{[\beta\delta]}{}_{\nu} = 0, \quad S^{\lambda}_{(\alpha|\beta|\gamma)\delta} A^{[\alpha|\mu\delta|\gamma]} = 0, \quad \dots \tag{5.55}$$

3. In D -dimensions, the number of independent components of a rank p tensor $T_{\alpha_1 \dots \alpha_p}$, and of the tensors $T_{(\alpha_1 \dots \alpha_p)}$ and $T_{[\alpha_1 \dots \alpha_p]}$ obtained by its symmetrization and antisymmetrization, are respectively

$$\mathcal{N} = D^p, \quad \mathcal{N}_{sym} = \binom{D+p-1}{p} = \frac{(D+p-1)!}{p!(D-1)!}, \quad \mathcal{N}_{antisym} = \binom{D}{p} = \frac{D!}{p!(D-p)!}. \tag{5.56}$$

For e.g. if $p = 2$, then $T_{\alpha_1 \dots \alpha_p} = T_{\alpha_1 \alpha_2}$ has $\mathcal{N} = D^2$ components, and the number of independent components of the symmetric tensor $T_{(\alpha_1 \alpha_2)}$ could be computed as

$$\mathcal{N}_{sym} \Big|_{p=2} = \mathcal{N}(\text{diag}) + \frac{1}{2} \mathcal{N}(\text{off-diag}) = D + \frac{(D^2 - D)}{2} = \frac{D(D+1)}{2}, \tag{5.57}$$

which matches with that obtained from the expression for $\mathcal{N}_{sym}$ in Eq. (5.56), viz.

$$\frac{(D+2-1)!}{2!(D-1)!} = \frac{D(D+1)}{2}.$$

Similarly, keeping in mind that no diagonal elements exist for the antisymmetric tensor $T_{[\alpha_1 \alpha_2]}$, the number of its independent components can be computed as

$$\mathcal{N}_{antisym} \Big|_{p=2} = \frac{1}{2} \mathcal{N}(\text{off-diag}) = \frac{(D^2 - D)}{2} = \frac{D(D-1)}{2}, \tag{5.58}$$

which matches with that one gets using the expression for $\mathcal{N}_{antisym}$ in Eq. (5.56). Moreover, since

$$\frac{D(D+1)}{2} + \frac{D(D-1)}{2} = D^2,$$

the tensors $T_{(\alpha_1 \alpha_2)}$ and $T_{[\alpha_1 \alpha_2]}$, due to symmetrization and antisymmetrization, add up to give the original tensor $T_{\alpha_1 \alpha_2}$. This is a unique feature of rank-2 tensors in a general D -dimensional spacetime.

4. The above expression for $\mathcal{N}_{antisym}$ in Eq. (5.56) is also suggestive from the point of view of its validity only for tensors of rank $p \leq D$. This is evident from the reduced expression:

$$\mathcal{N}_{antisym} = \frac{D!}{p!(D-p)!} = \frac{D(D-1)(D-2)\dots(D-p+1)}{p!}, \quad (5.59)$$

which vanishes whenever $p = D + 1$. Therefore, for any tensor whose rank exceeds the dimensionality, antisymmetrization identically leads to zero. For e.g. in 3-dimensions ($D = 3$) the maximum rank of a completely antisymmetric tensor is 3 (i.e. the tensor is $T_{[ijk]}$), and the tensor has *only one* non-vanishing independent component T_{123} . The other non-vanishing components are all equal to T_{123} in magnitude (e.g. $T_{231}, |T_{312}|$). Similarly, in (3+1)-dimensions ($D = 4$) the maximum rank of a completely antisymmetric tensor is 4 (i.e. the tensor is $T_{[\mu\nu\lambda\sigma]}$), and the only non-vanishing independent component of the tensor is T_{0123} . In fact, complete antisymmetry implies all the indices of a given tensor must be different. So when the rank equals the dimensionality, i.e. the number of tensor indices equals the number of dimensions, there is only one choice left for all the tensor indices to be unequal.

5.5.4 The Levi-Civita Tensor and the Generalized Kronecker Delta

In D -dimensions, the maximum ranked completely antisymmetric tensor is proportional to the *Levi-Civita tensor* defined by*

$$T_{[\alpha_1\alpha_2\dots\alpha_D]} \propto \epsilon_{\alpha_1\alpha_2\dots\alpha_D} = \begin{cases} -1 & \text{for } \alpha_1, \alpha_2, \dots, \alpha_D \text{ an even permutation of } 0, 1, \dots, D \\ +1 & \text{for } \alpha_1, \alpha_2, \dots, \alpha_D \text{ an odd permutation of } 0, 1, \dots, D \\ 0 & \text{otherwise.} \end{cases} \quad (5.60)$$

$$T^{[\alpha_1\alpha_2\dots\alpha_D]} \propto \epsilon^{\alpha_1\alpha_2\dots\alpha_D} = \begin{cases} +1 & \text{for } \alpha_1, \alpha_2, \dots, \alpha_D \text{ an even permutation of } 0, 1, \dots, D \\ -1 & \text{for } \alpha_1, \alpha_2, \dots, \alpha_D \text{ an odd permutation of } 0, 1, \dots, D \\ 0 & \text{otherwise.} \end{cases} \quad (5.61)$$

The following contraction identities exist between $\epsilon^{\mu\nu\lambda\sigma}$ and $\epsilon_{\mu\nu\lambda\sigma}$ in (3 + 1)-dimensions:

$$\epsilon^{\mu\nu\lambda\sigma}\epsilon_{\mu\nu\lambda\sigma} = -4! , \quad \epsilon^{\mu\nu\lambda\sigma}\epsilon_{\alpha\nu\lambda\sigma} = -3!\delta^\mu_\alpha , \quad \epsilon^{\mu\nu\lambda\sigma}\epsilon_{\alpha\beta\lambda\sigma} = -2!\delta^{\mu\nu}_{\alpha\beta} , \quad \epsilon^{\mu\nu\lambda\sigma}\epsilon_{\alpha\beta\gamma\sigma} = -\delta^{\mu\nu\lambda}_{\alpha\beta\gamma} , \quad (5.62)$$

where

$$\delta^{\mu\nu}_{\alpha\beta} = \begin{vmatrix} \delta^\mu_\alpha & \delta^\nu_\alpha \\ \delta^\mu_\beta & \delta^\nu_\beta \end{vmatrix}, \quad \delta^{\mu\nu\lambda}_{\alpha\beta\gamma} = \begin{vmatrix} \delta^\mu_\alpha & \delta^\nu_\alpha & \delta^\lambda_\alpha \\ \delta^\mu_\beta & \delta^\nu_\beta & \delta^\lambda_\beta \\ \delta^\mu_\gamma & \delta^\nu_\gamma & \delta^\lambda_\gamma \end{vmatrix}, \quad \dots, \quad (5.63)$$

are the *Generalized Kronecker Delta* tensors.

*Most appropriately, the definitions hold for the covariant and contravariant Levi-Civita tensor densities $\varepsilon_{\alpha_1\alpha_2\dots\alpha_D}$ and $\varepsilon^{\alpha_1\alpha_2\dots\alpha_D}$. Tensor densities are the most general entities which follow linear, homogeneous laws of transformation under the transformation of coordinates, and tensors form only a sub-class of tensor densities. One can distinguish a tensor from a tensor density considering that the latter involves in its transformation law an extra factor depending on a certain integral power W (called the *weight*) of the Jacobian $|\partial x/\partial x'|$:

$$\mathcal{T}'^{\mu\dots}_{\nu\dots} = \left| \frac{\partial x}{\partial x'} \right|^W \frac{\partial x'^\mu}{\partial x^\alpha} \dots \frac{\partial x^\beta}{\partial x'^\nu} \dots \mathcal{T}^{\alpha\dots}_{\beta\dots}.$$

However, since the Jacobian is identically equal to unity for the Lorentz transformations in Special Relativity, the above transformation law for the tensor density $\mathcal{T}^{\alpha\dots}_{\beta\dots}$ reduces to the transformation law for a tensor $T^{\alpha\dots}_{\beta\dots}$:

$$T'^{\mu\dots}_{\nu\dots} = \frac{\partial x'^\mu}{\partial x^\alpha} \dots \frac{\partial x^\beta}{\partial x'^\nu} \dots T^{\alpha\dots}_{\beta\dots}.$$

Accordingly, there is no distinction between the Levi-Civita tensor density $\varepsilon_{\alpha_1\alpha_2\dots\alpha_D}$ of weight $W = -1$ and the tensor $\epsilon_{\alpha_1\alpha_2\dots\alpha_D}$, and also between the Levi-Civita tensor density $\varepsilon^{\alpha_1\alpha_2\dots\alpha_D}$ of weight $W = +1$ and the tensor $\epsilon^{\alpha_1\alpha_2\dots\alpha_D}$, in Special Relativity.

5.5.5 Tensor Duality

In D -dimensions the total number of independent components of a completely antisymmetric tensor of rank p is the same as that of a completely antisymmetric tensor of rank $(D - p)$, since

$$\mathcal{N}_{antisym}(D, p) = \frac{D!}{p!(D-p)!} = \frac{D!}{(D-p)![D-(D-p)]!} = \mathcal{N}_{antisym}(D, D-p) . \quad (5.64)$$

This implies that there exists a *duality* between a rank p completely antisymmetric tensor and a rank $(D - p)$ completely antisymmetric tensor in D -dimensions. Such a duality relationship is given by

$$A_{\alpha_1 \alpha_2 \dots \alpha_p} = A_{[\alpha_1 \alpha_2 \dots \alpha_p]} = p! \epsilon_{\alpha_1 \alpha_2 \dots \alpha_p \alpha_{p+1} \dots \alpha_D} {}^* A^{\alpha_{p+1} \dots \alpha_D} , \quad (5.65)$$

where ${}^* A^{\alpha_{p+1} \dots \alpha_D}$ is called the (*Hodge*)-dual of $A_{\alpha_1 \alpha_2 \dots \alpha_p}$. The inverse duality relation is given by

$${}^* A_{\alpha_1 \alpha_2 \dots \alpha_p} = {}^* A_{[\alpha_1 \alpha_2 \dots \alpha_p]} = \frac{1}{p!} \epsilon_{\alpha_1 \alpha_2 \dots \alpha_p \alpha_{p+1} \dots \alpha_D} A^{\alpha_{p+1} \dots \alpha_D} . \quad (5.66)$$

The indices of $A_{\alpha\beta\dots}$ (or $A^{\alpha\beta\dots}$) and ${}^* A_{\alpha\beta\dots}$ (or ${}^* A^{\alpha\beta\dots}$) are raised (or lowered) in the usual way using the metric tensor. Hence, in 3-dimensions:

$$\begin{aligned} A_{ijk} = 3! \epsilon_{ijk} {}^* A &\implies A^{ijk} = 3! \epsilon^{ijk} {}^* A , & {}^* A_{ijk} = \frac{1}{3!} \epsilon_{ijk} A &\implies {}^* A^{ijk} = \frac{1}{3!} \epsilon^{ijk} A , \\ A_{ij} = 2! \epsilon_{ijk} {}^* A^k &\implies A^{ij} = 2! \epsilon^{ijk} {}^* A_k , & {}^* A_{ij} = \frac{1}{2!} \epsilon_{ijk} A^k &\implies {}^* A^{ij} = \frac{1}{2!} \epsilon^{ijk} A_k , \\ A_i = \epsilon_{ijk} {}^* A^{jk} &\implies A^i = \epsilon^{ijk} {}^* A_{jk} , & {}^* A_i = \epsilon_{ijk} A^{jk} &\implies {}^* A^i = \epsilon^{ijk} A_{jk} , \\ A = \epsilon_{ijk} {}^* A^{ijk} = \epsilon^{ijk} {}^* A_{ijk} , & & {}^* A = \epsilon_{ijk} A^{ijk} = \epsilon^{ijk} A_{ijk} , & \end{aligned} \quad (5.67)$$

whereas in $(3 + 1)$ -dimensions:

$$\begin{aligned} A_{\alpha\beta\gamma\delta} = 4! \epsilon_{\alpha\beta\gamma\delta} {}^* A &\implies A^{\alpha\beta\gamma\delta} = 4! \epsilon^{\alpha\beta\gamma\delta} {}^* A , & {}^* A_{\alpha\beta\gamma\delta} = \frac{1}{4!} \epsilon_{\alpha\beta\gamma\delta} A &\implies {}^* A^{\alpha\beta\gamma\delta} = \frac{1}{4!} \epsilon^{\alpha\beta\gamma\delta} A , \\ A_{\alpha\beta\gamma} = 3! \epsilon_{\alpha\beta\gamma\delta} {}^* A^\delta &\implies A^{\alpha\beta\gamma} = 3! \epsilon^{\alpha\beta\gamma\delta} {}^* A_\delta , & {}^* A_{\alpha\beta\gamma} = \frac{1}{3!} \epsilon_{\alpha\beta\gamma\delta} A^\delta &\implies {}^* A^{\alpha\beta\gamma} = \frac{1}{3!} \epsilon^{\alpha\beta\gamma\delta} A_\delta , \\ A_{\alpha\beta} = 2! \epsilon_{\alpha\beta\gamma\delta} {}^* A^{\gamma\delta} &\implies A^{\alpha\beta} = 2! \epsilon^{\alpha\beta\gamma\delta} {}^* A_{\gamma\delta} , & {}^* A_{\alpha\beta} = \frac{1}{2!} \epsilon_{\alpha\beta\gamma\delta} A^{\gamma\delta} &\implies {}^* A^{\alpha\beta} = \frac{1}{2!} \epsilon^{\alpha\beta\gamma\delta} A_{\gamma\delta} , \\ A_\alpha = \epsilon_{\alpha\beta\gamma\delta} {}^* A^{\beta\gamma\delta} &\implies A^\alpha = \epsilon^{\alpha\beta\gamma\delta} {}^* A_{\beta\gamma\delta} , & {}^* A_\alpha = \epsilon_{\alpha\beta\gamma\delta} A^{\beta\gamma\delta} &\implies {}^* A^\alpha = \epsilon^{\alpha\beta\gamma\delta} A_{\beta\gamma\delta} , \\ A = \epsilon_{\alpha\beta\gamma\delta} {}^* A^{\alpha\beta\gamma\delta} = \epsilon^{\alpha\beta\gamma\delta} {}^* A_{\alpha\beta\gamma\delta} , & & {}^* A = \epsilon_{\alpha\beta\gamma\delta} A^{\alpha\beta\gamma\delta} = \epsilon^{\alpha\beta\gamma\delta} A_{\alpha\beta\gamma\delta} . & \end{aligned} \quad (5.68)$$

5.6 Tensor Calculus

5.6.1 Tensor Differentiation: Grad, Div, Curl and D'Alembertian operators

Differentiation of tensors in a $(3 + 1)$ -dimensional space-time is analogous to that in the ordinary Euclidean 3-dimensional space. We consider only the special relativistic scenario, i.e. the $(3 + 1)$ -dimensional Minkowski space-time, in which the differential operators and operations are as follows:

Covariant and Contravariant Differential operators:

$$\boxed{\partial_\mu \equiv \frac{\partial}{\partial x^\mu} = \left(\frac{\partial}{\partial x^0}, \frac{\partial}{\partial x^i} \right) \equiv (\partial_0, \nabla)} \quad \text{and} \quad \boxed{\partial^\mu \equiv \frac{\partial}{\partial x_\mu} = \left(\frac{\partial}{\partial x_0}, \frac{\partial}{\partial x_i} \right) \equiv (\partial^0, -\nabla)} , \quad (5.69)$$

where we have used the notations:

$$x_0 = \eta_{0\beta} x^\beta = \eta_{00} x^0 = x^0 \equiv ct \implies \partial^0 \equiv \frac{\partial}{\partial x_0} = \frac{\partial}{\partial x^0} \equiv \partial_0 = \frac{1}{c} \frac{\partial}{\partial t} , \quad (5.70)$$

$$x_i = \eta_{i\beta} x^\beta = \eta_{ij} x^j = -\delta_{ij} x^j = -x^i \equiv -(\vec{x})^i \implies \partial^i \equiv \frac{\partial}{\partial x_i} = -\frac{\partial}{\partial x^i} \equiv -\partial_i = -(\nabla)^i , \quad (5.71)$$

∇ being the ordinary three-gradient operator.

Note that the above co(contra)variant operators [Eq. (5.69)] have component structures just opposite to the co(contra)variant four-vectors, viz. $A_\mu = (A_0, -\vec{A})$, and $A^\mu = (A^0, \vec{A})$, where $\vec{A}$ is the ordinary three-vector.

Covariant Gradient: In a (3 + 1)-dimensional space-time, the *covariant gradient* (or, the *four-gradient*) of a four-scalar $\phi(t, \vec{x})$, defined by

$$\partial_\mu \phi \equiv \frac{\partial \phi}{\partial x^\mu} = \left(\frac{\partial \phi}{\partial x^0}, \nabla \phi \right) = \left(\frac{1}{c} \frac{\partial \phi}{\partial t}, \nabla \phi \right), \quad (5.72)$$

gives a covariant vector. A similar operation, sometimes called the *contravariant gradient*, on the four-scalar $\phi(t, \vec{x})$, yields a contravariant vector

$$\partial^\mu \phi = g^{\mu\nu} \partial_\nu \phi \equiv \frac{\partial \phi}{\partial x_\mu} = \left(\frac{1}{c} \frac{\partial \phi}{\partial t}, -\nabla \phi \right). \quad (5.73)$$

Covariant Divergence: In a (3+1)-dimensional space-time, the *covariant divergence* (or, the *four-divergence*) of a (contravariant) four-vector $A^\mu(t, \vec{x})$, i.e. a contravariant tensor of rank 1, defined by

$$\partial_\mu A^\mu \equiv \frac{\partial A^\mu}{\partial x^\mu} = \frac{\partial A^0}{\partial x^0} + \frac{\partial A^i}{\partial x^i} = \frac{1}{c} \frac{\partial A^0}{\partial t} + \nabla \cdot \vec{A}, \quad (5.74)$$

gives a four-scalar, i.e. a tensor of rank 0. Similarly, the (covariant) divergence of a contravariant tensor $T^{\mu\nu}$ of rank 2, defined by

$$\partial_\mu T^{\mu\nu} \equiv \frac{\partial T^{\mu\nu}}{\partial x^\mu} = \frac{\partial T^{0\nu}}{\partial x^0} + \frac{\partial T^{i\nu}}{\partial x^i} = \left([\partial_0 T^{00} + \partial_i T^{i0}], [\partial_0 T^{0j} + \partial_i T^{ij}] \right), \quad (5.75)$$

yields a (contravariant) four-vector, i.e. a contravariant tensor of rank 1.

In general, we thus have for any contravariant tensor of rank m , the operation of (covariant) divergence giving a contravariant tensor of rank $(m - 1)$. An associated operation, sometimes called the *contravariant divergence* is similarly defined, e.g.

$$\partial_\mu A^\mu = \partial^\mu A_\mu, \quad \partial_\mu T^{\mu\nu} = \partial^\mu T_\mu{}^\nu, \quad \partial_\mu T^{\mu\nu\lambda} = \partial^\mu T_\mu{}^{\nu\lambda}, \quad \dots \quad (5.76)$$

Covariant Curl: In a (3 + 1)-dimensional space-time, the *covariant curl* (or, the *four-curl*) of a covariant vector $A_\nu(t, \vec{x})$, defined by

$$C^{\alpha\beta} = C^{[\alpha\beta]} = \frac{1}{2} \epsilon^{\alpha\beta\mu\nu} (\partial_\mu A_\nu - \partial_\nu A_\mu) = \epsilon^{\alpha\beta\mu\nu} \partial_\mu A_\nu. \quad (5.77)$$

gives a contravariant antisymmetric tensor of rank 2.

Note that this generalizes the ordinary (Euclidean) curl of a three-vector $A_k(\vec{x})$, which is itself a three-vector with the i -th component given as $(\partial_j A_k - \partial_k A_j)$, i.e.

$$(\nabla \times \vec{A})^i = \frac{1}{2} \epsilon^{ijk} (\partial_j A_k - \partial_k A_j) = \epsilon^{ijk} \partial_j A_k. \quad (5.78)$$

The D'Alembertian Operator: In a (3 + 1)-dimensional space-time, there is another useful operator, known as the *D'Alembertian operator*, defined as

$$\square^2 \equiv \partial_\alpha \partial^\alpha = \partial_0 \partial^0 + \partial_i \partial^i = \frac{1}{c^2} \frac{d^2}{dt^2} - \nabla^2. \quad (5.79)$$

This is the generalized version of the 3-dimensional Laplacian operator ($\nabla^2 = \nabla \cdot \nabla$), and is primarily used in the covariant formulation of wave equations.

5.6.2 The Four-volume and Tensor Integration

The invariant Four-volume element: The volume element in $(3 + 1)$ -dimensions is given by

$$d\mathcal{V} = d^4x \equiv dt d^3x . \quad (5.80)$$

Under a coordinate transformation $x^\mu \longrightarrow x'^\mu = x'^\mu(x^0, x^1, x^2, x^3)$ [cf. Eq. (5.1)], the volume element transforms to

$$d\mathcal{V}' = d^4x' = \left| \frac{\partial x'}{\partial x} \right| d^4x = \left| \frac{\partial x'}{\partial x} \right| d\mathcal{V} , \quad (5.81)$$

i.e. d^4x is actually a scalar density (or, a tensor density of rank 0). Taking now the determinant of both sides of the transformation equation of the metric tensor, viz.

$$g'_{\mu\nu} = \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} g_{\alpha\beta} \quad [\text{cf. Eq. (5.31)}] ,$$

we have

$$g' \equiv \det(g'_{\mu\nu}) = \left| \frac{\partial x}{\partial x'} \right|^2 \det(g_{\alpha\beta}) \equiv \left| \frac{\partial x}{\partial x'} \right|^2 g . \quad (5.82)$$

Since the determinant g of the metric tensor is negative in spaces having the Lorentzian signature $(+, -, -, -)$, therefore in such spaces

$$\sqrt{-g'} = \left| \frac{\partial x}{\partial x'} \right| \sqrt{-g} . \quad (5.83)$$

Substituting this in Eq. (5.81), and recalling that

$$\left| \frac{\partial x'}{\partial x} \right| \left| \frac{\partial x}{\partial x'} \right| = \mathcal{J}(x \rightarrow x') \mathcal{J}(x' \rightarrow x) = 1 \quad [\text{cf. Eq. (5.6)}] ,$$

we find

$$\sqrt{-g'} d\mathcal{V}' = \sqrt{-g} d\mathcal{V} , \quad (5.84)$$

i.e. the *invariant four-volume element* is a scalar (or, a tensor of rank 0) given by

$$\boxed{\sqrt{-g} d\mathcal{V} = \sqrt{-g} d^4x \equiv \sqrt{-g} dt d^3x} . \quad (5.85)$$

In Special Relativity, we have $g_{\alpha\beta}$ identified with the Minkowski metric tensor $\eta_{\alpha\beta}$, whence $g = \det(\eta_{\alpha\beta}) = -1$, and the invariant four-volume element is simply $d^4x = dt d^3x$ — a Lorentz scalar.

Tensor Integration: In principle, integration of tensors over the invariant four-volume yield tensor-like quantities. One defines surface and volume integrals by defining infinitesimal surface and volume elements in a $(3 + 1)$ -dimensional space-time, just as in the ordinary 3-dimensional Euclidean space. One also has the generalizations of the Euclidean-space *Divergence theorem*, *Stokes theorem* etc. in a $(3 + 1)$ -dimensional space-time. However, we skip for brevity discussing them in these lectures.

6 Special Relativistic Particle Dynamics

6.1 Four-velocity

In a covariant formalism one can construct a four vector, whose spatial components are proportional to the components of the three-velocity of a particle and the temporal component is proportional to the speed of light c . The contravariant projections of this *four-velocity* or *world velocity* vector are:

$$u^\alpha = \frac{dx^\alpha}{d\tau} = \frac{dx^\alpha}{dt/\gamma} = \gamma \left(\frac{dx^0}{dt}, \frac{dx^i}{dt} \right) = \gamma (c, \vec{v}). \quad (6.1)$$

Since the proper time τ is invariant under a Lorentz transformation (LT), the four-velocity u^α transforms like the coordinate differentials dx^α under LT. The norm of u^α is a universal constant, which being positive definite implies that u^α is a *time-like* four-vector:

$$u_\alpha u^\alpha = \gamma^2 (c^2 - |\vec{v}|^2) = c^2 > 0. \quad (6.2)$$

6.2 Four-acceleration

In a covariant formalism one can define the *four-acceleration*, or the *world acceleration* of a particle as the rate of change of the four-velocity w.r.to the proper time along the particle's world line. The contravariant projections of the four-acceleration are:

$$a^\alpha = \frac{du^\alpha}{d\tau} = \gamma \frac{du^\alpha}{dt} = \gamma (\dot{\gamma}c, \gamma \dot{\vec{v}} + \dot{\gamma} \vec{v}). \quad (6.3)$$

Now,

$$\dot{\gamma} = \frac{d}{dt} \left(1 - \frac{\vec{v} \cdot \vec{v}}{c^2} \right)^{-1/2} = \gamma^3 \frac{\vec{v} \cdot \dot{\vec{v}}}{c^2}. \quad (6.4)$$

So

$$a^\alpha = \gamma^2 \left(\gamma^2 \frac{\vec{v} \cdot \dot{\vec{v}}}{c}, \dot{\vec{v}} + \gamma^2 \frac{(\vec{v} \cdot \dot{\vec{v}}) \vec{v}}{c^2} \right). \quad (6.5)$$

One may explicitly verify the following:

$$\begin{aligned} (1) \quad a_\alpha a^\alpha &< 0 && [\text{i.e. } a^\alpha \text{ is a } \textit{space-like} \text{ four-vector}] ; \\ (2) \quad a_\alpha u^\alpha &= 0 && [\text{i.e. } u^\alpha \text{ and } a^\alpha \text{ are } \textit{orthogonal} \text{ to each other}]. \end{aligned} \quad (6.6)$$

6.3 Four-momentum and Energy

In a covariant formalism one can define the *four-momentum* of a particle as the product of the particle's rest mass m and its four-velocity. The spatial components of the four-momentum correspond to those for the relativistic three-momentum of the particle, and the temporal component is proportional to the total relativistic energy. The contravariant projections of the four-momentum are:

$$P^\alpha = m u^\alpha = m \frac{dx^\alpha}{d\tau} = \gamma (mc, m\vec{v}) = (\mathcal{M}c, \mathcal{M}\vec{v}) = \left(\frac{\mathcal{E}}{c}, \vec{p} \right), \quad (6.7)$$

where

$$\begin{aligned} \mathcal{M} &\equiv \gamma m : && \text{Relativistic mass,} \\ \vec{p} &= \mathcal{M}\vec{v} = \gamma m\vec{v} : && \text{Relativistic three-momentum,} \\ \mathcal{E} &= \mathcal{M}c^2 = \gamma mc^2 : && \text{Relativistic total (i.e., rest + kinetic) energy.} \end{aligned}$$

The above interpretations of $\vec{p}$ and $\mathcal{E}$ (respectively as the relativistic momentum and the relativistic total energy) are justified as follows:

- Expanding $\vec{p}$ and $\mathcal{E}$ in powers of (v/c) , one finds that in the non-relativistic limit $(v/c \rightarrow 0)$:

$$\begin{aligned} \vec{p} &\approx m\vec{v} : && \text{Non-relativistic three-momentum,} \\ \mathcal{E} &\approx mc^2 + \frac{1}{2}mv^2 : && \text{Rest energy + Non-relativistic kinetic energy.} \end{aligned} \quad (6.8)$$

- $\vec{p}$ and $\mathcal{E}$ can be shown to be the only functions of v whose conservations are Lorentz invariant.
- The conservation of the $\vec{p}$ under a Lorentz transformation (LT) necessarily implies the conservation of $\mathcal{E}$. This is proved in the following subsection.

The norm of the four-momentum P^α is given by

$$P_\alpha P^\alpha = \eta_{\alpha\beta} P^\alpha P^\beta = \eta_{00} (P^0)^2 + \eta_{ij} P^i P^j = \frac{\mathcal{E}^2}{c^2} - |\vec{p}|^2. \quad (6.9)$$

But, since

$$P_\alpha P^\alpha = m^2 u_\alpha u^\alpha = m^2 \gamma^2 (c^2 - v^2) = m^2 c^2, \quad (6.10)$$

we have

$$\boxed{\mathcal{E}^2 = |\vec{p}|^2 c^2 + m^2 c^4}. \quad (6.11)$$

Therefore, for massless particles the total energy is simply $\mathcal{E} = |\vec{p}| c$.

6.4 Conservation of Energy and Momentum

Since the four-momentum $P^\alpha = m dx^\alpha/d\tau$ transforms as a four vector under a LT (due to the fact that m and τ are invariants and dx^α is a four vector), in any reaction the change of the sum of the four-momenta P^α of all the involved particles (say n in number), also transforms as a four vector:

$$P'^\alpha = \Lambda^\alpha_\beta P^\beta \implies \Delta \sum_n P'^\alpha_{(n)} = \Lambda^\alpha_\beta \Delta \sum_n P^\beta_{(n)}. \quad (6.12)$$

Now, the conservation of the relativistic three-momentum $\vec{p}$ under a LT means that:

$$\Delta \sum_n \vec{p}_{(n)} = 0 \quad \text{in one inertial frame} \implies \Delta \sum_n \vec{p}'_{(n)} = 0 \quad \text{in all other inertial frames.}$$

i.e.,

$$\Delta \sum_n P^i_{(n)} = 0 \implies \Delta \sum_n P'^i_{(n)} = 0 = \Lambda^i_\beta \Delta \sum_n P^\beta_{(n)} = \Lambda^i_0 \Delta \sum_n P^0_{(n)} + \Lambda^i_j \Delta \sum_n P^j_{(n)}. \quad (6.13)$$

Since Λ^i_j are constants ($\forall i, j = 1, 2, 3$), $\Delta \sum_n P^i_{(n)} = 0$ implies that the last term on the r.h.s. of the above equation is automatically zero. So we are left with

$$\Lambda^i_0 \Delta \sum_n P^0_{(n)} = 0 \implies \boxed{\Delta \sum_n P^0_{(n)} = 0 = \frac{1}{c} \Delta \sum_n \mathcal{E}_{(n)}}, \quad (6.14)$$

as Λ^i_0 ($\forall i = 1, 2, 3$) are not necessarily zero.

Thus, the conservation of the relativistic three-momentum $\vec{p}$, implies conservation of the total relativistic energy $\mathcal{E}$, and both in turn imply conservation of the four-momentum P^α . In other words, the conditions

$$\begin{aligned} (i) \Delta \sum_n \vec{p}_{(n)} = 0 \quad \text{i.e.} \quad \sum_n \vec{p}_{(n)I} = \sum_n \vec{p}_{(n)F}, \quad (ii) \Delta \sum_n \mathcal{E}_{(n)} = 0 \quad \text{i.e.} \quad \sum_n \mathcal{E}_{(n)I} = \sum_n \mathcal{E}_{(n)F}, \\ \text{imply} \quad \Delta \sum_n P^\alpha_{(n)} = 0 \quad \text{i.e.} \quad \sum_n P^\alpha_{(n)I} = \sum_n P^\alpha_{(n)F} \end{aligned} \quad (6.15)$$

where the subscript I (F) denotes initial (final) configuration.

The conservation of the total relativistic energy, by definition, implies the conservation of the total relativistic mass:

$$\Delta \sum_n \mathcal{M}_{(n)} = 0 \quad \text{i.e.} \quad \sum_n \mathcal{M}_{(n)I} = \sum_n \mathcal{M}_{(n)F} \quad [\mathcal{M} = \gamma m] . \quad (6.16)$$

Illustrative Example: Compton effect

A photon hits a stationary particle of rest mass m as shown in Fig. 13. We need to determine the angle of deflection ϑ of the photon in terms of its initial and final energies ($\mathcal{E}_I$ and $\mathcal{E}_F$ respectively), or its initial and final frequencies (ν_I and ν_F respectively).

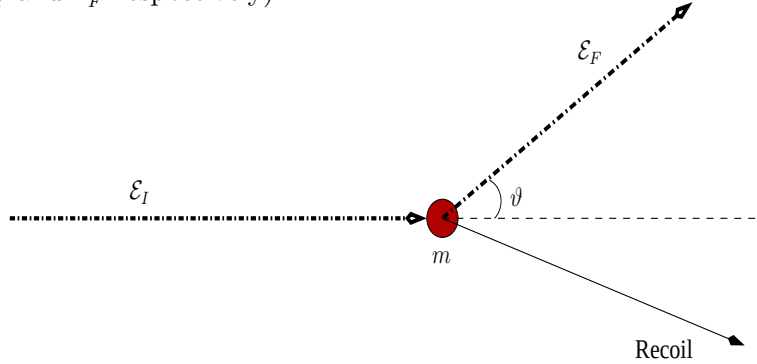


Figure 13: Illustration of the Compton effect.

Initial and final four-momenta ($P_{(I)}^\alpha$ and $P_{(F)}^\alpha$ respectively) of the photon are given by

$$P_{(I)}^\alpha = \left(\frac{\mathcal{E}_I}{c}, \vec{p}_I \right), \quad P_{(F)}^\alpha = \left(\frac{\mathcal{E}_F}{c}, \vec{p}_F \right), \quad (6.17)$$

where $\vec{p}_I$ and $\vec{p}_F$ are the initial and the final (relativistic) three-momenta of the photon. As, for the photon, $\mathcal{E}_I = c |\vec{p}_I|$ and $\mathcal{E}_F = c |\vec{p}_F|$, we have

$$P_{(I)}^\alpha = (|\vec{p}_I|, \vec{p}_I), \quad P_{(F)}^\alpha = (|\vec{p}_F|, \vec{p}_F). \quad (6.18)$$

For the massive particle, since its initial energy is equal to its rest energy mc^2 , its initial four-momentum ($\tilde{P}_{(I)}^\alpha$ say) is given as

$$\tilde{P}_{(I)}^\alpha = (mc, \vec{0}). \quad (6.19)$$

Let its final four-momentum be $\tilde{P}_{(F)}^\alpha$. Then since the norm of the four-momentum is Lorentz invariant, we have

$$\tilde{P}_{\alpha(F)} \tilde{P}_{(F)}^\alpha = \tilde{P}_{\alpha(I)} \tilde{P}_{(I)}^\alpha = m^2 c^2. \quad (6.20)$$

Now, by the conservation of four-momenta:

$$P_{(I)}^\alpha + \tilde{P}_{(I)}^\alpha = P_{(F)}^\alpha + \tilde{P}_{(F)}^\alpha \quad \implies \quad \tilde{P}_{(F)}^\alpha = (mc + |\vec{p}_I| - |\vec{p}_F|, \vec{p}_I - \vec{p}_F), \quad (6.21)$$

using Eqs. (6.18) and (6.19).

Again, by the Eq. (6.20)

$$\begin{aligned} \tilde{P}_{\alpha(F)} \tilde{P}_{(F)}^\alpha &= m^2 c^2 = (mc + |\vec{p}_I| - |\vec{p}_F|)^2 + |\vec{p}_I - \vec{p}_F|^2 \\ \implies 2mc(|\vec{p}_I| - |\vec{p}_F|) + (|\vec{p}_I| - |\vec{p}_F|)^2 &= |\vec{p}_I|^2 + |\vec{p}_F|^2 + 2 \vec{p}_I \cdot \vec{p}_F \\ \implies \frac{1}{|\vec{p}_I|} - \frac{1}{|\vec{p}_F|} &= \frac{2}{mc} \sin^2 \frac{\vartheta}{2}, \end{aligned} \quad (6.22)$$

where ϑ is the angle between $\vec{p}_I$ and $\vec{p}_F$.

Since, $\mathcal{E}_I = c |\vec{p}_I| = h \nu_I$ and $\mathcal{E}_F = c |\vec{p}_F| = h \nu_F$, h being the Planck's constant, we finally obtain

$$\boxed{\vartheta = 2 \sin^{-1} \left[\left\{ \frac{mc^2}{2h} \left(\frac{1}{\nu_F} - \frac{1}{\nu_I} \right) \right\}^{1/2} \right]}. \quad (6.23)$$

6.5 Four-force

In a covariant formalism, the *four-force* on a particle as the rate of change of the four-momentum w.r.to the proper time along the particle's world line. The contravariant projections of the four-force are:

$$\boxed{\mathcal{F}^\alpha = \frac{dP^\alpha}{d\tau} = \gamma \frac{dP^\alpha}{dt} = \gamma \left(\frac{\dot{\mathcal{E}}}{c}, \vec{\mathbf{F}} \right)}, \quad (6.24)$$

where

$$\vec{\mathbf{F}} = \dot{\vec{\mathbf{p}}} \equiv \frac{d\vec{\mathbf{p}}}{dt} \quad : \quad \text{Relativistic three-force.} \quad (6.25)$$

By defintion

$$\boxed{\mathcal{F}^\alpha = \mathcal{M}a^\alpha = \gamma m a^\alpha = \gamma m \frac{du^\alpha}{dt}}, \quad (6.26)$$

which can be regarded as a relativistic generalization to the Newton's second law of motion in classical mechanics.

Now, for a particle moving with instantaneous velocity $\vec{\mathbf{v}}$, the time derivative of the total energy is equal to the total power dissipated:

$$\dot{\mathcal{E}} = \vec{\mathbf{F}} \cdot \vec{\mathbf{v}} = \dot{\vec{\mathbf{p}}} \cdot \vec{\mathbf{v}}. \quad (6.27)$$

Therefore,

$$\boxed{\mathcal{F}^\alpha = \gamma \left(\frac{\dot{\mathcal{E}}}{c}, \vec{\mathbf{F}} \right) = \gamma \left(\frac{\vec{\mathbf{F}} \cdot \vec{\mathbf{v}}}{c}, \vec{\mathbf{F}} \right)}. \quad (6.28)$$

Moreover, since the components of the four-velocity are given by: $u^0 = \gamma c$ and $\vec{\mathbf{u}} = \gamma \vec{\mathbf{v}}$, we have

$$\boxed{\mathcal{F}^\alpha = \frac{1}{c} \left(\vec{\mathbf{F}} \cdot \vec{\mathbf{u}}, u^0 \vec{\mathbf{F}} \right)}. \quad (6.29)$$

7 Covariant Formulation of Electrodynamic Equations

7.1 Four-current Density

Consider a system of n charged particles, with fixed charges $q_{(n)}$ at locations $\vec{x}_{(n)}(t)$. The ordinary three-charge density and three-current density are defined respectively by

$$\begin{aligned} \rho(\vec{x}, t) &= \sum_n q_{(n)} \delta^3(\vec{x} - \vec{x}_{(n)}(t)) \\ \vec{j}(\vec{x}, t) &= \sum_n q_{(n)} \delta^3(\vec{x} - \vec{x}_{(n)}(t)) \frac{d\vec{x}_{(n)}(t)}{dt} \end{aligned} \quad (7.1)$$

where $\delta^3(\vec{x} - \vec{x}_{(n)})$ is the three-dimensional Dirac delta function. In analogy with the three-current density $\vec{j}(\vec{x}, t)$ one may define the *four-current density* $J^\alpha(\vec{x}, t)$ which encodes both ρ and $\vec{j}$ as:

$$J^\alpha(\vec{x}, t) = \sum_n q_{(n)} \delta^3(\vec{x} - \vec{x}_{(n)}(t)) \frac{dx_{(n)}^\alpha(t)}{dt}. \quad (7.2)$$

To verify that this is a four vector, we set $x_{(n)}^0(t) = ct$ ($\forall n$), and use the property of the delta function:

$$\int dx f(x) \delta(x - y) = f(y), \quad (7.3)$$

to write $J^\alpha(\vec{x}, t)$ as

$$\begin{aligned} J^\alpha(\vec{x}, t) &= \int dx_{(n)}^0 \sum_n q_{(n)} \delta(x^0 - x_{(n)}^0) \delta^3(\vec{x} - \vec{x}_{(n)}(t')) \frac{dx_{(n)}^\alpha(t')}{dx_{(n)}^0/c} \\ &= c \int cdt' \sum_n q_{(n)} \delta^4(x^\mu - x_{(n)}^\mu(t')) \frac{dx_{(n)}^\alpha(t')}{cdt'} \\ \Rightarrow J^\alpha(x^i, t) &= c \int d\tau \sum_n q_{(n)} \delta^4(x^\mu - x_{(n)}^\mu(\tau)) \frac{dx_{(n)}^\alpha(\tau)}{d\tau}. \end{aligned} \quad (7.4)$$

As the four-dimensional delta function $\delta^4(x^\mu - x_{(n)}^\mu(\tau))$ and the proper time τ are Lorentz scalars, whereas $dx_{(n)}^\alpha$ is a four vector, J^α is a four vector.

One may easily check from Eq. (7.2) that the temporal and spatial components of J^α are proportional to ρ and $\vec{j}$, given by Eqs. (7.1), respectively:

$$J^\alpha(x^i, t) = (c\rho(x^i, t), \vec{j}(x^i, t)). \quad (7.5)$$

It is also easy to verify, by using the continuity equation, that the four-current density is covariantly conserved, i.e., its four divergence is zero:

$$\partial_\alpha J^\alpha = \frac{\partial \rho}{\partial t} + \nabla \cdot \vec{j} = 0. \quad (7.6)$$

7.2 Covariant Lorentz Force Equation

In three dimensions, the electromagnetic force on a moving particle, with charge q and instantaneous velocity $\vec{v}$, is given by the well-known three-dimensional *Lorentz force* equation [cf. Eq. (1.22)]:

$$\vec{F}_L = q \left(\vec{E} + \frac{1}{c} \vec{v} \times \vec{B} \right). \quad (7.7)$$

In a covariant formalism, the four-force being given by Eq. (6.28) [or (6.29)], Eq. (7.7) generalizes to:

$$\boxed{\mathcal{F}_L^\alpha = \left(\frac{\gamma \vec{\mathbf{F}}_L \cdot \vec{\mathbf{v}}}{c}, \gamma \vec{\mathbf{F}}_L \right) = \frac{q}{c} \left(\vec{\mathbf{E}} \cdot \gamma \vec{\mathbf{v}}, \left[\gamma c \vec{\mathbf{E}} + \gamma \vec{\mathbf{v}} \times \vec{\mathbf{B}} \right] \right) = \frac{q}{c} \left(\vec{\mathbf{E}} \cdot \vec{\mathbf{u}}, \left[u^0 \vec{\mathbf{E}} + \vec{\mathbf{u}} \times \vec{\mathbf{B}} \right] \right)}. \quad (7.8)$$

This is often referred to as the *covariant Lorentz force equation*, which in compact form can be expressed as (q/c) times the contracted product of a second rank antisymmetric tensor $F^{\alpha\beta}$, constructed out of the electric and the magnetic field three-vectors $\vec{\mathbf{E}}$ and $\vec{\mathbf{B}}$, and the four-velocity vector u^β :

$$\boxed{\mathcal{F}_L^\alpha = \frac{q}{c} F^{\alpha\beta} u_\beta}. \quad (7.9)$$

Explicitly, the components of $F^{\alpha\beta}$, which is known as the *electromagnetic field tensor*, are given by

$$\boxed{F^{\alpha\beta} : \quad F^{00} = 0, \quad F^{0i} = -F^{i0} = -E^i, \quad F^{ij} = -F^{ji} = \epsilon^{ijk} B_k} \quad (7.10)$$

$$\Rightarrow F^{\alpha\beta} = -F^{\beta\alpha} = \begin{pmatrix} 0 & -E^1 & -E^2 & -E^3 \\ E^1 & 0 & -B^3 & B^2 \\ E^2 & B^3 & 0 & -B^1 \\ E^3 & -B^2 & B^1 & 0 \end{pmatrix} \equiv \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & -B_z & B_y \\ E_y & B_z & 0 & -B_x \\ E_z & -B_y & B_x & 0 \end{pmatrix}. \quad (7.11)$$

Let us now prove that $F^{\alpha\beta}$ is indeed a tensor of rank 2, i.e., Eq. (7.9) is a tensorial equation that makes the covariance of the Lorentz force explicit. Suppose that in a Lorentz frame Σ , the Eq. (7.9) holds. In another Lorentz frame Σ' , we have the transformed equation

$$\mathcal{F}_L'^\alpha = \frac{q}{c} F'^{\alpha\beta} u'_\beta. \quad (7.12)$$

Now,

$$\mathcal{F}_L'^\alpha = \frac{dP'^\alpha}{d\tau} = \frac{\partial x'^\alpha}{\partial x^\mu} \frac{dP^\mu}{d\tau} = \frac{\partial x'^\alpha}{\partial x^\mu} \mathcal{F}_L^\mu. \quad (7.13)$$

Also

$$u'_\beta = \frac{\partial x^\nu}{\partial x'^\beta} u_\nu \quad \text{or} \quad u_\nu = \frac{\partial x'^\beta}{\partial x^\nu} u'_\beta. \quad (7.14)$$

Substituting this in Eq. (7.13), and using Eqs. (7.9) and (7.12), we have

$$\begin{aligned} \mathcal{F}_L'^\alpha &= \frac{q}{c} F'^{\alpha\beta} u'_\beta = \frac{\partial x'^\alpha}{\partial x^\mu} \mathcal{F}_L^\mu = \frac{\partial x'^\alpha}{\partial x^\mu} \left(\frac{q}{c} F^{\mu\nu} u_\nu \right) = \frac{\partial x'^\alpha}{\partial x^\mu} \left(\frac{q}{c} F^{\mu\nu} \frac{\partial x'^\beta}{\partial x^\nu} u'_\beta \right) \\ \Rightarrow \left(F'^{\alpha\beta} - \frac{\partial x'^\alpha}{\partial x^\mu} \frac{\partial x'^\beta}{\partial x^\nu} F^{\mu\nu} \right) u'_\beta &= 0 \quad \Rightarrow \quad F'^{\alpha\beta} = \frac{\partial x'^\alpha}{\partial x^\mu} \frac{\partial x'^\beta}{\partial x^\nu} F^{\mu\nu}, \end{aligned} \quad (7.15)$$

as u'_β is arbitrary. Thus $F^{\alpha\beta}$ transforms like a rank-2 tensor.

7.3 Covariance of Maxwell's Equations

In a homogeneous, isotropic and non-dispersive medium, the Maxwell's equations for the electric and magnetic fields $[\vec{\mathbf{E}}(\vec{\mathbf{x}}, t), \vec{\mathbf{B}}(\vec{\mathbf{x}}, t)]$, due to the charge and current densities $\rho(\vec{\mathbf{x}}, t)$ and $\vec{\mathbf{j}}(\vec{\mathbf{x}}, t)$, are [cf. Eq. (1.28)]:

$$\boxed{\begin{aligned} \text{(i)} \quad \nabla \cdot \vec{\mathbf{E}} &= 4\pi\rho, & \text{(ii)} \quad \nabla \times \vec{\mathbf{B}} - \frac{\dot{\vec{\mathbf{E}}}}{c} &= \frac{4\pi}{c} \vec{\mathbf{j}}, \\ \text{(iii)} \quad \nabla \cdot \vec{\mathbf{B}} &= 0, & \text{(iv)} \quad \nabla \times \vec{\mathbf{E}} + \frac{\dot{\vec{\mathbf{B}}}}{c} &= 0 \end{aligned}}, \quad (7.16)$$

where the overhead dot $\{\dot{\cdot}\}$ denotes $\partial/\partial t = c \partial/\partial_0$.

In the indexed notation, the Maxwell's equations are given by

$$\boxed{\begin{array}{ll} \text{(i)} & \partial_i E^i = \frac{4\pi}{c} J^0, \quad \text{(ii)} \quad \epsilon^{ijk} \partial_j B_k + \partial_0 E^i = -\frac{4\pi}{c} J^i, \\ \text{(iii)} & \partial_i B^i = 0, \quad \text{(iv)} \quad \epsilon^{ijk} \partial_j E_k - \partial_0 B^i = 0 \end{array}}. \quad (7.17)$$

The first two of Eqs. (7.16) [or (7.17)], that involve the sources $J^0 = c\rho$ and $J^k = (\vec{j})^k$, $k = 1, 2, 3$, can be expressed in terms of the components of the electromagnetic field tensor $F^{\alpha\beta}$ [Eq. (7.10)] as

$$\text{(i)} \quad \partial_i F^{i0} = \frac{4\pi}{c} J^0, \quad \text{(ii)} \quad \partial_i F^{ik} + \partial_0 F^{0k} = \frac{4\pi}{c} J^k. \quad (7.18)$$

Similarly, the last two of Eqs. (7.16) [or (7.17)], that do not involve the sources, can be expressed in terms of the components of another second rank antisymmetric tensor $\tilde{F}^{\alpha\beta}$ (which like the electromagnetic field tensor $F^{\alpha\beta}$, is also constructed out of the components of $\vec{E}$ and $\vec{B}$), as

$$\text{(i)} \quad \partial_i \tilde{F}^{i0} = 0, \quad \text{(ii)} \quad \partial_i \tilde{F}^{ik} + \partial_0 \tilde{F}^{0k} = 0. \quad (7.19)$$

The components of $\tilde{F}^{\alpha\beta}$ are given explicitly by

$$\boxed{\tilde{F}^{\alpha\beta} : \quad \tilde{F}^{00} = 0, \quad \tilde{F}^{0i} = -\tilde{F}^{i0} = -B^i, \quad \tilde{F}^{ij} = -\tilde{F}^{ji} = -\epsilon^{ijk} E_k} \quad (7.20)$$

$$\Rightarrow \quad \tilde{F}^{\alpha\beta} = -\tilde{F}^{\beta\alpha} = \begin{pmatrix} 0 & -B^1 & -B^2 & -B^3 \\ B^1 & 0 & E^3 & -E^2 \\ B^2 & -E^3 & 0 & E^1 \\ B^3 & E^2 & -E^1 & 0 \end{pmatrix} \equiv \begin{pmatrix} 0 & -B_x & -B_y & -B_z \\ B_x & 0 & E_z & -E_y \\ B_y & -E_z & 0 & E_x \\ B_z & E_y & -E_x & 0 \end{pmatrix}. \quad (7.21)$$

The covariant tensors $F_{\alpha\beta}$ and $\tilde{F}_{\alpha\beta}$ can be obtained respectively from $F^{\alpha\beta}$ and $\tilde{F}^{\alpha\beta}$ as:

$$F_{\alpha\beta} = -F_{\beta\alpha} = \eta_{\alpha\mu} \eta_{\beta\nu} F^{\mu\nu} : \quad F^{\alpha\beta} \xrightarrow{\vec{E} \rightarrow -\vec{E}, \vec{B} \rightarrow \vec{B}} F_{\alpha\beta}, \quad (7.22)$$

$$\tilde{F}_{\alpha\beta} = -\tilde{F}_{\beta\alpha} = \eta_{\alpha\mu} \eta_{\beta\nu} \tilde{F}^{\mu\nu} : \quad \tilde{F}^{\alpha\beta} \xrightarrow{\vec{E} \rightarrow \vec{E}, \vec{B} \rightarrow -\vec{B}} \tilde{F}_{\alpha\beta}. \quad (7.23)$$

Using these, and the explicit forms of $F^{\alpha\beta}$ and $\tilde{F}^{\alpha\beta}$, given by Eqs. (7.11) and (7.21), one may verify that $\tilde{F}^{\alpha\beta}$ is indeed identical with the dual of $F^{\alpha\beta}$:

$$\tilde{F}^{\alpha\beta} \equiv {}^*F^{\alpha\beta} = \frac{1}{2} \epsilon^{\alpha\beta\mu\nu} F_{\mu\nu} \quad \text{or} \quad F^{\alpha\beta} = 2 \epsilon^{\alpha\beta\mu\nu} \tilde{F}_{\mu\nu} = 2 \epsilon^{\alpha\beta\mu\nu} {}^*F_{\mu\nu}. \quad (7.24)$$

Moreover, the above two Maxwell's equations involving the sources, viz., Eqs. (7.18 i, ii), whose l.h.s. contain the three-divergence or(and) time-derivative of the components of $F^{\alpha\beta}$, can be encoded in a single tensorial equation whose l.h.s. is the four-divergence of $F^{\alpha\beta}$. Similarly, the two source-free Maxwell's equations, viz., Eqs. (7.19 i, ii), can be encoded in a single tensorial equation whose l.h.s. is the four-divergence of ${}^*F^{\alpha\beta}$. Thus, in a covariant formalism the Maxwell's equations are expressed as:

$$\boxed{\begin{array}{l} \text{(i)} \quad \partial_\alpha F^{\alpha\beta} = \frac{4\pi}{c} J^\beta \\ \text{(ii)} \quad \partial_\alpha {}^*F^{\alpha\beta} = 0 \end{array}}, \quad (7.25)$$

where $J^\beta = (c\rho, \vec{j})$ is the four current density. Eq. (7.25 ii) is in fact an identity, known as the *Maxwell-Bianchi identity*, which can also be expressed as a *cyclic identity*:

$$\partial_\alpha {}^*F^{\alpha\beta} = \frac{1}{2} \epsilon^{\alpha\beta\mu\nu} \partial_\alpha F_{\mu\nu} = 0 \quad \Rightarrow \quad \boxed{\partial_\alpha F_{\mu\nu} + \partial_\mu F_{\nu\alpha} + \partial_\nu F_{\alpha\mu} = 0}. \quad (7.26)$$

In absence of sources ($\rho = 0, \vec{j} = 0$), the above two Maxwell's equations (7.25) in covariant form, are indistinguishable from each other under the transformation:

$$F^{\alpha\beta} \rightarrow {}^*F^{\alpha\beta} \quad \text{i.e.} \quad \vec{E} \rightarrow \vec{B}, \quad \vec{B} \rightarrow -\vec{E}. \quad (7.27)$$

Such a transformation is therefore called the *electromagnetic duality transformation*.

8 Electrodynamic Transformations Laws and Scalar Invariants

8.1 Transformation of Electromagnetic Fields

In Special Relativity, $F^{\alpha\beta}$ and $\star F^{\alpha\beta}$ being rank-2 tensors, follow the transformation rules

$$F'^{\alpha\beta} = \Lambda^\alpha_\mu \Lambda^\beta_\nu F^{\mu\nu}, \quad \star F'^{\alpha\beta} = \Lambda^\alpha_\mu \Lambda^\beta_\nu \star F^{\mu\nu}, \quad (8.1)$$

where Λ^α_μ are the Lorentz transformation matrix elements, which for a generic Lorentz boost, in an arbitrary direction, are given by (see §3.5 and the Eqs. (A.6) and (A.8) in the Appendix A):

$$\boxed{\begin{aligned} \Lambda^0_0 &= \gamma, & \Lambda^i_0 &= -\gamma \beta^i, & \Lambda^0_i &= -\gamma \delta_{ik} \beta^k = \gamma \beta_i, \\ \Lambda^i_j &= \delta^i_j + \frac{(\gamma-1)}{|\vec{\beta}|^2} \delta_{jk} \beta^i \beta^k = \delta^i_j - \frac{(\gamma-1)}{|\vec{\beta}|^2} \beta^i \beta_j \end{aligned}}. \quad (8.2)$$

Using these,

$$\begin{aligned} F'^{i0} = E^i &= \Lambda^i_\mu \Lambda^0_\nu F^{\mu\nu} = \Lambda^i_0 \Lambda^0_k F^{0k} + \Lambda^i_j \Lambda^0_0 F^{j0} + \Lambda^i_j \Lambda^0_k F^{jk} \\ &= \gamma (\vec{E} + \vec{\beta} \times \vec{B})^i - \gamma \left(\gamma - \frac{\gamma-1}{|\vec{\beta}|^2} \right) (\vec{\beta} \cdot \vec{E}) \beta^i, \end{aligned} \quad (8.3)$$

$$\begin{aligned} \star F'^{i0} = B^i &= \Lambda^i_\mu \Lambda^0_\nu \star F^{\mu\nu} = \Lambda^i_0 \Lambda^0_k \star F^{0k} + \Lambda^i_j \Lambda^0_0 \star F^{j0} + \Lambda^i_j \Lambda^0_k \star F^{jk} \\ &= \gamma (\vec{B} - \vec{\beta} \times \vec{E})^i - \gamma \left(\gamma - \frac{\gamma-1}{|\vec{\beta}|^2} \right) (\vec{\beta} \cdot \vec{B}) \beta^i. \end{aligned} \quad (8.4)$$

As

$$\gamma^2 = \frac{1}{1 - |\vec{\beta}|^2} \implies |\vec{\beta}|^2 = \frac{\gamma^2 - 1}{\gamma^2}, \quad (8.5)$$

we have

$$\gamma - \frac{\gamma-1}{|\vec{\beta}|^2} = \frac{\gamma}{\gamma+1}. \quad (8.6)$$

Hence, the expressions (8.3) for the transformed electric and magnetic fields reduce to

$$\boxed{\begin{aligned} \vec{E}' &= \gamma (\vec{E} + \vec{\beta} \times \vec{B}) - \frac{\gamma^2}{\gamma+1} (\vec{\beta} \cdot \vec{E}) \vec{\beta} \\ \vec{B}' &= \gamma (\vec{B} - \vec{\beta} \times \vec{E}) - \frac{\gamma^2}{\gamma+1} (\vec{\beta} \cdot \vec{B}) \vec{\beta} \end{aligned}}. \quad (8.7)$$

For a single Lorentz boost along, say, the $x^1 (\equiv x)$ axis, i.e., $\vec{\beta} = \beta \hat{x}$, Eqs. (8.7) give

$$\boxed{\begin{aligned} E'_x &= E_x & B'_x &= B_x \\ E'_y &= \gamma (E_y - \beta B_z) & B'_y &= \gamma (B_y + \beta E_z) \\ E'_z &= \gamma (E_z + \beta B_y) & B'_z &= \gamma (B_z - \beta E_y) \end{aligned}}. \quad (8.8)$$

So, a purely electric or a purely magnetic field in one reference frame, is in general a combination of electric and magnetic fields in another.

8.2 Electromagnetic Scalar invariants

In principle, one can construct three Lorentz scalars using the electromagnetic field tensor or (and) its dual. However, only two of these are actually linearly independent:

$$\boxed{\text{(i) } F_{\alpha\beta} F^{\alpha\beta} = - {}^*F_{\alpha\beta} {}^*F^{\alpha\beta} = 2 \left(|\vec{\mathbf{B}}|^2 - |\vec{\mathbf{E}}|^2 \right) \quad , \quad \text{(ii) } F_{\alpha\beta} {}^*F^{\alpha\beta} = {}^*F^{\alpha\beta} F_{\alpha\beta} = -4 \vec{\mathbf{E}} \cdot \vec{\mathbf{B}} .} \quad (8.9)$$

The consequences of these scalar invariants are enlisted as follows:

- If $|\vec{\mathbf{E}}| = |\vec{\mathbf{B}}|$ in one Lorentz frame Σ , then $|\vec{\mathbf{E}}'| = |\vec{\mathbf{B}}'|$ in every other Lorentz frame Σ' .
- If $\vec{\mathbf{E}} \perp \vec{\mathbf{B}}$ in one Lorentz frame Σ , then $\vec{\mathbf{E}}' \perp \vec{\mathbf{B}}'$ in every other Lorentz frame Σ' , unless $\vec{\mathbf{E}}'$ or $\vec{\mathbf{B}}'$ vanishes identically.
- If $\vec{\mathbf{E}} \parallel \vec{\mathbf{B}}$ in one Lorentz frame Σ , then the condition for $\vec{\mathbf{E}}'$ to be parallel to $\vec{\mathbf{B}}'$ in another Lorentz frame Σ' is that $|\vec{\mathbf{E}}||\vec{\mathbf{B}}| = |\vec{\mathbf{E}}'||\vec{\mathbf{B}}'|$. This follows from the fact that if ϑ is the angle between $\vec{\mathbf{E}}$ and $\vec{\mathbf{B}}$ in Σ , then the corresponding angle ϑ' between $\vec{\mathbf{E}}'$ and $\vec{\mathbf{B}}'$ in Σ' is given by

$$\vartheta' = \cos^{-1} \left[\frac{|\vec{\mathbf{E}}||\vec{\mathbf{B}}|}{|\vec{\mathbf{E}}'||\vec{\mathbf{B}}'|} \cos \vartheta \right] . \quad (8.10)$$

Therefore, when $\vartheta = 0$, the condition for $\vartheta' = 0$ is $|\vec{\mathbf{E}}||\vec{\mathbf{B}}| = |\vec{\mathbf{E}}'||\vec{\mathbf{B}}'|$.

9 Potential formulation of Electrodynamics and Gauge Conditions

9.1 Electromagnetic Four-potential

As pointed out in §7.3, the Maxwell-Bianchi identity, viz. Eq. (7.25 ii), is equivalent to vanishing of the sum of the cyclic permutations of $\partial_\mu F_{\nu\lambda}$:

$$\partial_\mu {}^*F^{\mu\nu} = 0 \quad \implies \quad \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} + \partial_\lambda F_{\mu\nu} = 0 \quad [\text{cf. Eq. (7.26)}] .$$

This has a unique solution:

$$\boxed{F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu} , \quad (9.1)$$

where $A^\mu := (\phi, \vec{A})$ is referred to as the *electromagnetic four-vector potential* (or simply the *four-potential*). Its temporal part is called the *electromagnetic scalar potential* ϕ , whereas the spatial part is the *electromagnetic three-vector potential* $\vec{A}$. The covariant vector A_μ has components $(\phi, -\vec{A})$.

Verification:

$$\begin{aligned} \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} + \partial_\lambda F_{\mu\nu} \\ = \partial_\mu \partial_\nu A_\lambda - \partial_\mu \partial_\lambda A_\nu + \partial_\nu \partial_\lambda A_\mu - \partial_\nu \partial_\mu A_\lambda + \partial_\lambda \partial_\mu A_\nu - \partial_\lambda \partial_\nu A_\mu = 0 . \end{aligned}$$

In the three-vector notation, the relations between the electric and magnetic fields ($\vec{E}$ and $\vec{B}$) and the scalar and three-vector potentials (ϕ and $\vec{A}$), can be obtained from the above equation (9.1) as follows:

$$F_{0i} = -E_i = \partial_0 A_i - \partial_i A_0 \quad \implies \quad \boxed{\vec{E} = -\nabla\phi - \frac{\dot{\vec{A}}}{c}} , \quad (9.2)$$

$${}^*F^{0i} = B^i = \frac{1}{2}\epsilon^{0ijk}F_{jk} = \frac{1}{2}\epsilon^{0ijk}(\partial_j A_k - \partial_k A_j) = \epsilon^{0ijk}\partial_j A_k \quad \implies \quad \boxed{\vec{B} = \nabla \times \vec{A}} . \quad (9.3)$$

Now, substituting Eq. (9.1) in the Maxwell's Eq. (7.25 i) [see §7.3], one obtains the evolution equation of the four-potential:

$$\begin{aligned} \partial_\alpha F^{\alpha\beta} &= \partial_\alpha (\partial^\alpha A^\beta - \partial^\beta A^\alpha) = \frac{4\pi}{c} J^\beta \\ \implies \partial_\alpha \partial^\alpha A^\beta - \partial_\alpha \partial^\beta A^\alpha &= \frac{4\pi}{c} J^\beta \\ \implies \boxed{\square^2 A^\beta - \partial^\beta (\partial_\alpha A^\alpha) = \frac{4\pi}{c} J^\beta} . \end{aligned} \quad (9.4)$$

9.2 Gauge Transformations

Under the transformation:

$$\boxed{A^\alpha \longrightarrow A'^\alpha = A^\alpha + \partial^\alpha \chi} , \quad (9.5)$$

where $\chi(\vec{x}, t)$ is an arbitrary scalar function of $\vec{x}$ and t , the r.h.s. of Eq. (9.4) is unaffected, whereas the l.h.s. transforms to

$$\begin{aligned} \square^2 A'^\beta - \partial^\beta \partial_\alpha A'^\alpha &= \square^2 A^\beta + \square^2 \partial^\beta \chi - \partial^\beta \partial_\alpha (A^\alpha + \partial^\alpha \chi) \\ &= \square^2 A^\beta + \partial^\beta \square^2 \chi - \partial^\beta \partial_\alpha A^\alpha - \partial^\beta \square^2 \chi = \square^2 A^\beta - \partial^\beta \partial_\alpha A^\alpha , \end{aligned}$$

i.e., the l.h.s., and hence the entire equation (9.4), is invariant. This implies that there is a freedom to choose the *gauge* (i.e., to impose a condition on the hitherto arbitrary scalar function χ) while solving the Eq. (9.4) for A^α , so as to determine the measurable quantities, viz. the fields $\vec{E}$ and $\vec{B}$, using the Eqs. (9.2) and (9.3). As such, the above transformation (9.5) is referred to as the *Gauge transformation*, the potential A^α as the *Gauge potential*, the fields $\vec{E}$ and $\vec{B}$ as the *Gauge fields*, the function χ as the *Gauge function*, and the condition on χ (required to be imposed) as the *Gauge condition*.

9.3 Gauge conditions: Lorentz and Coulomb gauges

In the following, we will discuss two popular gauge conditions which are of most importance in describing physical phenomena associated with the gauge fields.

9.3.1 Lorentz gauge condition

If we choose χ to be such that:

$$\square^2 \chi \equiv \partial_\alpha \partial^\alpha \chi = - \partial_\beta A^\beta, \quad (9.6)$$

then

$$\partial_\alpha A^\alpha \xrightarrow{A^\alpha \rightarrow A^\alpha + \partial^\alpha \chi} \partial_\alpha A^\alpha + \square^2 \chi = 0. \quad (9.7)$$

The above condition (9.6) is therefore equivalent to

$$\boxed{\partial_\alpha A^\alpha = 0} \quad \text{i.e.,} \quad \boxed{\nabla \cdot \vec{A} + \frac{\dot{\phi}}{c} = 0}. \quad (9.8)$$

This is known as the *Lorentz gauge condition*[†]. In this gauge, the above equation (9.4) reduces to

$$\boxed{\square^2 A^\alpha = \frac{4\pi}{c} J^\alpha}. \quad (9.9)$$

Recalling that $J^\alpha := (c\rho, \vec{j})$, we have the temporal and the spatial components of Eq. (9.9) given as

$$\boxed{\begin{aligned} (a) \quad \square^2 \phi &\equiv \left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 \right) \phi = 4\pi \rho \\ (b) \quad \square^2 \vec{A} &\equiv \left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 \right) \vec{A} = \frac{4\pi}{c} \vec{j} \end{aligned}}. \quad (9.10)$$

These are the *inhomogeneous wave equations* satisfied by the scalar potential ϕ and the components of the three-vector potential $\vec{A}$ in the Lorentz gauge.

9.3.2 Coulomb gauge condition

If we now choose χ to be such that:

$$\nabla^2 \chi = \nabla \cdot \vec{A} \quad \text{i.e.,} \quad - \partial_i \partial^i \chi = \partial_i A^i, \quad (9.11)$$

then

$$\partial_i A^i \xrightarrow{A^i \rightarrow A^i + \partial^i \chi} \partial_i A^i + \partial_i \partial^i \chi = \nabla \cdot \vec{A} - \nabla^2 \chi = 0. \quad (9.12)$$

The above condition (9.11) is therefore equivalent to

$$\boxed{\partial_i A^i = 0} \quad \text{i.e.,} \quad \boxed{\nabla \cdot \vec{A} = 0}. \quad (9.13)$$

This is known as the *Coulomb gauge condition*. In this gauge, Eq. (9.4) reduces to

$$\boxed{\square^2 A^\alpha - \partial^\alpha \left(\frac{\dot{\phi}}{c} \right) = \frac{4\pi}{c} J^\alpha}. \quad (9.14)$$

It is easy to verify that the temporal and spatial components of this equation are given by

$$\boxed{\begin{aligned} (a) \quad \nabla^2 \phi &= - 4\pi \rho \\ (b) \quad \square^2 \vec{A} + \frac{\nabla \dot{\phi}}{c} &= \frac{4\pi}{c} \vec{j} \end{aligned}}. \quad (9.15)$$

[†]Or, more appropriately the *Lorentz-Lorentz gauge condition*, after Hendrik Antoon Lorentz who, in the course of demonstrating the covariance of Maxwell's equations in 1903, rediscovered this condition already found in 1867 by Ludvig Valentin Lorentz.

Eq. (9.15a) is the *time-dependent Poisson equation* satisfied by the scalar potential ϕ and Eq. (9.15b) is a coupled second order differential equation in ϕ and the components of the three-vector potential $\vec{\mathbf{A}}$.

One way to work out the solution for $\vec{\mathbf{A}}$ is to solve first the equation (9.15 a) for ϕ , and then to substitute that solution in (9.15 b) and solve for $\vec{\mathbf{A}}$. However, this is a tedious and cumbersome procedure. There is in fact a easier way out:

Split the three-current density $\vec{\mathbf{j}}$ in longitudinal and transverse parts ($\vec{\mathbf{j}}_{\parallel}$ and $\vec{\mathbf{j}}_{\perp}$) as

$$\vec{\mathbf{j}} = \vec{\mathbf{j}}_{\parallel} + \vec{\mathbf{j}}_{\perp} \quad , \quad \left[\nabla \times \vec{\mathbf{j}}_{\parallel} = \mathbf{0} \quad , \quad \nabla \cdot \vec{\mathbf{j}}_{\perp} = 0 \right] . \quad (9.16)$$

Now, by the charge-current continuity equation (7.6)

$$\begin{aligned} \nabla \cdot \vec{\mathbf{j}} &= \nabla \cdot \vec{\mathbf{j}}_{\parallel} = -\dot{\rho} = -\frac{\partial}{\partial t} \left(-\frac{1}{4\pi} \nabla^2 \phi \right) \quad \longleftarrow \quad \text{by Eq. (9.15 a)} \\ \implies \nabla \cdot \vec{\mathbf{j}}_{\parallel} &= \frac{1}{4\pi} \nabla^2 \dot{\phi} = \frac{1}{4\pi} \nabla \cdot \nabla \dot{\phi} \\ \implies \nabla \cdot \left(\vec{\mathbf{j}}_{\parallel} - \frac{1}{4\pi} \nabla \dot{\phi} \right) &= 0 . \end{aligned} \quad (9.17)$$

Moreover, since $\nabla \times \vec{\mathbf{j}}_{\parallel} = 0$ and $\nabla \times \dot{\phi} = 0$ (in fact, curl of gradient of any scalar is zero), we have

$$\nabla \times \left(\vec{\mathbf{j}}_{\parallel} - \frac{1}{4\pi} \nabla \dot{\phi} \right) = 0 . \quad (9.18)$$

Hence,

$$\nabla \dot{\phi} = 4\pi \vec{\mathbf{j}}_{\parallel} . \quad (9.19)$$

Substituting this in Eq. (9.15 b) we finally obtain the inhomogeneous wave equation satisfied by the three-vector potential $\vec{\mathbf{A}}$ in the Coulomb gauge:

$$\boxed{\square^2 \vec{\mathbf{A}} \equiv \left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 \right) \vec{\mathbf{A}} = \frac{4\pi}{c} \vec{\mathbf{j}}_{\perp}} . \quad (9.20)$$

10 Wave Solutions in Electrodynamics — the Retarded Potentials

In this section, we shall focus on determining the solutions of the generic wave equations for the potentials ϕ and $\vec{A}$ in the Lorentz and Coulomb gauges. We shall first discuss the non-covariant approach of obtaining the solutions for ϕ and for the individual components of $\vec{A}$ in these gauges. The wave equation for ϕ in the Coulomb gauge being simply the time-dependent Poisson's equation (9.15a), it is easier to obtain its solution than in the Lorentz gauge in which ϕ satisfies the inhomogeneous wave equation (9.10a). However, on the whole, working in the Coulomb gauge is not of much advantage compared to working in the Lorentz gauge. The reason is that the components of $\vec{A}$ satisfy similar type of wave equations (9.15b) in the Coulomb gauge, as do the components of $\vec{A}$ as well as ϕ in the Lorentz gauge, viz. Eqs. (9.10). So, anyhow we have to solve for the inhomogeneous wave equation in order to obtain the complete set of solutions for the gauge potentials ϕ and $\vec{A}$, no matter whether we use the Coulomb or the Lorentz gauge condition. Now, an approximate solution of the inhomogeneous wave equation can be obtained without much difficulty, which we shall demonstrate here. Although such an approximation has a limited domain of validity, one can show that the solution obtained as such is the same as that obtained for the inhomogeneous wave equation in a rigorous exact way. We shall demonstrate the exact method in the covariant approach of solving the wave equation (9.9) for the four-potential A^μ in the Lorentz gauge. In fact, the Lorentz gauge is the only one useful for the determination of A^μ , as in the Coulomb gauge the corresponding equation (9.14) for A^μ requires intrinsically non-covariant way of getting solution due to the presence of the term involving $\dot{\phi}$. The covariant approach is much more elegant compared to the non-covariant approach. However, the importance of the latter is in the basic understanding of the development of the technique of solving for A^μ in a covariant way. One way to see this is that the covariant approach (which utilizes the Lorenz gauge condition) alludes to a direct generalization of the exact but non-covariant method of solving the time-dependent Poisson equation (satisfied by ϕ in the Coulomb gauge). This would be made clear as we go on in what follows.

10.1 Solutions for scalar and three-vector potentials — the Non-covariant approach

10.1.1 Solution ansatz in the Lorentz gauge

In the Lorentz gauge, the generic form of the inhomogeneous wave equation is given by

$$\boxed{\square^2 \psi(\vec{x}, t) = 4\pi f(\vec{x}, t)}, \quad (10.1)$$

where ψ and f are either ϕ and ρ respectively, or a component of $\vec{A}$ and the corresponding component of $\vec{j}/c$, respectively [see Eqs. (9.10)].

Let us expand the wave-functions and the sources in Fourier integrals:

$$(a) \quad \boxed{\psi(\vec{x}, t) = \int_{-\infty}^{\infty} d\omega \, \psi_\omega(\vec{x}) e^{-i\omega t}} \quad \text{and} \quad (b) \quad \boxed{f(\vec{x}, t) = \int_{-\infty}^{\infty} d\omega \, f_\omega(\vec{x}) e^{-i\omega t}}, \quad (10.2)$$

where ω is the *angular frequency*. $\psi_\omega(\vec{x})$, $f_\omega(\vec{x})$ are respectively the Fourier transforms of $\psi(\vec{x}, t)$, $f(\vec{x}, t)$:

$$(a) \quad \boxed{\psi_\omega(\vec{x}) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dt \, \psi(\vec{x}, t) e^{i\omega t}} \quad \text{and} \quad (b) \quad \boxed{f_\omega(\vec{x}) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dt \, f(\vec{x}, t) e^{i\omega t}}. \quad (10.3)$$

Plugging Eqs. (10.2) in Eq. (10.1), and using the *vacuum dispersion relation*

$$\omega = c k \quad , \quad k := \text{wave number} \quad , \quad (10.4)$$

we obtain the time-independent inhomogeneous wave equation, viz. the *Helmholtz equation*:

$$\boxed{(\nabla^2 + k^2) \psi_\omega(\vec{x}) = -4\pi f_\omega(\vec{x})}. \quad (10.5)$$

We next apply a well-known principle in wave optics, viz. the *Huygens-Fresnel principle*, which states:

“Given a wave, each point on its wavefront acts as a source for spherical wavelets of varying amplitude (weight). A linear superposition of the individual wavelets, from each of these point sources, gives rise to a new wavelet.”

According to this principle, the general solution of Eq. (10.5) is a weighted superposition of particular solutions of this equation with the term on the r.h.s replaced by the one due to a single point source:

$$\boxed{\psi_\omega(\vec{x}) = -4\pi \int d^3x' f_\omega(\vec{x}') G_\omega(\vec{x}, \vec{x}')} , \quad (10.6)$$

where $f_\omega(\vec{x}')$ is the wavelet amplitude at the source point $\vec{x}'$, and $G_\omega(\vec{x}, \vec{x}')$ is the *Green's function* or *Propagator* that satisfies the equation

$$\boxed{(\nabla^2 + k^2) G_\omega(\vec{x}, \vec{x}') = \delta^3(\vec{x} - \vec{x}')} . \quad (10.7)$$

Translational symmetry of space demands:

$$G_\omega(\vec{x}, \vec{x}') = G_\omega(\vec{x} - \vec{x}') \equiv G_\omega(\vec{R}) , \quad (10.8)$$

where we denote

$$\vec{R} \equiv \vec{x} - \vec{x}' . \quad (10.9)$$

Further, assuming spherical symmetry:

$$G_\omega(\vec{x} - \vec{x}') = G_\omega(|\vec{x} - \vec{x}'|) \quad i.e., \quad G_\omega(\vec{R}) = G_\omega(R) , \quad (10.10)$$

we can express Eq. (10.7) as

$$\boxed{(\nabla^2 + k^2) G_\omega(R) = \delta^3(\vec{R})} . \quad (10.11)$$

Thus, the original problem of finding solutions for the potentials in the Lorentz gauge boils down to the determination of the solution of the above Green's function equation (10.11). Before we work out such solution, let us resort, in the following subsection, to the corresponding scenario in the Coulomb gauge.

10.1.2 Solution ansatzes in the Coulomb gauge

In the Coulomb gauge, the wave equations are

$$(a) \quad \boxed{\nabla^2 \phi(\vec{x}, t) = -4\pi \rho(\vec{x}, t)} \quad \text{and} \quad (b) \quad \boxed{\square^2 \psi(\vec{x}, t) = 4\pi f(\vec{x}, t)} , \quad (10.12)$$

where, in the second equation, ψ and f denote a component of $\vec{A}$ and the corresponding component of $\vec{j}_\perp/c$, respectively [see Eqs. (9.15b)]. Expansion of the wave-function $\psi(\vec{x}, t)$ and the source $f(\vec{x}, t)$ in Eq. (10.12b) in Fourier integrals would lead to the same (Helmholtz) equation (10.5) whose solution could be obtained by determining the solution of the above Green's function equation (10.11). On the other hand, if we expand $\phi(\vec{x}, t)$ and $\rho(\vec{x}, t)$ in Eq. (10.12a) in Fourier integrals as

$$(a) \quad \boxed{\phi(\vec{x}, t) = \int_{-\infty}^{\infty} d\omega \phi_\omega(\vec{x}) e^{-i\omega t}} \quad \text{and} \quad (b) \quad \boxed{\rho(\vec{x}, t) = \int_{-\infty}^{\infty} d\omega \rho_\omega(\vec{x}) e^{-i\omega t}} , \quad (10.13)$$

where $\phi_\omega(\vec{x}), \rho_\omega(\vec{x})$ are respectively the Fourier transforms of $\phi(\vec{x}, t), \rho(\vec{x}, t)$:

$$(a) \quad \boxed{\phi_\omega(\vec{x}) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dt \phi(\vec{x}, t) e^{i\omega t}} \quad \text{and} \quad (b) \quad \boxed{\rho_\omega(\vec{x}) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dt \rho(\vec{x}, t) e^{i\omega t}} . \quad (10.14)$$

Then proceeding as before, and using Eq. (10.4) and the Huygens-Fresnel principle, we get

$$\boxed{\phi_\omega(\vec{x}) = -4\pi \int d^3x' \rho_\omega(\vec{x}') G_\omega(\vec{x}, \vec{x}')} . \quad (10.15)$$

Utilizing again the conditions of translational and spherical symmetries, we get the following equation for $G_\omega(\vec{x}, \vec{x}') = G_\omega(R)$:

$$\boxed{\nabla^2 G_\omega(R) = \delta^3(\vec{R})} . \quad (10.16)$$

In the next subsection we shall first work out the solution of this equation, and then determine the solution of the more general propagator equation (10.11).

10.1.3 Solutions of the Propagator equations

A. Solving $\nabla^2 G_\omega(R) = \delta^3(\vec{R})$ [cf. Eq. (10.16)]:

Let us expand

$$\boxed{G_\omega(R) = \frac{1}{(2\pi)^3} \int d^3k \mathcal{G}(\vec{k}) e^{-i\vec{k}\cdot\vec{R}}} , \quad (10.17)$$

where $\mathcal{G}(\vec{k})$ is the Fourier transform of $G_\omega(R)$ in the k -space:

$$\boxed{\mathcal{G}(\vec{k}) = \int d^3R G_\omega(\vec{R}) e^{i\vec{k}\cdot\vec{R}}} . \quad (10.18)$$

On the other hand, the Fourier transform of $\delta^3(\vec{R})$ is unity:

$$\boxed{\delta^3(\vec{R}) = \frac{1}{(2\pi)^3} \int d^3k e^{-i\vec{k}\cdot\vec{R}}} . \quad (10.19)$$

Substituting Eqs. (10.17) and (10.19) in the original Green's function equation (10.16), we have

$$\begin{aligned} \frac{1}{(2\pi)^3} \int d^3k \mathcal{G}(\vec{k}) (-i\vec{k}) \cdot (-i\vec{k}) e^{-i\vec{k}\cdot\vec{R}} &= \frac{1}{(2\pi)^3} \int d^3k e^{-i\vec{k}\cdot\vec{R}} \\ \Rightarrow \quad \boxed{\mathcal{G}(\vec{k}) \equiv \mathcal{G}(k) = -\frac{1}{k^2}} . \end{aligned} \quad (10.20)$$

Plugging this back in Eq. (10.17), we get

$$G_\omega(R) = -\frac{1}{(2\pi)^3} \int d^3k \frac{1}{k^2} e^{-i\vec{k}\cdot\vec{R}} . \quad (10.21)$$

Choosing now $\vec{k} = k\hat{\mathbf{z}}$ and using sperical polar coordinates (R, ϑ, φ) , in which

$$d^3k = k^2 dk \sin \vartheta d\vartheta d\varphi \quad \text{and} \quad \vec{k} \cdot \vec{R} = kR \cos \vartheta , \quad (10.22)$$

ϑ being the angle between the z -direction and $\vec{R}$, the above expression (10.21) becomes

$$\begin{aligned} G_\omega(R) &= -\frac{1}{(2\pi)^3} \int_0^\infty \int_0^\pi \int_0^{2\pi} k^2 dk \sin \vartheta d\vartheta d\varphi \frac{1}{k^2} e^{-ikR \cos \vartheta} \\ &= -\frac{1}{4\pi^2} \int_0^\infty dk \int_0^\pi \sin \vartheta d\vartheta e^{-ikR \cos \vartheta} . \end{aligned} \quad (10.23)$$

Let $\xi = \cos \vartheta$, so that $d\xi = -\sin \vartheta d\vartheta$. Then

$$\begin{aligned} G_\omega(R) &= \frac{1}{4\pi^2} \int_0^\infty dk \int_1^{-1} d\xi e^{-ikR\xi} = \frac{1}{4\pi^2} \int_0^\infty dk \left[-\frac{2 \sin(kR)}{kR} \right] \\ &= -\frac{1}{2\pi^2 R} \int_0^\infty d(kR) \frac{\sin(kR)}{kR} = -\frac{1}{2\pi^2 R} \frac{\pi}{2} , \end{aligned} \quad (10.24)$$

i.e.

$$\boxed{G_\omega(R) = -\frac{1}{4\pi R} \equiv -\frac{1}{4\pi |\vec{x} - \vec{x}'|}} . \quad (10.25)$$

B. Solving $(\nabla^2 + k^2) G_\omega(R) = \delta^3(\vec{R})$ [cf. Eq. (10.11)]:

Because of spherical symmetry (which implies $G_\omega(\vec{R}) = G_\omega(R)$), we can write the above equation in spherical polar coordinates as

$$\frac{1}{R^2} \frac{\partial}{\partial R} \left(R^2 \frac{\partial G_\omega(R)}{\partial R} \right) + k^2 G_\omega(R) = \delta^3(\vec{R}) \implies \boxed{\left(\frac{d^2}{dR^2} + k^2 \right) [R G_\omega(R)] = R \delta^3(\vec{R})}. \quad (10.26)$$

Away from the source point $R = |\vec{x} - \vec{x}'| = 0$, the delta function has no effect, and therefore $[R G_\omega(R)]$ satisfies the homogeneous equation

$$\underline{\underline{\left(\frac{d^2}{dR^2} + k^2 \right) [R G_\omega(R)] = 0}}. \quad (10.27)$$

The general solution of this equation is

$$\boxed{G_\omega(R) = C_+ G_\omega^+(R) + C_- G_\omega^-(R)}; \quad (10.28)$$

with $\underline{\underline{G_\omega^\pm(R) = \frac{e^{\pm i k R}}{R}}}$, $C_\pm :=$ Arbitrary constants .

Inserting this in the original equation (10.11), and integrating over a small volume V around $R = |\vec{x} - \vec{x}'| = 0$, the volume integrals can be approximated as

$$(C_+ + C_-) \int_V d^3 R \nabla^2 \left(\frac{1}{R} \right) + k^2 (C_+ + C_-) \int_V d^3 R \frac{1}{R} \approx \int_V d^3 R \delta^3(\vec{R}). \quad (10.29)$$

Now, the second integral on the l.h.s., viz.,

$$\int_V d^3 R \frac{1}{R} = \int_V R^2 dR \sin \vartheta d\vartheta d\varphi \cdot \frac{1}{R} \xrightarrow{R=|\vec{x}-\vec{x}'| \rightarrow 0} 0. \quad (10.30)$$

Moreover, an important result of vector calculus is that

$$\boxed{\nabla^2 \left(\frac{1}{R} \right) \equiv \nabla^2 \left(\frac{1}{|\vec{x} - \vec{x}'|} \right) = -4\pi \delta^3(\vec{R})}. \quad (10.31)$$

Therefore, the first integral on the l.h.s. of Eq. (10.29), viz.,

$$\int_V d^3 R \nabla^2 \left(\frac{1}{R} \right) = -4\pi \int_V d^3 R \delta^3(\vec{R}). \quad (10.32)$$

Substituting Eqs. (10.30) and (10.32) in Eq. (10.29), we have in the limit $R \rightarrow 0$:

$$\underline{\underline{C_+ + C_- = -\frac{1}{4\pi}}}. \quad (10.33)$$

Hence, the solution (10.28), which now takes the form

$$\boxed{G_\omega(R) = C [G_\omega^+(R) - G_\omega^-(R)] - \frac{G_\omega^-(R)}{4\pi} = \frac{K}{R} \sin(kR) - \frac{e^{-ikR}}{4\pi R}}; \quad (10.34)$$

with $\underline{\underline{C \equiv C_+, K = 2iC}}$,

can be treated as a complete solution of Eq. (10.11) [or, Eq. (10.26)], as long as we rely on the viability of the above approximation [Eq. (10.29)].

10.1.4 General Solutions for the Potentials

I. In the Lorentz gauge:

Substituting the solution (10.28) in Eq. (10.6) we have

$$\boxed{\psi_\omega(\vec{x}) = C_+ \psi_\omega^+(\vec{x}) + C_- \psi_\omega^-(\vec{x})}; \quad (10.35)$$

$$\text{with } \underline{\underline{\psi_\omega^\pm(\vec{x}) = -4\pi \int d^3x' f_\omega(\vec{x}') G_\omega^\pm(|\vec{x} - \vec{x}'|)}}. \quad (10.36)$$

These when inserted in Eq. (10.2a) give

$$\boxed{\psi(\vec{x}, t) = C_+ \psi^+(\vec{x}, t) + C_- \psi^-(\vec{x}, t)}; \quad (10.37)$$

$$\text{with } \underline{\underline{\psi^\pm(\vec{x}, t) = -4\pi \int d^3x' \int_{-\infty}^{\infty} d\omega f_\omega(\vec{x}') G_\omega^\pm(|\vec{x} - \vec{x}'|) e^{-i\omega t}}}. \quad (10.38)$$

Using now

$$\boxed{G_\omega^\pm(R) \equiv G_\omega^\pm(|\vec{x} - \vec{x}'|) = \frac{e^{\pm ik|\vec{x} - \vec{x}'|}}{|\vec{x} - \vec{x}'|}}, \quad (10.39)$$

we get

$$\underline{\underline{\psi^\pm(\vec{x}, t) = -4\pi \int d^3x' \int_{-\infty}^{\infty} d\omega \frac{f_\omega(\vec{x}')}{|\vec{x} - \vec{x}'|} e^{-i\omega t'_\pm}}}, \quad (10.40)$$

where

$$\underline{\underline{t'_\pm = t \mp \frac{k}{\omega} |\vec{x} - \vec{x}'| = t \mp \frac{1}{c} |\vec{x} - \vec{x}'| = t \mp \frac{R}{c}}}. \quad (10.41)$$

Finally, using Eq. (10.2b) we re-express $\psi(\vec{x}, t)$ in Eq. (10.37) as

$$\boxed{\psi(\vec{x}, t) = C_+ \psi^+(\vec{x}, t) + C_- \psi^-(\vec{x}, t)}; \quad (10.42)$$

$$\text{with } \underline{\underline{\psi^\pm(\vec{x}, t) = -4\pi \int d^3x' \frac{f(\vec{x}', t'_\pm)}{|\vec{x} - \vec{x}'|}}}. \quad (10.43)$$

This is the general solution of the inhomogeneous wave equation for the potentials ϕ, A^i in the Lorentz gauge. Note that the solution at time t at the location $\vec{x}$ is dependent on the behaviour of the source at some other time $t' = t'_+$ or t'_- at a location $\vec{x}'$. The particular solution $\psi^+(\vec{x}, t)$ is referred to as the *retarded solution*, because the argument of the source function f indicates that the effect at $\vec{x}$ at time t is caused by the action of the source at a distance $R = |\vec{x} - \vec{x}'|$ away, at an earlier or *retarded time* $t' = t'_+ = t - R/c \equiv t'_r$. The time difference $t - t'_r = R/c$ is the time of propagation of the wave disturbance from the (retarded) source point to the observation point. The particular solution $\psi^-(\vec{x}, t)$, on the other hand, is the *advanced solution* as it exhibits an effect observed at $\vec{x}$ at time t due the source at a distance $R = |\vec{x} - \vec{x}'|$ away, at a later or *advanced time* $t' = t'_- = t + R/c \equiv t'_a$.

By the demand of *causality* which requires that the potential at $(t, \vec{x})$ is set up by the source at an earlier time t'_r (at the location $\vec{x}'$), only the retarded solution is physically viable, i.e. $C_- = 0$, so that $C_+ = -1/(4\pi)$ by Eq. (10.33), whence

$$\boxed{\psi(\vec{x}, t) = \int d^3x' \frac{f(\vec{x}', t'_r)}{|\vec{x} - \vec{x}'|}}, \quad \underline{\underline{t'_r = t - \frac{|\vec{x} - \vec{x}'|}{c}}}. \quad (10.44)$$

The corresponding viable solutions for the potentials $(\phi, \vec{A})$ in the Lorentz gauge are therefore

$$\boxed{\phi(\vec{x}, t) = \int d^3x' \left[\frac{\rho(\vec{x}', t')}{|\vec{x} - \vec{x}'|} \right]_{ret}}, \quad \boxed{\vec{A}(\vec{x}, t) = \frac{1}{c} \int d^3x' \left[\frac{\vec{j}(\vec{x}', t')}{|\vec{x} - \vec{x}'|} \right]_{ret}}, \quad (10.45)$$

where $[]_{ret}$ denotes the quantity within is evaluated at the retarded time $t' = t'_r = t - |\vec{x} - \vec{x}'|/c$.

II. In the Coulomb gauge:

Substituting the solution (10.25), of Eq. (10.16), in Eq. (10.15) we have

$$\phi_{\omega}(\vec{x}) = \int d^3x' \frac{\rho_{\omega}(\vec{x}')}{|\vec{x} - \vec{x}'|}. \quad (10.46)$$

Inserting this in Eq. (10.13a), and finally using Eq. (10.13b), we get

$$\phi(\vec{x}, t) = \int d^3x' \int_{-\infty}^{\infty} d\omega \frac{\rho_{\omega}(\vec{x}')}{|\vec{x} - \vec{x}'|} e^{-i\omega t} = \int d^3x' \frac{\rho(\vec{x}', t)}{|\vec{x} - \vec{x}'|}. \quad (10.47)$$

Thus the scalar potential is just the instantaneous Coulomb potential due to the charge density $\rho(\vec{x}, t)$. Hence the name “Coulomb gauge”.

The equation satisfied by the three-vector potential $\vec{A}$ is similar to that it satisfies in the Lorentz gauge. The only difference being that in the Coulomb gauge, $\vec{A}$ is sourced by the transverse part $\vec{j}_{\perp}$ of the three-current density $\vec{j}$, rather than by $\vec{j}$ on the whole. Hence the Coulomb gauge is often called the “Transverse gauge”. In this gauge, therefore, the viable solution for $\vec{A}$ is

$$\vec{A}(\vec{x}, t) = \frac{1}{c} \int d^3x' \left[\frac{\vec{j}_{\perp}(\vec{x}', t')}{|\vec{x} - \vec{x}'|} \right]_{ret}. \quad (10.48)$$

10.2 Solutions for the four-potential — the Covariant approach

Let us now solve for the four-potential $A^{\mu}(x^{\lambda})$ on the whole. As seen from the evolution equation (9.14) in the Coulomb gauge, we require an intrinsically non-covariant method of solving for A^{μ} in the sense that we first need to solve for one component of A^{μ} , viz. the scalar potential ϕ , and then insert this solution back into Eq. (9.14) and solve for the remaining components of A^{μ} . So the Coulomb gauge is not suitable for obtaining the solution for A^{μ} in a covariant manner. On the other hand, in the Lorentz gauge, the wave equation satisfied by A^{μ} is given by

$$\square^2 A^{\mu}(x^{\lambda}) = \frac{4\pi}{c} J^{\mu}(x^{\lambda}). \quad (10.49)$$

Solving this doesn't require solving first for any of the components of A^{μ} , and therefore the Lorentz gauge is suitable for obtaining A^{μ} in a covariant way (hence, perhaps the name “Lorentz gauge”).

10.2.1 Potential expansion in terms of Green's function, and Solutions

In order to solve the above equation (10.49), we consider the following ansatz:

$$A^{\mu}(x^{\lambda}) = \frac{4\pi}{c} \int d^4x' D(x^{\lambda}, x'^{\lambda}) J^{\mu}(x'^{\lambda}), \quad (10.50)$$

where $D(x^{\lambda}, x'^{\lambda})$ is the four-dimensional Green's function, or propagator, which satisfies the equation:

$$\square_x^2 D(x^{\lambda}, x'^{\lambda}) = \delta^4(x^{\lambda} - x'^{\lambda}), \quad (10.51)$$

$\delta^4(x^{\lambda} - x'^{\lambda}) = \delta^3(\vec{x} - \vec{x}') \delta(t - t')$ being the four-dimensional delta function, and $\square_x^2$ is the D'Alembertian with spatial (Laplacian) part taken w.r.to the spatial components x^i of the field event position four-vector x^{λ} .

Now, in absence of boundary surfaces

$$D(x^{\lambda}, x'^{\lambda}) = D(x^{\lambda} - x'^{\lambda}), \quad (10.52)$$

whence Eq. (10.51) reduces to

$$\square_R^2 D(R^{\lambda}) = \delta^4(R^{\lambda}), \quad (10.53)$$

where

$$R^{\lambda} = x^{\lambda} - x'^{\lambda}. \quad (10.54)$$

Let us expand $D(R^\lambda)$ in a Fourier integral

$$D(R^\lambda) = \frac{1}{(2\pi)^4} \int d^4k \mathcal{D}(k^\lambda) e^{-ik_\alpha R^\alpha}, \quad (10.55)$$

where $\mathcal{D}(k^\lambda)$ is the Fourier transform of $D(R^\lambda)$ in the k^λ -space:

$$\mathcal{D}(k^\lambda) = \int d^4x D(R^\lambda) e^{ik_\alpha R^\alpha}. \quad (10.56)$$

Moreover, since the Fourier transform of the four dimensional delta function is unity (similar to the three dimensional delta function), i.e.

$$\delta^4(R^\lambda) = \frac{1}{(2\pi)^4} \int d^4k e^{-ik_\alpha R^\alpha}, \quad (10.57)$$

we have from Eq. (10.53)

$$\begin{aligned} \int d^4k \mathcal{D}(k^\lambda) \eta^{\mu\nu} \frac{\partial}{\partial R^\mu} \frac{\partial}{\partial R^\nu} (e^{-ik_\alpha R^\alpha}) &= \int d^4k e^{-ik_\alpha R^\alpha} \\ \Rightarrow \mathcal{D}(k^\lambda) \eta^{\mu\nu} (-k_\mu k_\nu) &= 1 \quad \Rightarrow \quad \boxed{\mathcal{D}(k^\lambda) = \mathcal{D}(\|k^\lambda\|) = -\frac{1}{k_\alpha k^\alpha}}. \end{aligned} \quad (10.58)$$

Plugging this in Eq. (10.55):

$$D(R^\lambda) = -\frac{1}{(2\pi)^4} \int \frac{d^4k}{k_\alpha k^\alpha} e^{-ik_\alpha R^\alpha}, \quad (10.59)$$

i.e.

$$\underline{\underline{D(R^\lambda) = -\frac{1}{(2\pi)^4} \int d^3k e^{i\vec{k} \cdot \vec{R}} I(k)}}, \quad \text{where} \quad \underline{\underline{I(k) = \int_{-\infty}^{\infty} dk_0 \frac{e^{-ik_0 R^0}}{k_0^2 - k^2}}}. \quad (10.60)$$

Evaluation of $I(k)$: The integrand having simple poles at $k_0 = \pm k$, we let k_0 to be complex-valued, i.e.

$$k_0 = \Re(k_0) + i \Im(k_0), \quad (10.61)$$

and perform a contour integration:

$$\underline{\underline{I_\Gamma(k) = \oint_\Gamma dk_0 \frac{e^{-ik_0 R^0}}{k_0^2 - k^2}}}, \quad (10.62)$$

over a suitably chosen closed contour Γ in the $\Re(k_0) - \Im(k_0)$ plane.

Since both the poles (at $\pm k$) lie on the $\Re(k_0)$ axis, we can choose either the contour Γ_R or the contour Γ_A shown in Fig. 10.

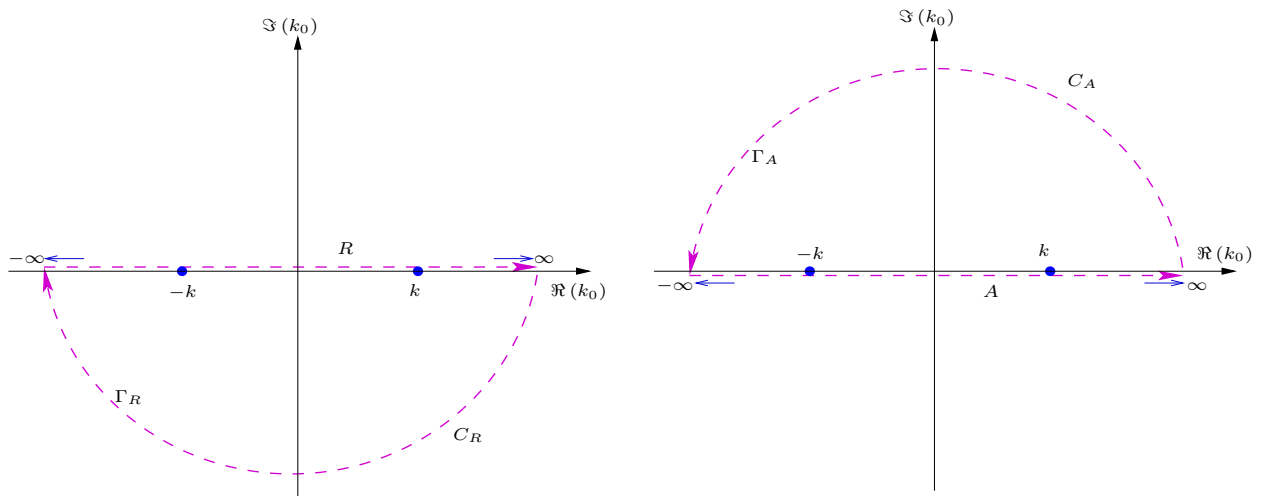


Figure 14: Illustration of the Contours.

- R and $A \implies$ Lines that graze the $\Re(k_0)$ axis from the top and the bottom respectively.
- C_R and $C_A \implies$ Semicircular arcs that enclose the open contours R and A in the lower and the upper half-planes respectively, at infinity.
- $\Gamma_R \implies$ Closed contour formed by the line R and the curve C_R .
- $\Gamma_A \implies$ Closed contour formed by the line A and the curve C_A .

Now, the exponential in the integrand of $I_\Gamma(k)$, Eq. (10.62), is factorized as

$$e^{-ik_0 R^0} = e^{-i[\Re(k_0) + i \Im(k_0)]R^0} = \underbrace{e^{-i \Re(k_0) \cdot R^0}}_{\text{Oscillatory factor}} \times \underbrace{e^{\Im(k_0) \cdot R^0}}_{\text{Growth or Decay factor}}. \quad (10.63)$$

For $\underline{\underline{R^0 > 0}}$: In the upper half-plane where $\Im(k_0) > 0$,

$$e^{-ik_0 R^0} \longrightarrow \infty \quad \text{as} \quad |\Im(k_0)| \longrightarrow \infty,$$

i.e. the integrand of $I_\Gamma(k)$, viz. Eq. (10.62), *diverges* (or, grows without limit) as we carry on increasing $\Im(k_0)$ in magnitude, so as to close the contour Γ_A at infinity, on the upper half-plane. Therefore, the contour Γ_A is *not permissible* for $R^0 > 0$.

In the lower half-plane where $\Im(k_0) < 0$,

$$e^{-ik_0 R^0} \longrightarrow 0 \quad \text{as} \quad |\Im(k_0)| \longrightarrow \infty,$$

i.e., the integrand of $I_\Gamma(k)$, Eq. (10.62), is *regular*, with *no contribution* from the curve C_R in the integral, as we carry on increasing $\Im(k_0)$ in magnitude, so as to close the contour Γ_R at infinity, on the upper half-plane. Therefore, the contour Γ_R is *permissible* for $R^0 > 0$.

Choosing the contour Γ_R we have

$$I_\Gamma(k) = \int_{-\infty}^{\infty} dk_0 \frac{e^{-ik_0 R^0}}{k_0^2 - k^2} + \int_{C_R} dk_0 \frac{e^{-ik_0 R^0}}{k_0^2 - k^2} \xrightarrow{0} \quad (10.64)$$

So,

$$\underline{\underline{I(k) \Big|_{R^0 > 0}}} = \Theta(R^0) \int_{-\infty}^{\infty} dk_0 \frac{e^{-ik_0 R^0}}{k_0^2 - k^2} = \Theta(R^0) \oint_{\Gamma_R} dk_0 \frac{e^{-ik_0 R^0}}{k_0^2 - k^2}. \quad (10.65)$$

For $\underline{\underline{R^0 < 0}}$: By a similar argument as above

$$\underline{\underline{I(k) \Big|_{R^0 < 0}}} = \Theta(-R^0) \int_{-\infty}^{\infty} dk_0 \frac{e^{-ik_0 R^0}}{k_0^2 - k^2} = -\Theta(-R^0) \oint_{\Gamma_A} dk_0 \frac{e^{-ik_0 R^0}}{k_0^2 - k^2}. \quad (10.66)$$

In the above, Θ is the *step function*:

$$\Theta(y) = \begin{cases} 1 & \text{for } y > 0 \\ 0 & \text{otherwise} \end{cases}. \quad (10.67)$$

Note also that the minus sign appears in the last equality of Eq. (10.66) due to the fact that the sense of integration of the contour Γ_A is counter-clockwise, i.e. opposite to clockwise sense of integration of the contour Γ_R . For the clockwise sense of integration, using the theorem of residues, we have

$$\begin{aligned} I_\Gamma(k) &= \oint_{\Gamma} dk_0 \frac{e^{-ik_0 R^0}}{k_0^2 - k^2} = -2\pi i \left[\text{Res}_{(k_0=k)} \left(\frac{e^{-ik_0 R^0}}{k_0^2 - k^2} \right) + \text{Res}_{(k_0=-k)} \left(\frac{e^{-ik_0 R^0}}{k_0^2 - k^2} \right) \right] \\ &= -2\pi i \left[(k_0 - k) \frac{e^{-ik_0 R^0}}{k_0^2 - k^2} \Big|_{k_0=k} + (k_0 + k) \frac{e^{-ik_0 R^0}}{k_0^2 - k^2} \Big|_{k_0=-k} \right] \\ &= -2\pi i \left[\frac{e^{-ikR^0}}{2k} + \frac{e^{ikR^0}}{-2k} \right] = \frac{\pi i}{k} (e^{ikR^0} - e^{-ikR^0}) = \frac{\pi i}{k} \cdot 2i \sin(kR^0). \end{aligned}$$

Or,

$$\underline{\underline{I_\Gamma(k) = \oint_\Gamma dk_0 \frac{e^{-ik_0 R^0}}{k_0^2 - k^2} = -\frac{2\pi}{k} \sin(kR^0) .}} \quad (10.68)$$

Substituting this in Eq. (10.60):

$$\underline{\underline{D(R^\lambda) \Big|_{R^0 > 0} \equiv D_R(R^\lambda) = \frac{\Theta(R^0)}{(2\pi)^3} \int \frac{d^3 k}{k} e^{i\vec{k} \cdot \vec{R}} \sin(kR^0) ,}} \quad (10.69)$$

$$\underline{\underline{D(R^\lambda) \Big|_{R^0 < 0} \equiv D_A(R^\lambda) = -\frac{\Theta(-R^0)}{(2\pi)^3} \int \frac{d^3 k}{k} e^{i\vec{k} \cdot \vec{R}} \sin(kR^0) .}} \quad (10.70)$$

Now,

$$\begin{aligned} D_R(R^\lambda) &= \frac{\Theta(R^0)}{(2\pi)^3} \int_0^\infty \int_0^\pi \int_0^{2\pi} k^2 dk \sin \vartheta d\vartheta d\varphi \frac{e^{i\vec{k} \cdot \vec{R}}}{k} \sin(kR^0) \\ &= \frac{\Theta(R^0)}{(2\pi)^3} \cdot 2\pi \int_0^\infty k dk \sin(kR^0) \int_0^\pi d\vartheta \sin \vartheta e^{ikR \cos \vartheta} \\ &= \frac{\Theta(R^0)}{4\pi^2} \int_0^\infty k dk \sin(kR^0) \int_{-1}^1 d\xi e^{ikR\xi} \quad \Leftarrow [\text{Substituting: } \cos \vartheta = \xi] \\ &= \frac{\Theta(R^0)}{4\pi^2} \int_0^\infty k dk \sin(kR^0) \left[\frac{2 \sin(kR)}{kR} \right] = \frac{\Theta(R^0)}{2\pi^2 R} \int_0^\infty dk \sin(kR^0) \sin(kR) \\ &= \frac{\Theta(R^0)}{2\pi^2 R} \cdot \frac{1}{2} \left\{ \int_0^\infty dk \cos[k(R^0 - R)] - \int_0^\infty dk \cos[k(R^0 + R)] \right\} \\ &= \frac{\Theta(R^0)}{4\pi^2 R} \cdot \pi \left[\delta(R^0 - R) - \cancel{\delta(R^0 + R)} \right] . \end{aligned}$$

Hence,

$$\boxed{D_R(R^\lambda) = \frac{\Theta(R^0)}{4\pi R} \delta(R^0 - R)} \quad \text{or, explicitly,} \quad \boxed{D_R(x^\lambda - x'^\lambda) = \frac{\Theta(x^0 - x'^0)}{4\pi R} \delta(x^0 - x'^0 - R)} . \quad (10.71)$$

And, similarly,

$$\boxed{D_A(R^\lambda) = \frac{\Theta(-R^0)}{4\pi R} \delta(R^0 + R)} \quad \text{or, explicitly,} \quad \boxed{D_A(x^\lambda - x'^\lambda) = \frac{\Theta(x'^0 - x^0)}{4\pi R} \delta(x^0 - x'^0 + R)} . \quad (10.72)$$

- D_R : *Retarded Green's function*, as the source's temporal position (t') is always earlier than that (t) of the observer's, i.e., $x'^0 = ct' < x^0 = ct$.
- D_A : *Advanced Green's function*, because the field computed at time t is earlier than the time t' at which the source is convoluted, i.e., $x^0 = ct < x'^0 = ct'$.

The function D_A is clearly *acausal*, and hence physically unacceptable.

10.2.2 Covariant formulation of Retarded and Advanced Green's functions

The above expressions for D_R and D_A are not generally covariant under proper Lorentz transformations. For a covariant formulation of the Green's functions let us first consider the function

$$f(R^0) = (R^0)^2 - R^2 = R_\lambda R^\lambda = \|R^\lambda\|^2 , \quad (10.73)$$

which vanishes at $R^0 = \pm R$.

Now, use a well-known identity for the delta function, viz. for a function $f(x)$ with zeroes at $x = x_{(i)}$,

$$\delta(f(x)) = \sum_{(i)} \frac{\delta(x - x_{(i)})}{|\partial f / \partial x|_{x=x_{(i)}}}. \quad (10.74)$$

Then

$$\delta(\|R^\lambda\|^2) = \delta(f(R^0)) = \frac{\delta(R^0 - R)}{|\partial f / \partial R^0|_{R^0=R}} + \frac{\delta(R^0 + R)}{|\partial f / \partial R^0|_{R^0=-R}} = \frac{\delta(R^0 - R)}{|2R^0|_{R^0=R}} + \frac{\delta(R^0 + R)}{|2R^0|_{R^0=-R}},$$

i.e.,

$$\delta(\|R^\lambda\|^2) = \frac{\delta(R^0 - R) + \delta(R^0 + R)}{2R}. \quad (10.75)$$

Therefore, Eq. (10.71) can be expressed as:

$$D_R(R^\lambda) = \frac{\Theta(R^0)}{4\pi R} \left[2R \delta(\|R^\lambda\|^2) - \delta(R^0 + R) \right], \quad (10.76)$$

and since $\Theta(R^0)$ eliminates $\delta(R^0 + R)$, we have

$$D_R(R^\lambda) = \frac{\Theta(R^0)}{2\pi} \delta(\|R^\lambda\|^2). \quad (10.77)$$

Similarly, Eq. (10.72) is expressed as:

$$D_A(R^\lambda) = \frac{\Theta(-R^0)}{2\pi} \delta(\|R^\lambda\|^2). \quad (10.78)$$

10.2.3 Retarded and Advanced Solutions for the Potential

Finally, in terms of the retarded and advanced Green's functions, the solution of the wave equation (10.49) is expressed either as:

$$A^\mu(x^\lambda) = A_{in}^\mu(x^\lambda) + \frac{4\pi}{c} \int d^4x' D_R(x^\lambda - x'^\lambda) J^\mu(x'^\lambda), \quad (10.79)$$

or as:

$$A^\mu(x^\lambda) = A_{out}^\mu(x^\lambda) + \frac{4\pi}{c} \int d^4x' D_A(x^\lambda - x'^\lambda) J^\mu(x'^\lambda). \quad (10.80)$$

where A_{in}^μ and A_{out}^μ are the solutions of the source-free (i.e., homogeneous) wave equation $\square^2 A^\mu(x^\lambda) = 0$.

The covariant forms of the Green's functions in Eqs. (10.77) and (10.78), as combinations of theta and delta functions, imply that they are non-zero only on backward or forward lightcones, i.e. when the *lightcone condition*, viz.

$$\underline{\underline{\|R^\lambda\|^2 = R_\lambda R^\lambda = (R^0)^2 - R^2 = 0}}, \quad (10.81)$$

is satisfied. Non-vanishing value at the backward lightcone is obviously for the retarded Green's function D_R , whereas non-vanishing value at the forward lightcone is for the advanced Green's function D_A .

Taking the view that whatever is observed at a certain instant of time is due to what has happened on or(and) in the past lightcone — the *causality argument* — and ignoring (for the time being) the solutions of the homogeneous wave equation, we have the only acceptable solution for the four-potential:

$$A^\mu(x^\lambda) = \frac{4\pi}{c} \int d^4x' D_R(x^\lambda - x'^\lambda) J^\mu(x'^\lambda) = \frac{1}{c} \int d^4x' \frac{\Theta(x^0 - x'^0)}{R} \delta(x^0 - x'^0 - R) J^\mu(x'^\lambda),$$

i.e.

$$A^\mu(x^\lambda) = \frac{2}{c} \int d^4x' \Theta(x^0 - x'^0) \delta(\|x^\lambda - x'^\lambda\|^2) J^\mu(x'^\lambda). \quad (10.82)$$

11 Radiation by Moving Charges

11.1 The Liénard-Wiechart potentials due to a moving point charge

Consider a particle of charge q in motion. Let τ denotes the proper time of the charge, $r^\lambda(\tau)$ its position four-vector, and $u^\lambda(\tau) = dr^\lambda/d\tau$ its four-velocity. The four-potential due to the charged particle is given (in Lorentz gauge) by

$$\boxed{A^\mu(x^\lambda) = \frac{4\pi}{c} \int d^4x' D_R(x^\lambda - x'^\lambda) J^\mu(x'^\lambda)} , \quad (11.1)$$

where $D_R(x^\lambda - x'^\lambda)$ is the retarded Green's function:

$$\underline{\underline{D_R(x^\lambda - x'^\lambda) = \frac{\Theta(x^0 - x'^0)}{4\pi R} \delta(x^0 - x'^0 - R) = \frac{\Theta(x^0 - x'^0)}{2\pi} \delta(\|x^\lambda - x'^\lambda\|^2)} , \quad (11.2)$$

and $J^\mu(x^\lambda)$ is the charge's four-current (defined in §7.1, see Eq. (7.4)):

$$\underline{\underline{J^\mu(x^\lambda) = q \int c d\tau u^\mu(\tau) \delta^4(x^\lambda - r^\lambda(\tau))}} . \quad (11.3)$$

Substituting Eqs. (11.2) and (11.3) in Eq. (11.1) we have

$$A^\mu(x^\lambda) = 4\pi q \int d^4x' \int d\tau u^\mu(\tau) \frac{\Theta(x^0 - x'^0)}{2\pi} \delta(\|x^\lambda - x'^\lambda\|^2) \delta^4(x'^\lambda - r^\lambda(\tau)) ,$$

i.e.,

$$\underline{\underline{A^\mu(x^\lambda) = 2q \int d\tau u^\mu(\tau) \Theta(x^0 - r^0(\tau)) \delta(\|x^\lambda - r^\lambda(\tau)\|^2)}} . \quad (11.4)$$

The integral contributes only at $\tau = \tau_0$, where τ_0 satisfies:

$$(1) \quad \underline{\underline{\|x^\lambda - r^\lambda(\tau_0)\|^2 = 0}} \quad \Longleftarrow \quad \text{Lightcone condition} , \quad (11.5)$$

$$\text{and} \quad (2) \quad \underline{\underline{x^0 > r^0(\tau_0)}} \quad \Longleftarrow \quad \text{Retardation requirement} . \quad (11.6)$$

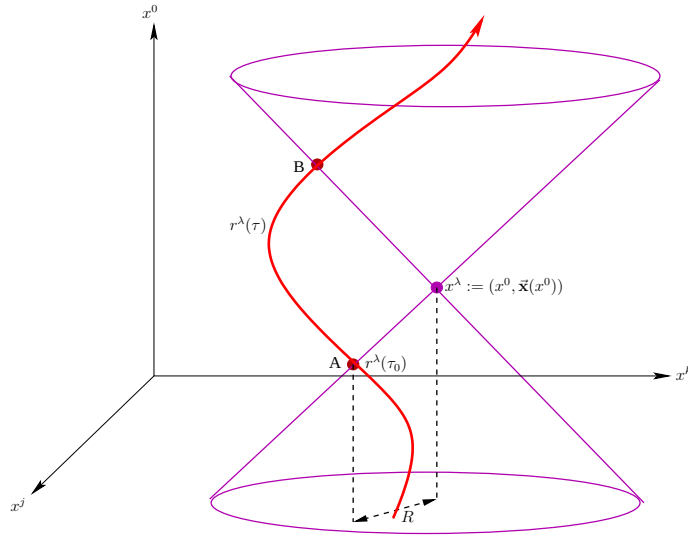


Figure 15: Illustration of the Lightcone condition and the Retardation requirement.

Let us refer to the Fig. 15. The retarded Green's function is by definition non-zero only on the backward lightcone of the observer. The charged particle's world line $r^\lambda(\tau)$ intersects the entire (past and future) lightcone at points A and B, earlier and later than x^0 respectively. The lightcone condition implies that the earlier point A is given by $r^\lambda(\tau_0)$, and this point is the only part of the path that contributes to the electromagnetic field at $x^\lambda := (x^0, \vec{x}(x^0))$, as per the retardation requirement.

Let us now denote

$$f(\tau) = \left\| x^\lambda - r^\lambda(\tau) \right\|^2, \quad (11.7)$$

which, as mentioned, has a zero of significance only at $\tau = \tau_0$. Applying the identity (10.74) mentioned in §10.2.2, viz.

$$\delta(f(x)) = \sum_{(i)} \frac{\delta(x - x_{(i)})}{|\partial f / \partial x|_{x=x_{(i)}}}, \quad \text{where } x_{(i)} \text{ are the zeroes of } f(x),$$

we have

$$\delta(f(\tau)) = \delta(\|x^\lambda - r^\lambda(\tau)\|^2) = \frac{\delta(\tau - \tau_0)}{|\partial f / \partial \tau|_{\tau=\tau_0}} = \frac{\delta(\tau - \tau_0)}{2[u_\lambda(\tau)(x^\lambda - r^\lambda(\tau))]_{\tau=\tau_0}}. \quad (11.8)$$

Substituting in Eq. (11.4):

$$\underline{\underline{A^\mu(x^\lambda) = q \int d\tau \frac{u^\mu(\tau) \delta(\tau - \tau_0)}{[u_\lambda(\tau)(x^\lambda - r^\lambda(\tau))]_{\tau=\tau_0}} = \left[\frac{q u^\mu(\tau)}{u_\lambda(\tau)(x^\lambda - r^\lambda(\tau))} \right]_{\tau=\tau_0}}}. \quad (11.9)$$

Again, the lightcone constraint [Eq. (11.5)] implies

$$|x^0 - r^0(\tau_0)|^2 - |\vec{x} - \vec{r}(\tau_0)|^2 = 0 \implies \underline{\underline{x^0 - r^0(\tau_0) = |\vec{x} - \vec{r}(\tau_0)| \equiv |\vec{R}| = R}}. \quad (11.10)$$

Therefore,

$$\left[u_\lambda(\tau) (x^\lambda - r^\lambda(\tau)) \right]_{\tau=\tau_0} = u_0 [x^0 - r^0(\tau_0)] - \vec{u} \cdot [\vec{x} - \vec{r}(\tau_0)] = u_0 R - \vec{u} \cdot \vec{R}, \quad (11.11)$$

i.e.

$$\left[u_\lambda(\tau) (x^\lambda - r^\lambda(\tau)) \right]_{\tau=\tau_0} = (u_0 - \vec{u} \cdot \hat{n}) R, \quad \text{where } \hat{n} = \frac{\vec{R}}{R}. \quad (11.12)$$

If $\vec{v}$ be the velocity of the moving charge, then since

$$u_0 = u^0 = \gamma c \quad \text{and} \quad \vec{u} = \gamma \vec{v} = \gamma c \vec{\beta}, \quad \text{where} \quad \vec{\beta} = \frac{\vec{v}}{c} \quad \text{and} \quad \gamma = \left(1 - |\vec{\beta}|^2\right)^{-1/2},$$

Eq. (11.12) reduces to

$$\left[u_\lambda(\tau) (x^\lambda - r^\lambda(\tau)) \right]_{\tau=\tau_0} = \left[\gamma c (1 - \vec{\beta} \cdot \hat{n}) R \right]_{ret}, \quad (11.13)$$

where the subscript “ret” implies that the quantity inside the brackets [] is to be evaluated at the retarded time t'_r , given by

$$\underline{\underline{c t'_r = r^0(\tau_0) = x^0 - R = c t - R}}. \quad (11.14)$$

Plugging Eq. (11.13) back in Eq. (11.9) we get the final expression for the four potential:

$$\boxed{A^\mu(x^\lambda) = \left[\frac{q u^\mu}{\gamma c (1 - \vec{\beta} \cdot \hat{n}) R} \right]_{ret}}, \quad (11.15)$$

the temporal and spatial components of which give respectively the scalar and three-vector potentials, commonly referred to as the *Liénard-Wiechart potentials* due to the moving point charge:

$$\boxed{\begin{aligned} \phi(\vec{x}, t) &= \left[\frac{q u^0}{\gamma c (1 - \vec{\beta} \cdot \hat{n}) R} \right]_{ret} = \left[\frac{q}{\kappa R} \right]_{ret} \\ \vec{A}(\vec{x}, t) &= \left[\frac{q \vec{u}}{\gamma c (1 - \vec{\beta} \cdot \hat{n}) R} \right]_{ret} = \left[\frac{q \vec{\beta}}{\kappa R} \right]_{ret} \end{aligned}}, \quad \text{where} \quad \underline{\underline{\kappa = 1 - \vec{\beta} \cdot \hat{n}}}. \quad (11.16)$$

In what follows, we shall consider the retarded brackets []_{ret} to be implicit, and express the Liénard-Wiechart potentials simply as

$$\boxed{\phi = \frac{q}{s}, \quad \vec{A} = \frac{q \vec{\beta}}{s}}, \quad \text{where} \quad \underline{\underline{s = \kappa R = (1 - \vec{\beta} \cdot \hat{n}) R = R - \vec{R} \cdot \vec{\beta}}}. \quad (11.17)$$

11.2 Retarded Electric and Magnetic fields due to a moving charge

In principle, one can work out the expressions for the electric and magnetic fields, $\vec{E}$ and $\vec{B}$, directly from the Liénard-Wiechart potentials (11.17). However, this essentially non-covariant approach requires tedious, albeit straightforward, calculations all the way through. As an alternative, one can go back to the expression (11.4) for the four-potential A^μ , given as an integral over the charge's proper time τ , and compute the electromagnetic field tensor $F^{\mu\nu}$ therefrom, and finally work out its components, viz. the expressions for $\vec{E}$ and $\vec{B}$ fields. This (covariant) approach, although requires a lesser amount of work in getting upto $F^{\mu\nu}$, is not of much help anyhow, because getting the expressions for $\vec{E}$ and $\vec{B}$ again involves tedious calculations. In what follows, we shall resort to the first (i.e. non-covariant) approach of obtaining $\vec{E}$ and $\vec{B}$ from the Liénard-Wiechart potentials (11.17). We will also outline briefly the alternative (covariant) approach of getting $F^{\mu\nu}$ from the four-potential expression (11.4), and the $\vec{E}$ and $\vec{B}$ fields therefrom (leaving however the detailed steps for the reader).

Now, it should be noted that, when an observer at a certain location, say $\vec{x}$, measures the $\vec{E}$ and $\vec{B}$ fields at some instant of time t , the fields are not due to the source charge at the same instant t , but at a former instant $t'_r = t - R/c$, where R/c is the time taken by the signal to travel from the (retarded) source location, say $\vec{x}'_r$, to the field location $\vec{x}$. As the potentials ϕ and $\vec{A}$ are expressed in terms of the retarded time t'_r (by virtue of the retarded bracket $[\]_{ret}$, that is implicit in the expressions (11.17)), we need to find the relations between the time derivatives and the (spatial) gradients in two different frames of the time — one of the (retarded) source's, i.e. t'_r , and the other of the observer's, i.e. t .

11.2.1 Time and space derivatives in the frames of the retarded source and the observer

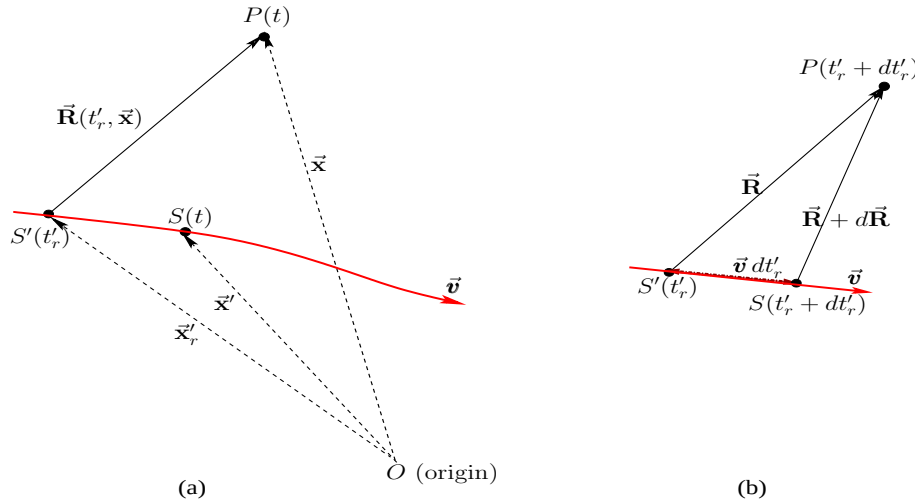


Figure 16: Relation between time and space derivatives with respect to the field point and the retarded source point.

Refer first to the Fig. 16 a. Here,

- $P(t, \vec{x}) \implies$ The field point.
- $S(t, \vec{x}') \implies$ Location of the source when the fields are computed.
- $S'(t'_r, \vec{x}'_r) \implies$ Actual (retarded) location of the source which causes the fields.

Since, the position four vector of the retarded source $r^\lambda := (ct'_r, \vec{x}'_r)$, with $t'_r = t - R/c$, R being the distance between the retarded source point S' and the field point P , we have

$$R = c(t - t'_r) \implies \underline{\underline{\frac{\partial R}{\partial t} = c \left(1 - \frac{\partial t'_r}{\partial t} \right)}}. \quad (11.18)$$

Now, refer to the Fig. 16 b. Here we suppose the interval $t - t'_r = dt'_r$, in which time $\vec{R}$ changes to $\vec{R} + d\vec{R}$, and the source is displaced from its retarded position $S'(t'_r, \vec{x}'_r)$ to the position $S(t'_r + dt'_r, \vec{x}')$ by an amount $\vec{v} dt'_r$ (with velocity $\vec{v}$). Then

$$\vec{R} = \vec{v} dt'_r + \vec{R} + d\vec{R} \implies \frac{\partial \vec{R}}{\partial t'_r} \Big|_{\vec{x}} = -\vec{v}. \quad (11.19)$$

Also,

$$R^2 = \vec{R} \cdot \vec{R} \implies 2R \frac{\partial R}{\partial t'_r} = 2\vec{R} \cdot \frac{\partial \vec{R}}{\partial t'_r} \implies \frac{\partial R}{\partial t'_r} = -\hat{n} \cdot \vec{v}, \quad (11.20)$$

using Eq. (11.19). Hence,

$$\frac{\partial R}{\partial t} = \frac{\partial R}{\partial t'_r} \frac{\partial t'_r}{\partial t} = -(\hat{n} \cdot \vec{v}) \frac{\partial t'_r}{\partial t}. \quad (11.21)$$

Comparing with Eq. (11.18):

$$\frac{\partial t'_r}{\partial t} = \frac{1}{1 - \hat{n} \cdot \vec{\beta}} \equiv \frac{1}{\kappa} \implies \boxed{\frac{\partial}{\partial t} = \frac{1}{\kappa} \frac{\partial}{\partial t'_r}}. \quad (11.22)$$

Again, since $R = R(t'_r, \vec{x})$ we have

$$dR = \frac{\partial R}{\partial t'_r} \Big|_{\vec{x}} dt'_r + \frac{\partial R}{\partial x^i} \Big|_{t'_r} dx^i \implies \frac{\partial R}{\partial x^i} \Big|_t = \left(\frac{\partial R}{\partial t'_r} \right) \frac{\partial t'_r}{\partial x^i} \Big|_t + \frac{\partial R}{\partial x^i} \Big|_{t'_r},$$

i.e.,

$$\frac{\partial R}{\partial x^i} = \left(\frac{\partial R}{\partial t'_r} \right) \nabla'_r t'_r + \nabla'_r R, \quad \text{where} \quad (\nabla'_r)^i \equiv \frac{\partial}{\partial x^i} \Big|_{t'_r}. \quad (11.23)$$

But

$$R^2 = |\vec{x} - \vec{x}'_r|^2 = \sum_i (x^i - x'^i_r)^2 \implies 2R \frac{\partial R}{\partial x^i} \Big|_{t'_r} = 2(x^i - x'^i_r) \Big|_{t'_r},$$

as x'^i_r ($\forall i$) are held fixed at $t'_r = \text{constant}$. So,

$$\frac{\nabla'_r R}{R} = \frac{\vec{R}}{R} = \hat{n}. \quad (11.24)$$

Moreover,

$$R = c(t - t'_r) \implies \frac{\nabla R}{R} = -c \nabla t'_r. \quad (11.25)$$

Substituting Eqs. (11.24) and (11.25) in Eq. (11.23), and using Eq. (11.20), we have

$$-c \nabla t'_r = -\hat{n} \cdot \vec{v} \nabla t'_r + \hat{n} \implies \nabla t'_r = -\frac{\hat{n}}{c(1 - \vec{\beta} \cdot \hat{n})} = -\frac{\hat{n}}{c\kappa}. \quad (11.26)$$

Plugging this again in Eq. (11.23), we finally obtain

$$\boxed{\nabla = \nabla'_r - \frac{\hat{n}}{c\kappa} \frac{\partial}{\partial t'_r}}. \quad (11.27)$$

11.2.2 Electric and Magnetic fields from the Liénard-Wiechart potentials

The electric and magnetic fields can be worked out straightaway from the Liénard-Wiechart potentials (11.17) as

$$\boxed{\begin{aligned} \vec{E} &= -\nabla\phi - \frac{1}{c} \frac{\partial A}{\partial t} = -\nabla\left(\frac{q}{s}\right) - \frac{1}{c} \frac{\partial}{\partial t} \left(\frac{q\vec{\beta}}{s}\right) = q \left[\frac{\nabla s}{s^2} - \frac{1}{c} \frac{\partial}{\partial t} \left(\frac{\vec{\beta}}{s}\right) \right] \\ \vec{B} &= \nabla \times \vec{A} = \nabla \times \left(\frac{q\vec{\beta}}{s}\right) = q \left[\frac{\nabla \times \vec{\beta}}{s} + \frac{\vec{\beta} \times \nabla s}{s^2} \right] \end{aligned}}. \quad (11.28)$$

where (in the second line, i.e., for $\vec{B}$) we have used the vector identity:

$$\nabla \times (\psi \vec{U}) = \psi \nabla \times \vec{U} - \vec{U} \times \nabla \psi, \quad [\psi : \text{scalar}, \vec{U} : \text{vector}]. \quad (11.29)$$

Now, the retarded brackets $[\]_{ret}$ being implicit, we need to work out the expressions for $\vec{E}$ and $\vec{B}$ in terms of time derivatives and (spatial) gradients defined in the retarded source frame, i.e. $\partial/\partial t'_r$ and ∇'_r . Substituting therefore the relations (11.22) and (11.27) in the above expression for $\vec{E}$ in Eq. (11.28), we have

$$\frac{\vec{E}}{q} = \frac{1}{s^2} \left(\nabla' s - \frac{\hat{n}}{c\kappa} \frac{\partial s}{\partial t'} \right) - \frac{1}{c\kappa} \frac{\partial}{\partial t'} \left(\frac{\vec{\beta}}{s} \right) = \frac{\nabla' s}{s^2} - \frac{(\hat{n} - \vec{\beta}) \dot{s}}{c\kappa s^2} - \frac{\dot{\vec{\beta}}}{c\kappa s}, \quad (11.30)$$

where, for brevity, we have omitted the subscript 'r' for retardation, i.e., from now on we will use t' for t'_r , $\partial/\partial t'$ for $\partial/\partial t'_r$ and ∇' for ∇'_r . The overhead dot $\{\dot{\cdot}\}$ implies $\partial/\partial t' \equiv \partial/\partial t'_r$. From the expression of s , given in Eq. (11.17), we evaluate:

$$\begin{aligned} \nabla' s &= \nabla' (R - \vec{R} \cdot \vec{\beta}) = \hat{n} + \frac{1}{c} \nabla' \left(\vec{R} \cdot \frac{\partial \vec{R}}{\partial t'} \right) \left[\text{as } \nabla' R = \hat{n}, \quad \vec{\beta} = \frac{\vec{v}}{c} = -\frac{1}{c} \frac{\partial \vec{R}}{\partial t'} \right] \\ &= \hat{n} + \frac{1}{c} \nabla' \left(R \frac{\partial R}{\partial t'} \right) = \hat{n} + \frac{1}{c} (\nabla' R) \frac{\partial R}{\partial t'} \left[\text{as } \vec{R} \cdot \frac{\partial \vec{R}}{\partial t'} = \frac{\partial R}{\partial t'}, \text{ and } \nabla' \frac{\partial}{\partial t'} R = \frac{\partial}{\partial t'} \nabla' R = 0 \right] \\ &= \hat{n} + \frac{1}{c} \frac{\partial R}{\partial t'} \hat{n} = \hat{n} + \frac{1}{c} \frac{\partial \vec{R}}{\partial t'} = \hat{n} + \frac{1}{c} (-\vec{v}) \end{aligned}$$

i.e.

$$\underline{\underline{\nabla' s = \hat{n} - \vec{\beta}}}. \quad (11.31)$$

and

$$\begin{aligned} \dot{s} &\equiv \frac{\partial s}{\partial t'} = \frac{\partial}{\partial t'} (R - \vec{R} \cdot \vec{\beta}) = \frac{\partial R}{\partial t'} - \frac{\partial R}{\partial t'} \cdot \vec{\beta} - \vec{R} \cdot \frac{\partial \vec{\beta}}{\partial t'} \\ &= -\hat{n} \cdot \vec{v} + \vec{v} \cdot \vec{\beta} - \vec{R} \cdot \dot{\vec{\beta}} \left[\text{as } \frac{\partial R}{\partial t'} = -\hat{n} \cdot \vec{v}, \quad \frac{\partial \vec{R}}{\partial t'} = -\vec{v} \right] \end{aligned}$$

i.e.

$$\underline{\underline{\dot{s} = -c \left(\hat{n} \cdot \vec{\beta} - \beta^2 + \frac{\vec{R} \cdot \dot{\vec{\beta}}}{c} \right)}}. \quad (11.32)$$

Substituting Eqs. (11.31) and (11.32) in Eq. (11.30), and using the relation $\kappa = s/R = 1 - \hat{n} \cdot \vec{\beta}$, we have

$$\begin{aligned} \frac{\vec{E}}{q} &= \frac{\hat{n} - \vec{\beta}}{s^2} - \left(\frac{\hat{n} - \vec{\beta}}{s^3} \right) R \left(\hat{n} \cdot \vec{\beta} - \beta^2 + \frac{\vec{R} \cdot \dot{\vec{\beta}}}{c} \right) - \frac{R \dot{\vec{\beta}}}{c s^2} \\ &= \frac{\hat{n} - \vec{\beta}}{s^3} (s + \vec{R} \cdot \vec{\beta} - R\beta^2) + \frac{1}{c s^3} \left[(\vec{R} - R\vec{\beta}) \vec{R} \cdot \dot{\vec{\beta}} - (R - \vec{R} \cdot \vec{\beta}) R \dot{\vec{\beta}} \right]. \end{aligned} \quad (11.33)$$

Since, $s = R - \vec{R} \cdot \vec{\beta}$, first term on the r.h.s. reduces to

$$\frac{\hat{n} - \vec{\beta}}{s^3} (R - R\beta^2),$$

whereas, using the bac-cab rule of vector algebra, viz.

$$\underline{\underline{\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B} (\vec{A} \cdot \vec{C}) - \vec{C} (\vec{A} \cdot \vec{B})}}, \quad (11.34)$$

the second term on the r.h.s. simplifies to

$$\frac{1}{c s^3} \vec{R} \times [(\vec{R} - R\vec{\beta}) \times \dot{\vec{\beta}}].$$

Therefore, the final expression for the electric field is

$$\boxed{\vec{E} = \frac{qR}{s^3} (1 - \beta^2) (\hat{n} - \vec{\beta}) + \frac{qR^2}{c s^3} \hat{n} \times [(\hat{n} - \vec{\beta}) \times \dot{\vec{\beta}}]}. \quad (11.35)$$

As to the magnetic field, the first term in the expression for $\vec{B}$ in (11.28) involves $\nabla \times \vec{\beta}$, which is can be reduced as follows:

$$\begin{aligned} (\nabla \times \vec{\beta})^i &= -\epsilon^{ijk} \frac{\partial}{\partial x^j} \beta_k = -\epsilon^{ijk} \frac{\partial t'}{\partial x^j} \dot{\beta}_k \quad [\text{as } \beta^i = \beta^i(t') \quad \forall i] \\ &= \epsilon^{ijk} \frac{n_j}{c\kappa} \dot{\beta}_k = \frac{1}{c\kappa} (\dot{\vec{\beta}} \times \hat{n})^i \quad \left[\text{as } \nabla t' = -\frac{\hat{n}}{c\kappa} \iff \frac{\partial t'}{\partial x^i} = -\frac{n_i}{c\kappa} \right], \end{aligned}$$

i.e.

$$\underline{\underline{\nabla \times \vec{\beta} = \frac{\dot{\vec{\beta}} \times \hat{n}}{c\kappa}}}. \quad (11.36)$$

Substituting now the relations (11.22) and (11.27), and the result (11.36), in the expression for $\vec{B}$ in Eq. (11.28), we have

$$\underline{\underline{\frac{\vec{B}}{q} = \frac{\dot{\vec{\beta}} \times \hat{n}}{c\kappa s} + \frac{1}{s^2} \vec{\beta} \times \left(\nabla' s - \frac{\hat{n}}{c\kappa} \dot{s} \right)}}, \quad (11.37)$$

which when simplified using the expressions of $s = \kappa R$ [Eq. (11.17)], $\nabla' s$ [Eq. (11.31)], and $\dot{s}$ [Eq. (11.32)], yields the final expression for the magnetic field as

$$\boxed{\vec{B} = \frac{qR}{s^3} (1 - \beta^2) \vec{\beta} \times \hat{n} + \frac{qR^2}{c s^3} \hat{n} \times \left\{ \hat{n} \times \left[(\hat{n} - \vec{\beta}) \times \dot{\vec{\beta}} \right] \right\} = \hat{n} \times \vec{E}}. \quad (11.38)$$

Returning back to the notation of the retarded brackets, the above expressions (11.35) and (11.38) for the electric and the magnetic fields are given as

$$\boxed{\vec{E} = \left[\frac{q}{\kappa^3 R^2} (1 - \beta^2) (\hat{n} - \vec{\beta}) + \frac{q}{c \kappa^3 R^2} \hat{n} \times \left\{ (\hat{n} - \vec{\beta}) \times \dot{\vec{\beta}} \right\} \right]_{ret}, \quad \vec{B} = [\hat{n} \times \vec{E}]_{ret}}, \quad (11.39)$$

where the overhead dot $\{\cdot\}$ now denotes $\partial/\partial t$ (since the subscript “ret” already implies that everything is determined at the retarded time $t' \equiv t'_r = t - R/c$).

11.2.3 Electromagnetic field tensor and components from generic four potential expression

Let us go back to the expression (11.4) for the four-potential A^μ , given as an integral over the proper time of the moving charge. Since the electromagnetic field tensor is given by $F^{\alpha\mu} = \partial^\alpha A^\mu - \partial^\mu A^\alpha$, we first determine

$$\begin{aligned} \partial^\alpha A^\mu(x^\lambda) &= 2q \partial^\alpha \int d\tau u^\mu(\tau) \Theta(x^0 - r^0(\tau)) \delta(\|x^\lambda - r^\lambda(\tau)\|^2) \\ &= 2q \int d\tau u^\mu(\tau) \left[\Theta(x^0 - r^0(\tau)) \partial^\alpha \delta(\|x^\lambda - r^\lambda(\tau)\|^2) + \delta(\|x^\lambda - r^\lambda(\tau)\|^2) \partial^\alpha \Theta(x^0 - r^0(\tau)) \right]. \end{aligned} \quad (11.40)$$

As the derivative of the Theta function w.r.to its argument is equal to the delta function of the same argument[†], i.e.

$$\underline{\underline{\frac{d}{dx} \Theta(x) = \delta(x)}}, \quad (11.41)$$

we can express

$$\partial^\alpha \Theta(x^0 - r^0(\tau)) = \delta_0^\alpha \delta(x^0 - r^0(\tau)). \quad (11.42)$$

Moreover, since

$$\delta(\|x^\lambda - r^\lambda(\tau)\|^2) = \delta(|x^0 - r^0(\tau)|^2 - |\vec{x} - \vec{r}(\tau)|^2) = \delta(|x^0 - r^0(\tau)|^2 - R^2),$$

the delta function in Eq. (11.42) reduces $\delta(\|x^\lambda - r^\lambda(\tau)\|^2)$ in the second term on the r.h.s. of Eq. (11.40) to $\delta(-R^2)$, which is zero for $R \neq 0$. So, considering $R \neq 0$ we write the above Eq. (11.40) as

$$\partial^\alpha A^\mu(x^\lambda) = 2q \int d\tau u^\mu(\tau) \Theta(x^0 - r^0(\tau)) \partial^\alpha \delta(\|x^\lambda - r^\lambda(\tau)\|^2). \quad (11.43)$$

[†] **Proof:** We have

$$\int_{-a}^a \left[\frac{d\Theta(x)}{dx} - \delta(x) \right] dx = \Theta(a) + \Theta(-a) - 1 = 1 + 0 - 1 = 0.$$

Now, the limits $\pm a$ being arbitrary, one can always equate the integrand to zero. Hence the proof.

Denoting again

$$f(\tau) = \|x^\lambda - r^\lambda(\tau)\|^2 \quad [\text{cf. Eq. (11.7)}] ,$$

we have

$$\begin{aligned} \partial^\alpha \delta(\|x^\lambda - r^\lambda(\tau)\|^2) &\equiv \partial^\alpha \delta(f(\tau)) = \partial^\alpha f \cdot \frac{d\tau}{df} \cdot \frac{d}{d\tau} \delta(f(\tau)) \\ &= \frac{x^\alpha - r^\alpha(\tau)}{[x^\lambda - r^\lambda(\tau)] u_\lambda(\tau)} \frac{d}{d\tau} \delta(\|x^\lambda - r^\lambda(\tau)\|^2) , \end{aligned} \quad (11.44)$$

as

$$\partial^\alpha f(\tau) = 2[x^\alpha - r^\alpha(\tau)] \quad \text{and} \quad \frac{df(\tau)}{d\tau} = -2[x^\lambda - r^\lambda(\tau)] \frac{dr_\lambda(\tau)}{d\tau} = -2[x^\lambda - r^\lambda(\tau)] u_\lambda(\tau) . \quad (11.45)$$

Substituting Eq. (11.44) in Eq. (11.43), we get

$$\partial^\alpha A^\mu(x^\lambda) = -2q \int d\tau \frac{[x^\alpha - r^\alpha(\tau)] u^\mu(\tau)}{[x^\lambda - r^\lambda(\tau)] u_\lambda(\tau)} \cdot \Theta(x^0 - r^0(\tau)) \cdot \frac{d}{d\tau} \delta(\|x^\lambda - r^\lambda(\tau)\|^2) . \quad (11.46)$$

Performing an integration by parts

$$\begin{aligned} \partial^\alpha A^\mu(x^\lambda) &= -2q \left[\frac{(x^\alpha - r^\alpha) u^\mu}{(x^\lambda - r^\lambda) u_\lambda} \right] \Theta(x^0 - r^0) \delta(\|x^\lambda - r^\lambda\|^2) \\ &\quad + 2q \int d\tau \frac{d}{d\tau} \left\{ \left[\frac{(x^\alpha - r^\alpha) u^\mu}{(x^\lambda - r^\lambda) u_\lambda} \right] \Theta(x^0 - r^0) \right\} \delta(\|x^\lambda - r^\lambda\|^2) \\ &= 2q \int d\tau \frac{d}{d\tau} \left[\frac{(x^\alpha - r^\alpha) u^\mu}{(x^\lambda - r^\lambda) u_\lambda} \right] \Theta(x^0 - r^0) \delta(\|x^\lambda - r^\lambda\|^2) \\ &\quad + 2q \int d\tau \frac{d\Theta(x^0 - r^0)}{d\tau} \left[\frac{(x^\alpha - r^\alpha) u^\mu}{(x^\lambda - r^\lambda) u_\lambda} \right] \delta(\|x^\lambda - r^\lambda\|^2) . \end{aligned} \quad (11.47)$$

0 , at the boundaries

As the derivative of the Theta function again takes out the delta function for $R \neq 0$, the second term on the r.h.s. vanishes, and we have

$$\partial^\alpha A^\mu(x^\lambda) = 2q \int d\tau \frac{d}{d\tau} \left\{ \frac{[x^\alpha - r^\alpha(\tau)] u^\mu(\tau)}{[x^\lambda - r^\lambda(\tau)] u_\lambda(\tau)} \right\} \cdot \Theta(x^0 - r^0(\tau)) \cdot \delta(\|x^\lambda - r^\lambda(\tau)\|^2) . \quad (11.48)$$

The integrand is now similar to that in Eq. (11.4), with $u^\mu(\tau)$ being replaced by the derivative term in Eq. (11.48). The end result should also be similar [to Eq. (11.9)]:

$$\partial^\alpha A^\mu(x^\lambda) = \underline{\underline{\left[\frac{q}{[x^\lambda - r^\lambda(\tau)] u_\lambda(\tau)} \cdot \frac{d}{d\tau} \left(\frac{[x^\alpha - r^\alpha(\tau)] u^\mu(\tau)}{[x^\lambda - r^\lambda(\tau)] u_\lambda(\tau)} \right) \right]_{ret}}} , \quad (11.49)$$

whence

$$F^{\alpha\beta}(x^\lambda) = \left[\frac{q}{[x^\lambda - r^\lambda(\tau)] u_\lambda(\tau)} \cdot \frac{d}{d\tau} \left(\frac{[x^\alpha - r^\alpha(\tau)] u^\beta(\tau) - [x^\beta - r^\beta(\tau)] u^\alpha(\tau)}{[x^\lambda - r^\lambda(\tau)] u_\lambda(\tau)} \right) \right]_{ret} . \quad (11.50)$$

Although this expression is manifestly covariant, it is often useful to reduce this and get the expressions for the (retarded) electric and magnetic field vectors $\vec{E}$ and $\vec{B}$. This is however a tedious (albeit straightforward) exercise. One needs to make use of the following relations:

$$x^\alpha - r^\alpha := (R, R\hat{n}) , \quad u^\alpha := \gamma c \left(1, \vec{\beta} \right) , \quad \text{and} \quad \frac{du^\alpha}{d\tau} := \gamma^2 c \left[\gamma^2 \vec{\beta} \cdot \dot{\vec{\beta}}, \dot{\vec{\beta}} + \gamma^2 \vec{\beta} \left(\vec{\beta} \cdot \dot{\vec{\beta}} \right) \right] , \quad (11.51)$$

to work out the derivative factor in Eq. (11.50), and after a series of simplifications finally determine F^{0i} and F^{ij} . The result would be given by the equations (11.39), which we leave for the reader to verify.

11.2.4 Acceleration-dependent parts of the fields and Electromagnetic Radiation

Let us now decompose the electric and magnetic fields, given by Eqs. (11.39), into the acceleration-dependent and the acceleration-independent parts (denoted by subscripts a and v respectively):

$$\boxed{\begin{aligned} \vec{E} &= \vec{E}_a + \vec{E}_v ; \quad \vec{B} = \vec{B}_a + \vec{B}_v , \\ \vec{E}_a &= \frac{q}{cR} \frac{\hat{n} \times [(\hat{n} - \vec{\beta}) \times \dot{\vec{\beta}}]}{(1 - \hat{n} \cdot \vec{\beta})^3} ; \quad \vec{E}_v = \frac{q}{R^2} \frac{(1 - \beta^2)(\hat{n} - \vec{\beta})}{(1 - \hat{n} \cdot \vec{\beta})^3} , \quad \vec{B}_a = \hat{n} \times \vec{E}_a ; \quad \vec{B}_v = \hat{n} \times \vec{E}_v \end{aligned}} . \quad (11.52)$$

For convenience, we have omitted the retarded brackets $[]_{ret}$, considering them to be implicit.

Now, the acceleration-dependent fields satisfy:

$$(i) \quad \underline{\underline{\vec{B}_a \cdot \vec{E}_a = 0}} = \underline{\underline{\hat{n} \cdot \vec{B}_a = \hat{n} \cdot \vec{E}_a}} \quad \text{and} \quad (ii) \quad \underline{\underline{|\vec{B}_a| \sim |\vec{E}_a| \sim \frac{1}{R}}} , \quad (11.53)$$

whereas, the acceleration-independent fields satisfy:

$$(i) \quad \underline{\underline{\vec{B}_v \cdot \vec{E}_v = 0}} = \underline{\underline{\hat{n} \cdot \vec{B}_v}} \quad \text{but} \quad \underline{\underline{\hat{n} \cdot \vec{E}_v \neq 0}} \quad \text{and} \quad (ii) \quad \underline{\underline{|\vec{B}_v| \sim |\vec{E}_v| \sim \frac{1}{R^2}}} . \quad (11.54)$$

The acceleration-dependent terms are the only ones responsible for electromagnetic radiation. This is evident from the following:

- $\vec{E}_a, \vec{B}_a$ form an orthogonal set including the propagation vector, which in this case is $\hat{n} = \vec{R}/R$. $\vec{E}_v, \vec{B}_v$ and $\hat{n}$, however, do not form an orthogonal set.
- For the acceleration-dependent fields, the magnitude of the Poynting vector is given by

$$|\vec{S}|_a = \frac{c}{4\pi} |\vec{E}_a \times \vec{B}_a| = \frac{c}{4\pi} |\vec{E}_a|^2 \sim \frac{1}{R^2} , \quad (11.55)$$

so that the amount of energy crossing a spherical closed surface of radius R is equal to

$$|\vec{S}|_a \cdot 4\pi R^2 \sim \frac{1}{R^2} \cdot 4\pi R^2 \sim 1 \quad (\text{independent of } R) . \quad (11.56)$$

Therefore the total energy remains the same as the radius R is varied — *Conservation of Energy*.

For the acceleration-independent fields, the magnitude of the Poynting vector is

$$|\vec{S}|_v = \frac{c}{4\pi} |\vec{E}_v \times \vec{B}_v| = \frac{c}{4\pi} |\vec{E}_v|^2 \sim \frac{1}{R^4} , \quad (11.57)$$

so that $|\vec{S}|_v$ integrated over a closed spherical surface of radius R is

$$|\vec{S}|_v \cdot 4\pi R^2 \sim \frac{1}{R^4} \cdot 4\pi R^2 \sim \frac{1}{R^2} . \quad (11.58)$$

As $R \rightarrow \infty$, the total energy goes to zero — *not in support of Radiation*.

11.3 Power radiated by an accelerated charge

11.3.1 Non-relativistic radiant power formula — the Larmor result

For non-relativistic motion, $\beta = v/c \ll 1$, and one can approximate the acceleration-dependent fields [cf Eqs. (11.52)] as:

$$\underline{\underline{\vec{E}_a \approx \frac{q}{cR} \hat{n} \times (\hat{n} \times \dot{\vec{\beta}})}} \quad \text{and} \quad \underline{\underline{\vec{B}_a = \hat{n} \times \vec{E}_a}} . \quad (11.59)$$

The instantaneous energy flux is given by the Poynting vector

$$\vec{S} = \frac{c}{4\pi} \vec{E}_a \times \vec{B}_a = \frac{c}{4\pi} |\vec{E}_a|^2 \hat{n} = \frac{c}{4\pi} \frac{q^2}{c^2 R^2} \left| \hat{n} \times (\hat{n} \times \dot{\vec{\beta}}) \right|^2 \hat{n} . \quad (11.60)$$

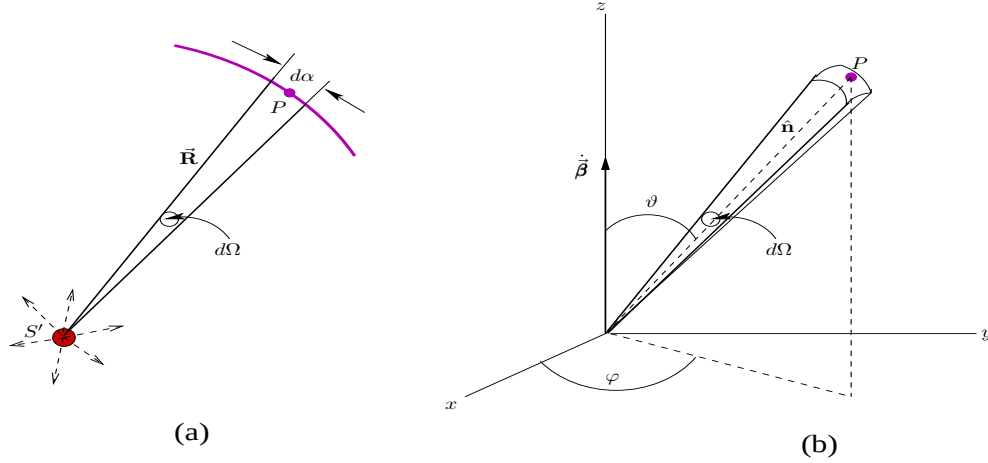


Figure 17: Power radiated by an accelerated point charge.

Amount of radiation energy crossing unit area in unit time is therefore

$$\frac{d\mathcal{P}}{d\alpha} = |\vec{\mathcal{S}}| = \frac{q^2 \dot{\beta}^2}{4\pi c R^2} \sin^2 \vartheta, \quad (11.61)$$

where ϑ is the angle between the observation direction $\hat{n}$ and the acceleration direction $\vec{\beta}$ (see Fig. 17).

Now, $d\alpha = R^2 d\Omega$, where $d\Omega$ is the solid angle subtended by the infinitesimal area element $d\alpha$ at the retarded source location S' . Hence, the power radiated per unit solid angle is given by

$$\frac{d\mathcal{P}}{d\Omega} = |\vec{\mathcal{S}}| \cdot R^2 = \frac{q^2 \dot{\beta}^2}{4\pi c} \sin^2 \vartheta, \quad (11.62)$$

Choosing $\vec{\beta}$ to be in the z -direction, we have $d\Omega = \sin \vartheta d\vartheta d\varphi$, whence the instantaneous power radiated is evaluated as

$$\mathcal{P} = \frac{q^2 \dot{\beta}^2}{4\pi c} \int_0^\pi \int_0^{2\pi} \sin^3 \vartheta d\vartheta d\varphi = \frac{q^2 \dot{\beta}^2}{4\pi c} \cdot 2\pi \cdot \frac{4}{3} \implies \boxed{\mathcal{P} = \frac{2}{3} \frac{q^2 \dot{\beta}^2}{c} = \frac{2}{3} \frac{q^2 \dot{v}^2}{c^3}}. \quad (11.63)$$

This is the *Larmor formula* for the total power radiated by a non-relativistic accelerated charge. It is also convenient to express this as

$$\boxed{\mathcal{P} = \frac{2}{3} \frac{q^2}{m^2 c^3} \left[\frac{d\vec{p}_{NR}}{dt} \cdot \frac{d\vec{p}_{NR}}{dt} \right]}, \quad (11.64)$$

where $\vec{p}_{NR} = m\vec{v}$ is the non-relativistic three-momentum, m being the rest mass.

11.3.2 Relativistic radiant power formula — the Liénard result

For relativistic motion, one may generalize the above Larmor expression (11.64) for the total power radiated $\mathcal{P}$, taking into consideration the following criteria:

- $\mathcal{P}$ should be a Lorentz scalar,
- $\mathcal{P}$ should reduce to Eq. (11.63) [or (11.64)] in the limit of small velocities $\beta = v/c \rightarrow 0$.

A suggested generalization of Eq. (11.64) is

$$\mathcal{P} = \frac{2}{3} \frac{q^2}{m^2 c^3} \left[-\frac{dP^\mu}{d\tau} \frac{dP_\mu}{d\tau} \right] = \frac{2}{3} \frac{q^2}{m^2 c^3} \left[-\frac{1}{c^2} \left(\frac{d\mathcal{E}}{d\tau} \right)^2 + \left(\frac{d\vec{p}}{d\tau} \right)^2 \right], \quad (11.65)$$

where τ is the charge's proper time [$d\tau = dt/\gamma$, $\gamma = (1 - \beta^2)^{-1/2}$], and $P^\mu := (\mathcal{E}, \vec{p})$ is the four momentum [$\vec{p} = \gamma m\vec{v}$ being the relativistic three-momentum and $\mathcal{E} = \gamma mc^2$ is the total relativistic energy]. The above

expression (11.65) is, by construction, Lorentz invariant. To see whether it reduces to Eq. (11.63) [or (11.64)] in the limit $\beta = v/c \rightarrow 0$, we proceed in the following way. We have

$$\mathcal{E}^2 = p^2 c^2 + m^2 c^4 \quad \implies \quad 2\mathcal{E} \frac{d\mathcal{E}}{d\tau} = 2pc^2 \frac{dp}{d\tau} \quad \implies \quad \underline{\underline{\frac{d\mathcal{E}}{d\tau} = \frac{p}{\mathcal{E}} c^2 \frac{dp}{d\tau} = v \frac{dp}{d\tau}}}, \quad (11.66)$$

as,

$$\frac{p}{\mathcal{E}} = \frac{\gamma m v}{\gamma m c^2} = \frac{v}{c^2}. \quad (11.67)$$

Therefore, Eq. (11.65) reduces to

$$\underline{\underline{\mathcal{P} = \frac{2}{3} \frac{q^2}{m^2 c^3} \left[\left(\frac{d\vec{p}}{d\tau} \right)^2 - \beta^2 \left(\frac{dp}{d\tau} \right)^2 \right]}}. \quad (11.68)$$

In the non-relativistic limit ($\beta = v/c \rightarrow 0$):

$$d\tau \rightarrow dt, \quad \vec{p} \rightarrow \vec{p}_{NR}, \quad \text{whence} \quad \mathcal{P} \rightarrow \frac{2}{3} \frac{q^2}{m^2 c^3} \left(\frac{d\vec{p}_{NR}}{dt} \right)^2 : \quad \text{the Larmor formula}. \quad (11.69)$$

So, Eq. (11.65) [or (11.68)] indeed gives the total power radiated by a relativistic accelerated charge.

Now, since $\vec{p} = \gamma m \vec{v} = \gamma m c \vec{\beta}$, we have

$$\frac{d\vec{p}}{d\tau} = \gamma \frac{d\vec{p}}{dt} = \gamma m c \frac{d}{dt} (\gamma \vec{\beta}) = \gamma m c (\dot{\gamma} \vec{\beta} + \gamma \dot{\vec{\beta}}). \quad (11.70)$$

Substituting

$$\dot{\gamma} \equiv \frac{d\gamma}{dt} = \frac{d}{dt} (1 - \vec{\beta} \cdot \vec{\beta})^{-1/2} = \gamma^3 \vec{\beta} \cdot \dot{\vec{\beta}} = \gamma^3 \beta \dot{\beta} \cos \chi, \quad (11.71)$$

where χ is the angle between $\vec{\beta}$ and $\dot{\vec{\beta}}$, and squaring, one obtains (after some algebraic simplifications):

$$\left(\frac{d\vec{p}}{d\tau} \right)^2 = \gamma^6 m^2 c^2 \dot{\beta}^2 (1 - \beta^2 \sin^2 \chi) + \gamma^2 \dot{\gamma}^2 m^2 c^2. \quad (11.72)$$

Moreover,

$$\mathcal{E} = \gamma m c^2 \quad \implies \quad \frac{d\mathcal{E}}{d\tau} = \gamma \frac{d\mathcal{E}}{dt} = \gamma m c^2 \dot{\gamma}. \quad (11.73)$$

Substituting Eqs. (11.72) and (11.73) in Eq. (11.65) and simplifying, one finally gets

$$\boxed{\mathcal{P} = \frac{2}{3} \frac{q^2}{c} \gamma^6 \left[(\dot{\vec{\beta}})^2 - (\vec{\beta} \times \dot{\vec{\beta}})^2 \right]}. \quad (11.74)$$

This is the *Liénard's expression* for the total power radiated by an accelerated charge in relativistic motion.

11.4 Charged particle accelerators

The major area of the above formalism is that of the charged particle accelerators. The radiation losses in these accelerators provide the limiting factor in the maximum practical energy attainable.

Since for a given applied force $\vec{F} = d\vec{p}/dt$, the radiative power $\mathcal{P} \sim 1/m^2$, the radiative effects are optimized for the particles with the least rest mass, i.e. the electrons ($m \equiv m_e$, the electron's rest mass; $q \equiv e$, the electronic charge).

11.4.1 Linear accelerator

The common examples are the Van de Graff generator and the Cockroft and Walton's chamber. The particles are accelerated along their directions of motion, i.e.

$$\vec{v} \parallel \dot{\vec{v}} \quad \implies \quad \vec{\beta} \times \dot{\vec{\beta}} = 0, \quad (11.75)$$

where $\vec{v}$ denotes the particle velocity, and $\dot{\vec{v}} = d\vec{v}/dt$ is the acceleration.

Expressing the momentum $\vec{p} = p\hat{p}$, we have for the linear accelerator, the unit vector $\hat{p}$ being constant,

$$\frac{d\vec{p}}{d\tau} = \frac{dp}{d\tau}\hat{p} \implies \left(\frac{d\vec{p}}{d\tau}\right)^2 = \left(\frac{dp}{d\tau}\right)^2. \quad (11.76)$$

Substituting Eq. (11.76) in Eq. (11.68) we get the total power emitted by an accelerated electron:

$$\mathcal{P} = \frac{2}{3} \frac{e^2}{m_e^2 c^3} (1 - \beta^2) \left(\frac{dp}{d\tau}\right)^2 \implies \underline{\underline{\mathcal{P} = \frac{2}{3} \frac{e^2}{m_e^2 c^3} F^2}}, \quad (11.77)$$

where $F = dp/dt = |\vec{F}|$ is the magnitude of the applied force, and we have used the relations $(1 - \beta^2) = 1/\gamma^2$ and $dp/d\tau = \gamma dp/dt$.

Now, the infinitesimal change in the electron's energy $\delta\mathcal{E}$ corresponding to an infinitesimal displacement $\delta\vec{x}$ by the application of the force $\vec{F}$ is given by

$$\delta\mathcal{E} = \vec{F} \cdot \delta\vec{x} = F \delta x \implies \underline{\underline{F = \frac{dp}{dt} = \frac{d\mathcal{E}}{dx}}}. \quad (11.78)$$

Hence, for linear motion, the power emitted depends only on the external forces that determine the rate of change of the particle's energy with distance, and not on the actual energy or momentum of the particle.

$$\boxed{\mathcal{P} = \frac{2}{3} \frac{e^2}{m_e^2 c^3} \left(\frac{d\mathcal{E}}{dx}\right)^2}. \quad (11.79)$$

The quantity of interest is the ratio of power emitted and the power supplied by the external sources, i.e.

$$\frac{\mathcal{P}}{d\mathcal{E}/dt} = \frac{\text{Radiation loss per unit time}}{\text{Energy gain per unit time}} = \frac{2}{3} \frac{e^2}{m_e^2 c^3} \frac{(d\mathcal{E}/dx)^2}{(d\mathcal{E}/dx)(dx/dt)}. \quad (11.80)$$

Or, since $dx/dt = v = c\beta$ is the speed of the electron, we have

$$\boxed{\frac{\mathcal{P}}{d\mathcal{E}/dt} = \frac{2}{3} \frac{e^2}{m_e^2 c^4 \beta} \left(\frac{d\mathcal{E}}{dx}\right)}. \quad (11.81)$$

For relativistic particles ($\beta \sim 1$), this ratio is given by

$$\lim_{\beta \rightarrow 1} \frac{\mathcal{P}}{d\mathcal{E}/dt} = \frac{2}{3} \frac{e^2}{m_e^2 c^4} \left(\frac{d\mathcal{E}}{dx}\right) = \underline{\underline{\frac{2}{3} \frac{r_e}{\mathcal{E}_e} \left(\frac{d\mathcal{E}}{dx}\right)}}, \quad (11.82)$$

where

$$\begin{aligned} r_e &= \frac{e^2}{m_e c^2} = 2.82 \times 10^{-15} \text{ metres} \quad : \quad \text{Classical electron radius,} \\ \mathcal{E}_e &= m_e c^2 = 0.511 \text{ MeV} \quad : \quad \text{Rest energy of the electron.} \end{aligned} \quad (11.83)$$

Using the values of r_e and $\mathcal{E}_e$, one has

$$\underline{\underline{\lim_{\beta \rightarrow 1} \frac{\mathcal{P}}{d\mathcal{E}/dt} \sim 10^{-14} \left(\frac{d\mathcal{E}}{dx}\right)}}, \quad (11.84)$$

which means that the radiation loss is significant (i.e. $\mathcal{P} \sim d\mathcal{E}/dt$) for relativistic electrons ($\beta \sim 1$) only when $d\mathcal{E}/dx \sim 10^{14}$ MeV/m. However, in standard linear accelerators, the energy gradient $d\mathcal{E}/dx \lesssim 50$ MeV/m and the energy losses are negligible.

11.4.2 Circular accelerator

The common examples are the Betatron, the Cyclotron, the Synchrotron and the Synchrocyclotron. The particles are accelerated along a circular path, i.e. the velocity $\vec{v}$ is tangential, whereas the centripetal acceleration $\dot{\vec{v}}$ is along the radius. Thus

$$\vec{v} \perp \dot{\vec{v}} \implies |\vec{\beta} \times \dot{\vec{\beta}}| = \beta \dot{\beta}. \quad (11.85)$$

The general setup is that in which the particle's momentum $\vec{p}$ changes rapidly in direction, but not in magnitude, i.e.

$$\frac{dp}{d\tau} \sim \text{small} \implies \frac{d\mathcal{E}}{d\tau} \sim \text{small}, \quad \text{whereas} \quad \frac{d\vec{p}}{d\tau} \text{ is significant.} \quad (11.86)$$

Therefore,

$$\underline{\underline{\left(\frac{d\vec{p}}{d\tau}\right)^2}} \gg \frac{1}{c^2} \left(\frac{d\mathcal{E}}{d\tau}\right)^2, \quad (11.87)$$

as the energy gain per revolution, $d\mathcal{E}/d\tau$, already being small, the square of that amount divided by c^2 is even more negligible compared to $(d\vec{p}/d\tau)^2$.

The expression (11.65) for the radiated power is approximated to

$$\mathcal{P} \approx \frac{2}{3} \frac{e^2}{m_e^2 c^3} \left(\frac{d\vec{p}}{d\tau}\right)^2 \implies \underline{\underline{\mathcal{P} \approx \frac{2}{3} \frac{e^2 \gamma^2}{m_e^2 c^3} F^2}}, \quad (11.88)$$

where

$$\underline{\underline{F \equiv |\vec{F}| = \left|\frac{d\vec{p}}{dt}\right| = \frac{\gamma m_e v^2}{r}}}, \quad (11.89)$$

is the magnitude of the centrifugal force, r being the radius of the orbit, and γm_e is the effective (relativistic) mass of the electron in motion.

Plugging Eq. (11.89) back in Eq. (11.88), and using $v = c\beta$, we obtain the power emitted in a circular accelerator given approximately as

$$\boxed{\mathcal{P} \approx \frac{2}{3} \frac{e^2 c \beta^4 \gamma^4}{r^2}}. \quad (11.90)$$

The radiative energy loss per revolution is

$$\delta\mathcal{E} = \mathcal{P} \delta t, \quad \text{with} \quad \delta t = \frac{2\pi r}{v} = \frac{2\pi r}{c\beta} : \quad \text{Time period for each revolution.} \quad (11.91)$$

Using Eq. (11.90), and the relation $\gamma = \mathcal{E}/(m_e c^2)$, we have

$$\underline{\underline{\delta\mathcal{E} \approx \frac{4\pi}{3} \frac{e^2 \beta^3 \gamma^4}{r} = \frac{4\pi}{3} \frac{e^2 \beta^3 \mathcal{E}^4}{(m_e c^2)^4 r}}}. \quad (11.92)$$

For relativistic electrons ($\beta \sim 1$), this ratio is given by

$$\underline{\underline{\lim_{\beta \rightarrow 1} \delta\mathcal{E} = \frac{4\pi}{3} \frac{r_e}{\mathcal{E}_e^3} \frac{\mathcal{E}^4}{r}}}. \quad (11.93)$$

Or, using the values of r_e and $\mathcal{E}_e$ given in Eq. (11.83),

$$\underline{\underline{\lim_{\beta \rightarrow 1} \delta\mathcal{E} [\text{MeV}] = 8.85 \times 10^{-2} \frac{(\mathcal{E} [\text{GeV}])^4}{r [\text{metres}]}}}}. \quad (11.94)$$

In the first electron synchrotron: $r \sim 1$ m, and the maximum value of supplied energy $\mathcal{E}_{\max} \approx 0.3$ GeV. By Eq. (11.94), the maximum value of the energy emitted per revolution by relativistic electrons is

$$\delta\mathcal{E}_{\max} = 8.85 \times 10^{-2} \times \frac{(0.3)^4}{1} \approx 0.7 \text{ KeV}, \quad (11.95)$$

which is less than, but not negligible compared to energy gain, that typically corresponds to a voltage of a few Kilovolts per turn.

In the 10 GeV Cornell electron synchrotron: $r \sim 100$ m. The radiation loss per turn is therefore

$$\delta\mathcal{E}_{\max} = 8.85 \times 10^{-2} \times \frac{10^4}{100} = 8.85 \text{ MeV}, \quad (11.96)$$

whereas the radio-frequency voltage is 10.5 Megavolts at 10 GeV, which just provides the requisite power to overcome the radiation loss, so as to maintain the electron beams at fixed energy as they circulate. In this context, the maximum magnetic field of the electromagnet, which is used to bend the particle's track into a circle, is ~ 3.3 KiloGauss.

11.5 A comparison of the linear and circular acceleration cases

Let us recall the *Liénard's expression* (11.74) for the total power radiated by a relativistic accelerated charge. This can be expressed as

$$\boxed{\mathcal{P} = \frac{2q^2}{3c} \gamma^6 \dot{\beta}^2 (1 - \beta^2 \sin^2 \chi)}, \quad (11.97)$$

where χ is the angle between $\vec{\beta}$ and $\dot{\vec{\beta}}$.

I. For a given (fixed) non-relativistic force:

$$\vec{F}_{NR} = \frac{d\vec{p}_{NR}}{dt} = \frac{d}{dt}(m\vec{v}) : \text{fixed} \implies \dot{\vec{v}} : \text{fixed} \implies \underline{\underline{\dot{\vec{\beta}} : \text{fixed}}}. \quad (11.98)$$

(a) **Linear acceleration:** implies $\chi = 0$, whence by Eq. (11.97) the total radiated power is

$$\underline{\underline{\mathcal{P}_L = \frac{2q^2}{3c} \gamma^6 \dot{\beta}^2}}. \quad (11.99)$$

(a) **Circular acceleration:** implies $\chi = \pi/2$, whence by Eq. (11.97) the total radiated power is

$$\mathcal{P}_C = \frac{2q^2}{3c} \gamma^6 \dot{\beta}^2 (1 - \beta^2) \implies \underline{\underline{\mathcal{P}_C = \frac{2q^2}{3c} \gamma^4 \dot{\beta}^2}}. \quad (11.100)$$

Therefore, for a fixed $\vec{F}_{NR}$, we have in the non-relativistic limit ($\beta \rightarrow 0$, i.e. $\gamma \rightarrow 1$):

$$\underline{\underline{\frac{\mathcal{P}_C}{\mathcal{P}_L} \sim \frac{1}{\gamma^2} \xrightarrow{\beta \rightarrow 0} 1}}. \quad (11.101)$$

II. For a given (fixed) force:

$$\vec{F} = \frac{d\vec{p}}{dt} = \frac{d}{dt}(\gamma m \vec{v}) : \text{fixed} \implies \frac{d}{dt}(\gamma \vec{v}) : \text{fixed} \implies \underline{\underline{\frac{d}{dt}(\gamma \vec{\beta}) : \text{fixed}}}. \quad (11.102)$$

Since

$$\frac{d}{dt}(\gamma \vec{\beta}) = \gamma \dot{\vec{\beta}} + \beta \dot{\gamma} = \gamma \dot{\vec{\beta}} (1 + \gamma^2 \beta^2 \cos \chi), \quad (11.103)$$

by the use of Eq. (11.71), we have after some simplification

$$\underline{\underline{\dot{\beta}^2 = [\gamma^2 (1 + 2\gamma^2 \beta^2 \cos^2 \chi + \gamma^4 \beta^4 \cos^2 \chi)]^{-1} \left[\frac{d}{dt}(\gamma \vec{\beta}) \right]^2}}. \quad (11.104)$$

(a) **Linear acceleration:** implies $\chi = 0$, whence Eq. (11.104) reduces to

$$\dot{\beta}^2 = [\gamma (1 + \gamma^2 \beta^2)]^{-2} \left[\frac{d}{dt}(\gamma \vec{\beta}) \right]^2 = \gamma^{-6} \left[\frac{d}{dt}(\gamma \vec{\beta}) \right]^2, \quad (11.105)$$

as $\gamma^2 \beta^2 = \gamma^2 - 1$. The radiated power is then given by

$$\underline{\underline{\mathcal{P}_L = \frac{2q^2}{3c} \left[\frac{d}{dt}(\gamma \vec{\beta}) \right]^2}}. \quad (11.106)$$

(a) **Circular acceleration:** implies $\chi = \pi/2$, whence Eq. (11.104) reduces to

$$\dot{\beta}^2 = \gamma^{-2} \left[\frac{d}{dt}(\gamma \vec{\beta}) \right]^2. \quad (11.107)$$

The radiated power in this case is

$$\underline{\underline{\mathcal{P}_C}} = \frac{2q^2}{3c} \gamma^2 \left[\frac{d}{dt} (\gamma \vec{\beta}) \right]^2. \quad (11.108)$$

Thus, for a fixed $\vec{F}$, we have in the extreme relativistic limit ($\beta \rightarrow 1$, i.e. $\gamma \rightarrow \infty$):

$$\underline{\underline{\frac{\mathcal{P}_C}{\mathcal{P}_L}}} \sim \gamma^2 \xrightarrow{\beta \rightarrow 1} \infty. \quad (11.109)$$

Appendices

A The general Lorentz Boost Transformation matrix elements

The general Lorentz boost with velocity $\vec{v}$ in an arbitrary direction is the resultant of three boosts (v^1, v^2, v^3) along the (x^1, x^2, x^3) directions respectively. In order to determine the elements of the corresponding Lorentz transformation matrix, viz. Λ^μ_ν ($\forall \mu, \nu = 0, 1, 2, 3$), let us resort to the following:

I. Lorentz transformation equations (in differential form):

$$\boxed{dx'^\mu = \Lambda^\mu_\nu dx^\nu = \Lambda^\mu_0 dx^0 + \Lambda^\mu_i dx^i} . \quad (\text{A.1})$$

II. Invariance of the metric under Lorentz transformations:

$$\begin{aligned} ds^2 = ds'^2 &\implies \eta_{\mu\nu} dx^\mu dx^\nu = \eta_{\alpha\beta} dx'^\alpha dx'^\beta = \eta_{\alpha\beta} (\Lambda^\alpha_\mu dx^\mu) (\Lambda^\beta_\nu dx^\nu) \quad [\text{by Eqs. (A.1)}] \\ &\implies \boxed{\eta_{\mu\nu} = \eta_{\alpha\beta} \Lambda^\alpha_\mu \Lambda^\beta_\nu} \end{aligned} \quad (\text{A.2})$$

Let us now suppose a particle to be at rest w.r.to the frame Σ , i.e.,

$$dx^i = 0 \quad (i = 1, 2, 3) .$$

Then, since the origin of the frame Σ' moves relative to the frame Σ with constant velocity $\vec{v}$, the particle is seen to move with velocity $-\vec{v}$ in the frame Σ' , i.e.,

$$\frac{dx'^i}{dt'} = -v^i \quad (i = 1, 2, 3)$$

From Eqs. (A.1) we get

$$\begin{aligned} dx'^0 &= \Lambda^0_0 dx^0, & dx'^i &= \Lambda^i_0 dx^0 \quad (i = 1, 2, 3), \\ \implies \underline{\underline{\Lambda^i_0}} &= \underline{\underline{-\beta^i \Lambda^0_0}}, & \left[\text{where } \beta^i &= \frac{v^i}{c} = -\frac{1}{c} \frac{dx'^i}{dt'} = -\frac{dx'^i}{dx'^0} \right]. \end{aligned} \quad (\text{A.3})$$

Moreover, by setting $\mu = \nu = 0$ in Eq. (A.2), we have

$$\eta_{00} = \eta_{\alpha\beta} \Lambda^\alpha_0 \Lambda^\beta_0 = \eta_{00} (\Lambda^0_0)^2 + \eta_{ij} \Lambda^i_0 \Lambda^j_0 \implies \underline{\underline{(\Lambda^0_0)^2 - \Lambda^i_0 \Lambda^j_0 \delta_{ij} = 1}} . \quad (\text{A.5})$$

Solving Eqs. (A.3) and (A.5) simultaneously, one obtains:

$$\boxed{\Lambda^0_0 = \gamma \quad , \quad \Lambda^i_0 = -\gamma \beta^i} , \quad (\text{A.6})$$

where

$$\gamma = (1 - \delta_{ij} \beta^i \beta^j)^{-1/2} = (1 + \eta_{ij} \beta^i \beta^j)^{-1/2} = (1 + \beta^i \beta_i)^{-1/2} = \left(1 - |\vec{\beta}|^2\right)^{-1/2} . \quad (\text{A.7})$$

The other elements of the LT matrix Λ^μ_ν are not uniquely determined. One convenient choice is however

$$\boxed{\Lambda^0_i = -\gamma \delta_{ij} \beta^j = \gamma \beta_i \quad , \quad \Lambda^i_j = \delta^i_j + \frac{(\gamma - 1)}{|\vec{\beta}|^2} \delta_{jk} \beta^i \beta^k = \delta^i_j - \frac{(\gamma - 1)}{|\vec{\beta}|^2} \beta^i \beta_j} . \quad (\text{A.8})$$

We leave it to the reader to verify that these indeed satisfy Eq. (A.2).

B The general expression for Relativistic Velocity Addition

Refer back to the Fig 6. In order to determine the particle's velocity $\vec{v}''$ w.r.to the frame Σ in terms of the relative frame velocity $\vec{v}$ and the particle's velocity $\vec{v}'$ w.r.to Σ' are known, let us re-write Eq. (3.45), derived in §3.7, as

$$\Lambda^\mu_\nu(\vec{\beta}'') = \Lambda^\mu_0(\vec{\beta}') \Lambda^0_\nu(\vec{\beta}) + \Lambda^\mu_i(\vec{\beta}') \Lambda^i_\nu(\vec{\beta}), \quad (\text{B.1})$$

and recall from Appendix §A, that the transformation matrix elements for a generic Lorentz boost, in an arbitrary direction, are given by the equations (A.6) and (A.8). Use these to obtain the (0 - 0) and (i - 0) component equations of (B.1) as

$$\Lambda^0_0(\beta''^k) = \Lambda^0_0(\beta'^k) \Lambda^0_0(\beta^k) + \Lambda^0_i(\beta'^k) \Lambda^i_0(\beta^k) \implies \underline{\underline{\gamma'' = \gamma' \gamma (1 + \delta_{ik} \beta^i \beta'^k)}}, \quad (\text{B.2})$$

$$\begin{aligned} \Lambda^i_0(\beta''^k) &= \Lambda^i_0(\beta'^k) \Lambda^0_0(\beta^k) + \Lambda^i_j(\beta'^k) \Lambda^j_0(\beta^k) \\ \implies \underline{\underline{\beta''^i &= \frac{\gamma' \gamma}{\gamma''} \left[(\beta'^i + \beta^i) - \left(1 - \frac{1}{\gamma'} \right) \left(\beta^i - \frac{\delta_{jk} \beta^j \beta'^k}{|\vec{\beta}'|^2} \beta^i \right) \right]}}. \end{aligned} \quad (\text{B.3})$$

Note however that

$$\delta_{il} \beta'^l \left(\beta^i - \frac{\delta_{jk} \beta^j \beta'^k}{|\vec{\beta}'|^2} \beta^i \right) = \vec{\beta} \cdot \vec{\beta}' - \frac{(\vec{\beta} \cdot \vec{\beta}') (\vec{\beta}' \cdot \vec{\beta}')}{|\vec{\beta}'|^2} = 0. \quad (\text{B.4})$$

Since $\delta_{il} \beta'^l$ is arbitrary and not necessarily zero, we have

$$\underline{\underline{\beta^i - \frac{\delta_{jk} \beta^j \beta'^k}{|\vec{\beta}'|^2} \beta^i = 0.}} \quad (\text{B.5})$$

Substituting this in Eq. (B.3), and using Eq. (B.2), gives

$$\boxed{\beta''^i = \frac{\beta'^i + \beta^i}{1 + \delta_{jk} \beta^j \beta'^k}} \implies \boxed{\vec{v}'' = \frac{\vec{v}' + \vec{v}}{1 + (\vec{v}' \cdot \vec{v}/c^2)}}. \quad (\text{B.6})$$

This is the general formula for the relativistic velocity addition.